

Public School Teachers' Pension and Retirement Fund of Chicago

Actuarial Valuation Report as of June 30, 2020





October 12, 2020

Board of Trustees
Public School Teachers' Pension and Retirement Fund of Chicago
425 S. Financial Place, Suite 1400
Chicago, Illinois 60605-10000

**Re: Public School Teachers' Pension and Retirement Fund of Chicago Actuarial Valuation as of
June 30, 2020**

Dear Members of the Board:

The results of the June 30, 2020, Annual Actuarial Valuation of the Public School Teachers' Pension and Retirement Fund of Chicago (commonly known as the Chicago Teachers' Pension Fund, "CTPF" or "Fund") are presented in this report.

This report was prepared at the request of the Board and is intended for use by the Fund and those designated or approved by the Board. This report may be provided to parties other than the Fund only in its entirety and only with the permission of the Board. GRS is not responsible for unauthorized use of this report.

The purposes of the actuarial valuation are to measure the Fund's funding progress, and to determine the contribution requirements for the fiscal year beginning July 1, 2021, and ending June 30, 2022. The employer's contribution requirement has been determined in accordance with Illinois State Statutes, in particular under 40 ILCS Sections 5/17-127, 5/17-127.2, and 5/17-129. Information required by Governmental Accounting Standards Board ("GASB") Statement Nos. 67 and 68 is provided in a separate report. This report should not be relied on for any purpose other than the purposes described herein. Determinations of financial results, associated with the benefits described in this report, for purposes other than those identified above may be significantly different.

The contribution requirement in this report is determined using the actuarial assumptions and methods disclosed in Section G of this report. This report includes risk metrics beginning on page 17, but does not include a more robust assessment of the risks if future experience deviates from the actuarial assumptions. Additional assessment of risks was outside the scope of this assignment.

This actuarial valuation assumed the continuing ability of the plan sponsor to make the contributions necessary to fund this plan. A determination regarding whether or not the plan sponsor is actually able to do so is outside our scope of expertise and was not performed.

The findings in this report are based on data and other information through June 30, 2020. The actuarial valuation was based upon information furnished by CTPF staff, concerning Retirement Fund benefits, financial transactions, plan provisions and active members, terminated members, retirees and beneficiaries. We checked for internal reasonability and year-to-year consistency, but did not audit the data. We are not responsible for the accuracy or completeness of the information provided by CTPF staff.

This report was prepared using actuarial assumptions adopted by the Board as authorized under the Illinois Pension Code. The actuarial assumptions used for the June 30, 2020 actuarial valuation are based on an experience study for the five-year period from July 1, 2012 through June 30, 2017, and the 2020 Actuarial Assumption Study, performed by GRS. All actuarial assumptions used in this report are reasonable for the purposes of this actuarial valuation. Additional information about the actuarial assumptions is included in Section G of this report entitled Actuarial Methods and Assumptions.

The benefit provisions for members hired on or after January 1, 2011, were changed under Public Act 96-0889. Public Act 100-0023 created a third tier of benefits for new members provided that adoption by a resolution or ordinance occurs. Given this uncertainty, GRS has not valued the benefits provided under Public Act 100-0023. Members hired on or after this date and the assumed new hires in the projections were valued under Public Act 96-0889 benefit provisions.

Although the statutory contribution requirements were met, in our opinion the statutory funding method generates a contribution requirement that is less than a reasonable actuarially determined contribution ("ADC"). Meeting the statutory requirement does not mean that the undersigned agree that adequate actuarial funding has been achieved. We recommend the adherence to a funding policy, such as the Board policy used to calculate the ADC under GASB Statement Nos. 67 and 68, which funds the normal cost of the plan as well as an amortization payment that seeks to pay off 100 percent of the unfunded accrued liability over a closed period of 30 years, beginning July 1, 2013.

This report was prepared using our proprietary valuation model and related software which in our professional judgment has the capability to provide results that are consistent with the purposes of the valuation. We performed tests to ensure that the model reasonably represents that which is intended to be modeled. We are relying on the GRS actuaries and Internal Software, Training, and Processes Team who developed and maintain the model.

This report reflects the impact of COVID-19 through June 30, 2020. However, this report does not reflect the longer term and still developing future impact of COVID-19, which is likely to further influence demographic experience and economic expectations. We will continue to monitor these developments and their impact.

This report has been prepared by actuaries who have substantial experience valuing public employee retirement systems. To the best of our knowledge the information contained in this report is accurate and fairly presents the actuarial position of the CTPF as of the actuarial valuation date. All calculations have been made in conformity with generally accepted actuarial principles and practices, with the Actuarial Standards of Practice issued by the Actuarial Standards Board and with applicable statutes.



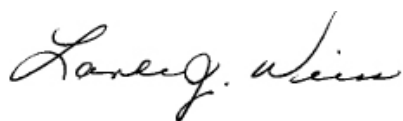
Lance J. Weiss and Amy Williams are Members of the American Academy of Actuaries and meet the American Academy of Actuaries Qualification Standards to render the actuarial opinions contained herein.

The signing actuaries are independent of the plan sponsor.

Gabriel, Roeder, Smith & Company will be pleased to review this actuarial valuation and report with the Board of Trustees and to answer any questions pertaining to the actuarial valuation.

Respectfully submitted,

GABRIEL, ROEDER, SMITH & COMPANY



Lance J. Weiss, EA, MAAA, FCA
Senior Consultant and Team Leader



Amy Williams, ASA, MAAA, FCA
Senior Consultant

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SECTION A

EXECUTIVE SUMMARY

Executive Summary

Actuarial Valuation Date:	June 30, 2020	June 30, 2019
Required Employer Contributions for Fiscal Year Ending:	June 30, 2022	June 30, 2021
Estimated Required Employer Contributions: - Required Board of Education Contributions - Additional Board of Education Contributions (0.58 percent of pay) - Additional State Contributions (0.544 percent of pay) - State Contributions Pursuant to P.A. 100-0465 (Normal Cost)^a - Total Required Employer Contributions - Percentage of Projected Capped Payroll	\$ 653,694,000 13,486,000 12,649,000 264,848,000 \$ 944,677,000 40.629%	\$ 605,852,000 13,149,000 12,333,000 254,560,000 \$ 885,894,000 39.076%
Actuarial Valuation Date:	June 30, 2020	June 30, 2019
Actuarial Information for Fiscal Year Ending:	June 30, 2021	June 30, 2020
Actuarially Determined Contribution^b as of the Actuarial Valuation Date: - Annual Amount - Percentage of Projected Capped Payroll for Upcoming Year	\$ 1,219,857,317 53.80%	\$ 1,147,030,511 52.07%
Membership: - Number of - Active Members^c - Members Receiving Payments - Vested Former Members Eligible for Deferred Benefits - Non-vested Former Members Eligible for Refunds Only - Total - Covered Payroll as of the Actuarial Valuation Date - Projected Capped Payroll for Upcoming Year - Projected Capped Payroll for Upcoming Year + 1 - Annualized Benefit Payments	30,091 28,015 10,024 21,260 89,390 \$ 2,262,053,256 \$ 2,267,288,286 \$ 2,325,129,719 \$ 1,488,748,930	29,295 28,317 9,926 20,621 88,159 \$ 2,196,918,127 \$ 2,203,055,500 \$ 2,267,106,915 \$ 1,469,042,542
Assets: - Market Value of Assets (MVA) - Actuarial Value of Assets (AVA) - Approximate Return on Market Value of Assets - Approximate Return on Actuarial Value of Assets - Ratio – Actuarial Value of Assets to Market Value of Assets	\$ 10,937,062,021 \$ 11,240,208,045 4.07% 7.06% 102.77%	\$ 11,038,837,459 \$ 11,021,811,634 4.75% 5.92% 99.85%
Basic Results: - Total Normal Cost Amount (Including Admin. Expenses) - Employer's Normal Cost Amount (Including Admin. Expenses) - Employer's Normal Cost Amount (Including Admin. Expenses and Health Insurance Subsidy)^d - Actuarial Accrued Liability (AAL) - Unfunded Actuarial Accrued Liability (UAAL) - Funded Ratio based on Actuarial Value of Assets - UAAL as % of Covered Payroll - Funded Ratio based on Market Value of Assets	\$ 399,729,564 \$ 195,673,618 \$ 260,673,618 \$ 24,073,482,607 \$ 12,833,274,562 46.69% 567.33% 45.43%	\$ 382,994,837 \$ 184,719,842 \$ 249,719,842 \$ 23,252,163,307 \$ 12,230,351,673 47.40% 556.70% 47.47%

^a State Normal Cost contribution represents the projected employer Normal Cost for fiscal years 2021 and 2022, including \$65 million for the health insurance subsidy.

^b The policy adopted by the Board calculates the Actuarially Determined Contribution (ADC) as the Normal Cost plus a 30-year level percent of payroll closed-period (beginning June 30, 2013) amortization of the Unfunded Accrued Liability. As of June 30, 2020, the remaining amortization period is 23 years. The ADC is used for financial reporting purposes only.

^c Active count excludes members expected to be hired to replace retirements and terminations that occurred in June.

^d Used for calculation of the ADC.



SECTION B

SUMMARY OF THE *ACTUARIAL VALUATION*

Summary of the Actuarial Valuation

Introduction

The law governing the Public School Teachers' Pension and Retirement Fund of Chicago requires the Actuary, as the technical advisor to the Board of Trustees to determine the amount of Board of Education contributions required for each fiscal year.

“The Board Shall determine the amount of Board of Education contributions required for each fiscal year on the basis of the actuarial tables and other assumptions adopted by the Board and the recommendations of the actuary, in order to meet the minimum contribution requirements of subsections (a) and (b). Annually, on or before February 28, the Board shall certify to the Board of Education the amount of the required Board of Education contribution for the coming fiscal year. The certification shall include a copy of the actuarial recommendations upon which it is based. (40 ILCS Section 5/17 - 129(c)).”

Gabriel, Roeder, Smith & Company has been retained by the CTPF Board of Trustees to perform an actuarial valuation as of June 30, 2020. In this report, we present the results of the actuarial valuation and the appropriation requirements under Public Act 96-0889, Public Act 90-0655, Public Act 91-0357, and Public Act 100-0465 for the fiscal year ending June 30, 2022.

Accounting information required by GASB Statement Nos. 67 and 68 is provided in a separate report.

The actuarial valuation was completed based upon membership and financial data provided by the administrative staff of the CTPF. The cost method used to determine the benefit liabilities for statutory funding is the Projected Unit Credit Cost Method as required by statute. For actuarial valuation purposes, as well as for projection purposes, the actuarial value of assets is based on a four-year smoothing method.

Assumptions and Methods

The actuarial assumptions used for the June 30, 2020, actuarial valuation were adopted by the Board during the September 17, 2020, Board meeting, and were based on the recommendations from the 2020 Actuarial Assumptions Study. The changes in actuarial assumptions include:

- Decreased the investment return assumption from 7.00 percent to 6.75 percent;
- Decreased the price inflation assumption from 2.50 percent to 2.25 percent;
- Decreased the wage inflation assumption from 3.00 percent to 2.75 percent;
- Decreased the salary increase assumption based on the lower wage inflation assumption of 2.75 percent
- Decreased the assumed increase to the pay cap for members hired on and after January 1, 2011 from 1.250 percent to 1.125 percent; and
- Decreased the assumed cost of living adjustment rate from 1.250 percent to 1.125 percent for members hired on and after January 1, 2011.

All other actuarial assumptions remain unchanged from the previous actuarial valuation and were adopted by the Board (including CPS's requested modifications) during the September 20, 2018, Board meeting, and were based on the recommendations from the experience study for the five-year period



Summary of the Actuarial Valuation

from July 1, 2012 through June 30, 2017, and the requested modifications of Chicago Public Schools (“CPS”).

The actuarial assumptions can be found in Section G of the report.

Report Highlights

The employer’s contribution requirement for FY 2022 is \$944.7 million. The 2019 actuarial valuation had projected the statutory contribution would increase from \$885.9 million for FY 2021 to \$909.1 million for FY 2022. The key reasons for the \$29.6 million increase in the employer’s contribution requirement of \$944.7 million over the projected amount from the prior actuarial valuation of \$909.1 million are the change in the actuarial assumptions and unfavorable investment experience on the market value of assets.

Over the past 10 years, CTPF experienced investment gains on a market value basis compared to the actuarial assumption in fiscal years 2011, 2013, 2014, 2017 and 2018 and investment losses in fiscal years 2012, 2015, 2016, 2019, and 2020. The return on market value for the year ending June 30, 2020, was approximately 4.07 percent compared to a return of 4.75 percent in FY 2019. The average market value investment return over the most recent 10 years has been approximately 8.59 percent. Table 10 on page 38 provides historical investment returns (on an MVA and AVA basis) over the past 25 years.

The funded ratio decreased from 47.5 percent as of June 30, 2019 to 45.4 percent as of June 30, 2020, based on the market value of assets, and decreased from 47.4 percent as of June 30, 2019, to 46.7 percent as of June 30, 2020, based on the actuarial value of assets. There are net deferred asset losses of \$298.4 million which will be recognized in the actuarial value of assets over the next three years.

The funded ratio and unfunded actuarial accrued liability are useful for assessing the need for and amount of future contributions other than normal cost contributions. They are not appropriate, however, for assessing the sufficiency of plan assets to cover the estimated cost of settling the plan's benefit obligations.

Experience during 2020

The Fund assets earned approximately 4.07 percent on a market value basis during FY 2020 which was less than the investment return assumption of 7.00 percent for FY 2020. The Fund assets earned approximately 7.06 percent on an actuarial value of assets basis during FY 2020, due to recognition of net deferred investment gains under the asset smoothing method. Since 7.06 percent is greater than the assumed rate of investment return of 7.00 percent for FY 2020, there was an asset gain of \$36.18 million on the actuarial value of assets.

There was also a net gain of \$173.43 million from actuarial liabilities, which is comprised of a gain of approximately \$55.35 million from demographic experience, and a gain of \$118.08 million from lower than expected pay increases.



Summary of the Actuarial Valuation

The total gain from liabilities for the Fund is calculated as follows (dollars in millions):

1. Actuarial Accrued Liability ("AAL") - Prior Year (Pensions Only)	\$	23,252.16
2. Total Normal Cost - Prior Year ^a		382.99
3. Benefits and Administrative Expenses Paid in FY 2020 ^b		(1,541.25)
4. Interest on the above items, 1, 2 and 3		<u>1,587.80</u>
5. Expected AAL 06/30/2020 (1+2+3+4)		23,681.70
6. Impact of Change in Actuarial Assumptions and Methods		<u>565.21</u>
7. Expected AAL 06/30/2020 After Assumption Changes (5+6)		24,246.91
8. Actual AAL 06/30/2020		<u>24,073.48</u>
9. Actuarial (Gain)/Loss on Liabilities (8-7) (Pensions Only)	\$	(173.43)

^aTotal Normal Cost from the previous actuarial valuation includes both employee and employer portion. The employee portion is based on actual contributions.

^bIncludes refund of insurance premiums.

Numbers may not add due to rounding.

CTPF experienced an overall actuarial gain of \$209.61 million. The total net actuarial gain is the total of the gain from assets and the net gain from liabilities. The total actuarial gain for the year is as follows (dollars in millions):

1. Actuarial (Gain)/Loss on Assets	\$	(36.18)
2. Actuarial (Gain)/Loss on Liabilities		<u>(173.43)</u>
3. Total Actuarial (Gain)/Loss (1+2)		(209.61)

The experience of the population determines the liability gain or loss for the year. There was a gain on salaries, due to lower salary increases than assumed. From the last year to this year, there were small gains or losses on retirement, disability experience, active mortality and retiree mortality. There was a gain due to termination experience, and there was a new entrant loss. New entrant losses will occur each year but are offset by additional contributions to the assets. Deviations from other assumptions generated an actuarial gain.

See Table 4 (page 24), Section C, for detail of the gains and losses by source.

Asset Information

The market value of the assets of the Fund that are available for benefits decreased from \$11,038.8 million as of June 30, 2019, to \$10,937.1 million as of June 30, 2020. The actuarial value of assets as of June 30, 2020, is \$11,240.2 million, which is \$303.1 million greater than the market value of assets. This difference is due to the continuing recognition of deferred investment gains and losses. Twenty-five percent of these gains and losses are recognized each year. There are net deferred asset losses of \$298.4 million which will be recognized in the actuarial value of assets over the next three years (a \$94.0 million loss in FY 2021, a \$129.5 million loss in FY 2022 and a \$74.9 million loss in FY 2023).



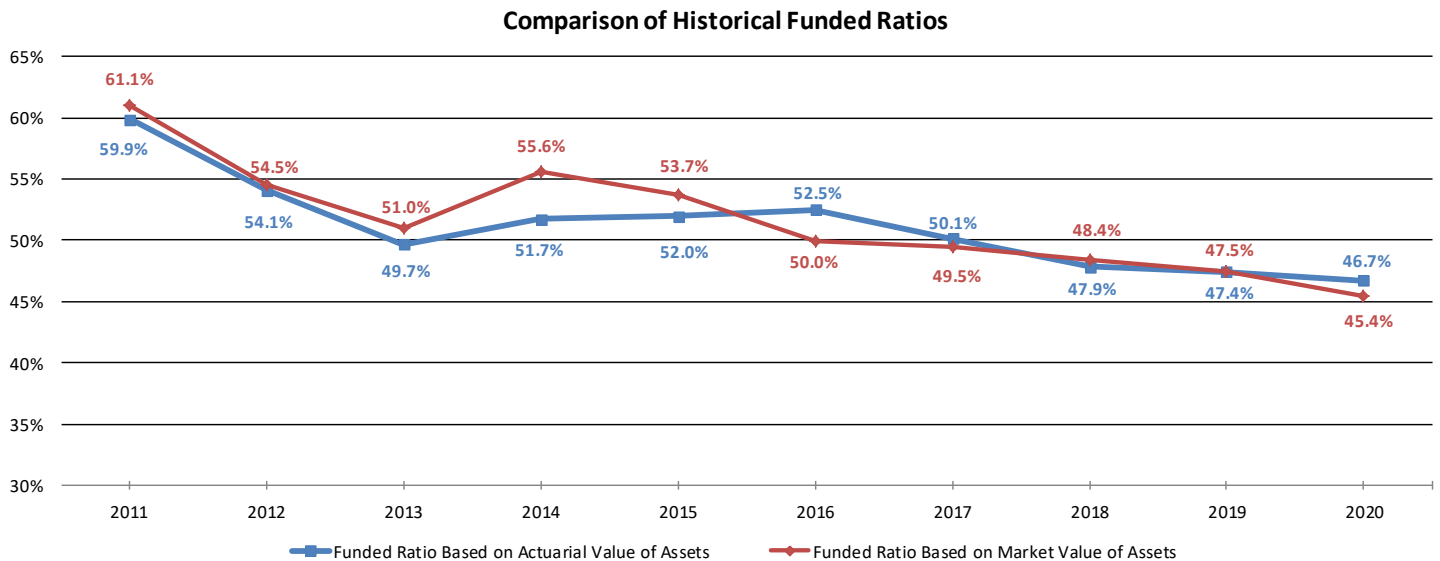
Summary of the Actuarial Valuation

The detailed determinations of asset values utilized in this valuation and the change in assets in the last year are set out in Section E.

Funding Status

The funding status of CTPF is measured by the Funded Ratio. The Funded Ratio is the ratio of the assets available for benefits compared to the actuarial accrued liability of the Fund. Thus, it reflects the portion of benefits earned to date by CTPF members, which are covered by current Fund assets. A funded ratio of 100 percent means that all of the benefits earned to date by CTPF members are covered by assets. By monitoring changes in the funded ratio each year we can determine whether or not funding progress is being made.

Below is a comparison of funded ratios determined on a market value basis and an actuarial value basis over the last 10 years.



Appropriation Requirements under P.A. 90-0655, P.A. 91-0357, P.A. 96-0889 and P.A. 100-0465

The law governing the Fund under P.A. 96-0889 provides that:

For fiscal years 2014 through 2059, the minimum contribution to the Fund to be made by the Board of Education in each fiscal year shall be an amount determined by the Fund to be sufficient to bring the total assets of the Fund up to 90 percent of the total actuarial liabilities of the Fund by the end of fiscal year 2059. In making these determinations, the required Board of Education contribution shall be calculated each year as a level percentage of the applicable employee payrolls over the years remaining to and including fiscal year 2059 and shall be determined under the Projected Unit Credit actuarial cost method. Beginning in fiscal year 2060, the minimum Board of Education contribution for each fiscal year shall be the amount needed to maintain the total assets of the Fund at 90 percent of the total actuarial liabilities of the Fund.

The above calculation provides the basis for calculating the appropriation requirements under P.A. 96-0889. Beginning in State fiscal year 1999, P.A. 90-0655 provides additional State contributions of 0.544 percent of the Fund's total teacher payroll to the Fund to offset the portion of the cost of benefit increases enacted under P.A. 90-0582, except that no additional contributions are required if the Board has certified in the previous fiscal year that the Fund is at least 90 percent funded.

Pursuant to P.A. 91-0357, beginning on and after July 1, 1999, the Board of Education shall make additional contributions of 0.58 percent of the Fund's total teacher payroll to the Fund to offset the portion of the cost of benefit increases enacted under P.A. 90-0582, except that no additional contributions are required if the Board has certified in the previous fiscal year that the Fund is at least 90 percent funded.

Pursuant to P.A. 100-0465, beginning with fiscal year 2018, the State shall contribute for each fiscal year an amount to be determined by the Fund, equal to the employer normal cost for that fiscal year, plus the retiree health insurance reimbursement subsidy. In addition, the Board of Education's property tax levy capped rate was increased from 0.383 percent to 0.567 percent, with proceeds dedicated solely for CTPF.

According to Section 17-129(b)(vii) of the Illinois Pension Code, any contribution by the State to or for the benefit of the Fund, shall be a credit against any contribution required to be made by the Board of Education.



Employer Contribution Requirement for Fiscal Year 2022

The funded ratio as of the June 30, 2020 actuarial valuation on an actuarial value basis is 46.7 percent. Therefore, additional contributions by the Board of Education and State will be required for fiscal year 2022. The projected payroll for fiscal year 2022 is \$2,325,129,719. Based on the projected payroll for fiscal year 2022, and the additional State and Board of Education contribution rates of 0.544 percent and 0.58 percent of payroll, respectively, the additional State and Board of Education contributions for fiscal year 2022 are as follows:

Development of Additional Contributions under Section 17-127 and 17-127.2 of the Illinois Pension Code	Fiscal Year 2022	Fiscal Year 2021
Projected Total Capped Payroll	\$ 2,325,129,719	\$ 2,267,106,915
Additional State Contributions under Section 17-127 of the Illinois Pension Code (% of Projected Capped Payroll)	\$ 12,649,000 0.544%	\$ 12,333,000 0.544%
Additional Board of Education Contributions under Section 17-127.2 of the Illinois Pension Code (% of Projected Capped Payroll)	\$ 13,486,000 0.580%	\$ 13,149,000 0.580%

Pursuant to P.A. 100-0465, the State shall contribute for each fiscal year an amount to be determined by the Fund, equal to the employer normal cost for that fiscal year, plus the retiree health insurance reimbursement subsidy. The following table provides the development of the State contribution requirement under P.A. 100-0465:

Development of Normal Cost State Contributions under Section 17-127(d)(1) of the Illinois Pension Code	Fiscal Year 2022	Fiscal Year 2021*
Total Normal Cost	\$ 384,449,000	\$ 376,410,000
Projected Administrative Expenses	24,661,000	23,320,000
Total Normal Cost Including Administrative Expenses	\$ 409,110,000	\$ 399,730,000
Expected Employee Contributions	209,262,000	204,056,000
Employer Normal Cost	\$ 199,848,000	\$ 195,674,000
Health Insurance Subsidy	65,000,000	65,000,000
State Contributions under Section 17-127(d)(1) of the Illinois Pension Code	\$ 264,848,000	\$ 260,674,000

*Based on June 30, 2020 actuarial valuation results. Fiscal year 2021 numbers are presented for illustrative and comparative purposes only. They are only used to develop the Actuarially Determined Contribution (ADC).



Employer Contribution Requirement for Fiscal Year 2022

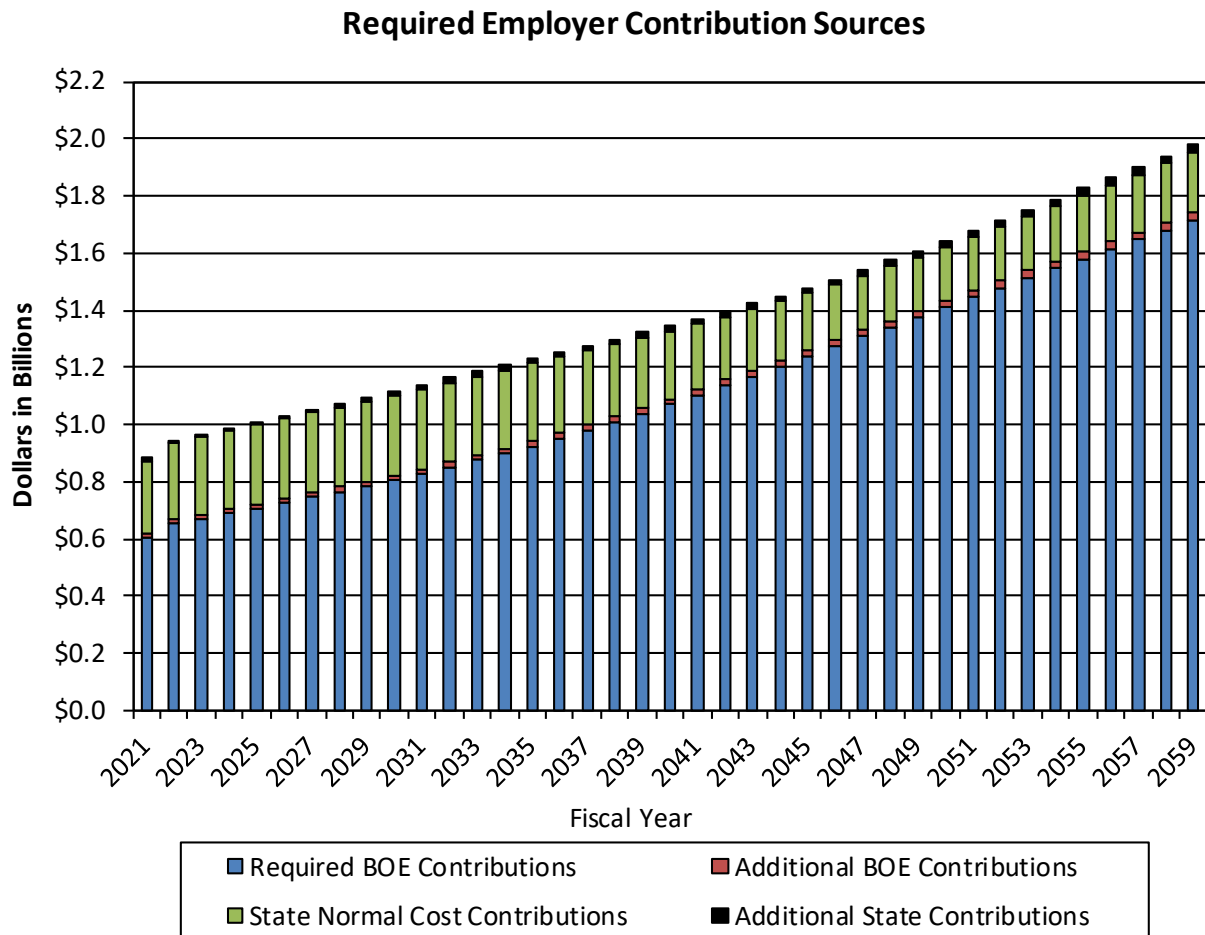
Pursuant to P.A. 96-0889, the Board of Education contribution requirement in each fiscal year shall be an amount determined by the Fund to be sufficient to bring the total assets of the Fund up to 90 percent of the total actuarial liabilities of the Fund by the end of fiscal year 2059. In making these determinations, the required Board of Education contribution shall be calculated each year as a level percentage of the applicable employee payrolls over the years remaining to and including fiscal year 2059 and shall be determined under the Projected Unit Credit actuarial cost method. Based on the funding projections provided in Section D of this report, the Board of Education's required contribution for fiscal year 2022 is equal to \$653,694,000, (net of Additional State and Board of Education Contributions).

The fiscal year ending June 30, 2021, and June 30, 2022, certified contribution requirements and projected future year required contribution amounts are shown below:

Fiscal Year Ending June 30,	Required Board of Education Contributions	Additional Board of Education Contributions	Additional State Contributions	State Contributions Pursuant to P.A. 100-0465	Total Required Employer Contributions
2021	\$ 605,852,000	\$ 13,149,000	\$ 12,333,000	\$ 254,560,000	\$ 885,894,000
2022	653,694,000	13,486,000	12,649,000	264,848,000	944,677,000
2023	671,117,000	13,803,000	12,946,000	269,005,000	966,871,000
2024	689,059,000	14,118,000	13,242,000	272,577,000	988,996,000
2025	707,478,000	14,431,000	13,535,000	275,424,000	1,010,868,000
2026	726,144,000	14,741,000	13,826,000	277,863,000	1,032,574,000
2027	745,310,000	15,050,000	14,116,000	279,767,000	1,054,243,000
2028	765,039,000	15,357,000	14,404,000	280,936,000	1,075,736,000
2029	785,207,000	15,660,000	14,688,000	281,453,000	1,097,008,000
2030	806,268,000	15,968,000	14,977,000	281,347,000	1,118,560,000
2031	828,304,000	16,282,000	15,271,000	280,673,000	1,140,530,000

Employer Contribution Requirement for Fiscal Year 2022

The following graph details the projected employer contribution requirements by Source for fiscal years 2021 through 2059.



Method of Calculation for Appropriation Requirements

The actuarial valuation results are based on the Projected Unit Credit actuarial cost method, the data provided and actuarial assumptions used for the June 30, 2020 actuarial valuation. In order to determine projected contribution amounts, the following additional assumptions were used:

- Total employer contributions of \$885,894,000 for fiscal year 2021.
- Administrative expenses of \$18,375,799 for fiscal year 2020, as provided by the Fund. For fiscal year 2021, the budgeted administrative expense amount of \$23,319,842, as provided by Staff. Thereafter, administrative expenses are assumed to increase 5.75 percent annually for the first 14 years and then increase in line with projected capped payroll after 14 years.
- New entrants whose average age is 32.69 and average pay is \$49,385 (2020 dollars).
- The active member population is assumed to remain level at 30,197 (includes 106 expected new hires) for all years of the 39-year projection. The projection is based on assuming that new active members are hired to replace the current members who leave active membership (through termination, retirement, death or disability). As shown in Table 12 on page 42, the number of active members decreased by about 10 percent between 2008 and 2017, which is an average annualized decrease of about 1.0 percent. The number of actives increased by less than 1.0 percent in 2018, by 1.2 percent in 2019 and by 2.7 percent in 2020. We will continue to review the assumption regarding the projected active member population annually.
- Projected benefits for members hired on or after January 1, 2011, are based on the new provisions established in P.A. 96-0889.
- Additional State contributions of 0.544 percent of pay are assumed to occur mid-year.
- State contributions of the employer's normal cost (includes administrative expenses and \$65 million health insurance subsidy) are assumed to occur mid-year.
- Additional Board of Education contributions of 0.58 percent of pay are assumed to occur end of year.
- 55 percent of the Board of Education's previous year's special tax levy is assumed to occur March 1st each year. The March 1st payment as provided by CTPF is equal \$263,523,630 in fiscal year 2020 and is assumed to increase three percent per year.
- The remaining Board of Education required contribution is assumed to occur end of year.

The average increase in total uncapped payroll for the 39-year projection period is approximately 2.75 percent per year. It is important to note that benefits for new hires are based on capped payroll which is ultimately projected to grow at 1.125 percent per year. All results in this actuarial valuation assume that employer contributions will be made on capped pay.

Recommendations and Future Considerations

Measuring the statutory contribution against a policy such as the Actuarially Determined Contribution (“ADC”) helps evaluate the funding adequacy of the current statutory funding method. Therefore, the Board adopted a policy to calculate the ADC. Under this policy, the ADC is calculated as the Normal Cost plus a 30-year level percent of payroll closed-period amortization of the Unfunded Accrued Liability as of June 30, 2013. The remaining amortization period as of June 30, 2020, actuarial valuation is 23 years.

A key objective of the ADC is to accrue costs over the working lifetime of plan members to ensure that benefit obligations are satisfied and intergenerational equity is promoted. Although the ADC is solely an accounting provision, in certain circumstances it could represent a reasonable annual funding target and therefore is used by some plan sponsors as their “de facto” funding requirement. Note that the statutory funding policy differs significantly from the ADC approach, and results in “back-loading,” meaning that contributions are deferred into the future. Back-loading could result in an underfunding of the fund.

The ADC for fiscal years 2020 and 2021, as well as the statutory employer contribution for fiscal years 2020 and 2021, are shown below as a percentage of projected capped payroll. The ADC for 2020 and statutory employer contribution for 2021 are based on the results of the June 30, 2019, actuarial valuation.

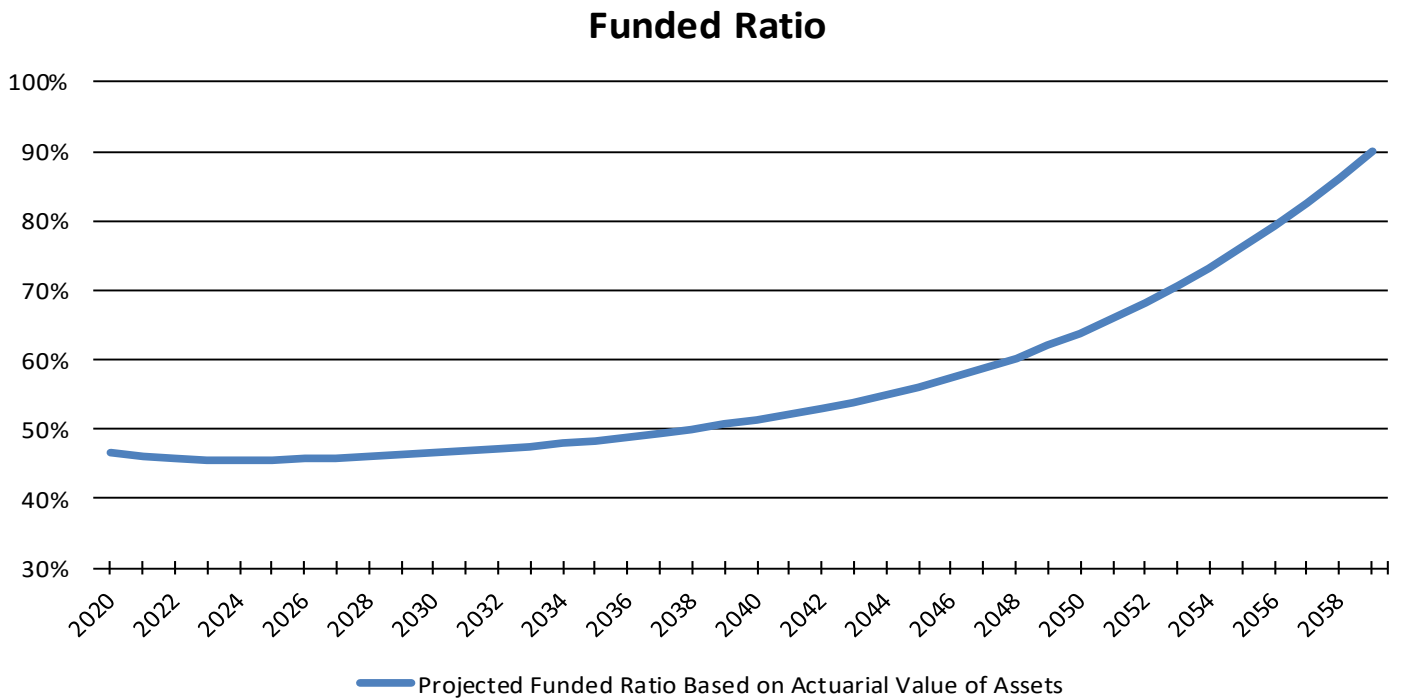
Actuarial Valuation Date:	June 30, 2020	June 30, 2019
Actuarially Determined Contributions for Fiscal Year Ending:	June 30, 2021	June 30, 2020
1. Actuarial Accrued Liability	\$ 24,073,482,607	\$ 23,252,163,307
2. Actuarial Value of Assets	\$ 11,240,208,045	\$ 11,021,811,634
3. Unfunded Actuarial Accrued Liability (1. – 2.)	\$ 12,833,274,562	\$ 12,230,351,673
4. Employer Normal Cost (Including Administrative Expenses and Health Insurance Subsidy)	\$ 260,673,618	\$ 249,719,842
5. Employer Normal Cost Adjusted for Contribution Timing	\$ 264,917,480	\$ 254,477,830
6. Amount to Amortize the Unfunded Liability over a 30-year Closed-period, Beginning July 1, 2013, as a Level Percentage of Payroll	\$ 954,939,837	\$ 892,552,681
7. Actuarially Determined Contribution Requirement [5. + 6.]	\$ 1,219,857,317	\$ 1,147,030,511
8. Projected Capped Payroll For Fiscal Year	\$ 2,267,288,286	\$ 2,203,055,500
9. Actuarially Determined Contribution as a Percentage of Projected Capped Payroll [7./8.]	53.80%	52.07%
10. Total Required Employer Contribution Including Health Insurance Subsidy	\$ 885,894,000	\$ 854,500,000
11. Total Required Employer Contribution as a Percentage of Projected Capped Payroll [10./8.]	39.07%	38.79%
12. Total Required Employer Contribution as a Percentage of Actuarially Determined Contribution [10./7.]	72.62%	74.50%

The fiscal year 2021 Actuarially Determined Contribution is based on an amortization factor which reflects 23 years remaining in the amortization period, an interest rate of 6.75 percent, an annualized assumed rate of increase in total capped payroll of 2.00 percent (which is consistent with the projected increase in total payroll from the projections used to calculate the statutory contribution requirements). The Employer Normal Cost and Amortization Payment are adjusted for expected contribution timing.



Recommendations and Future Considerations

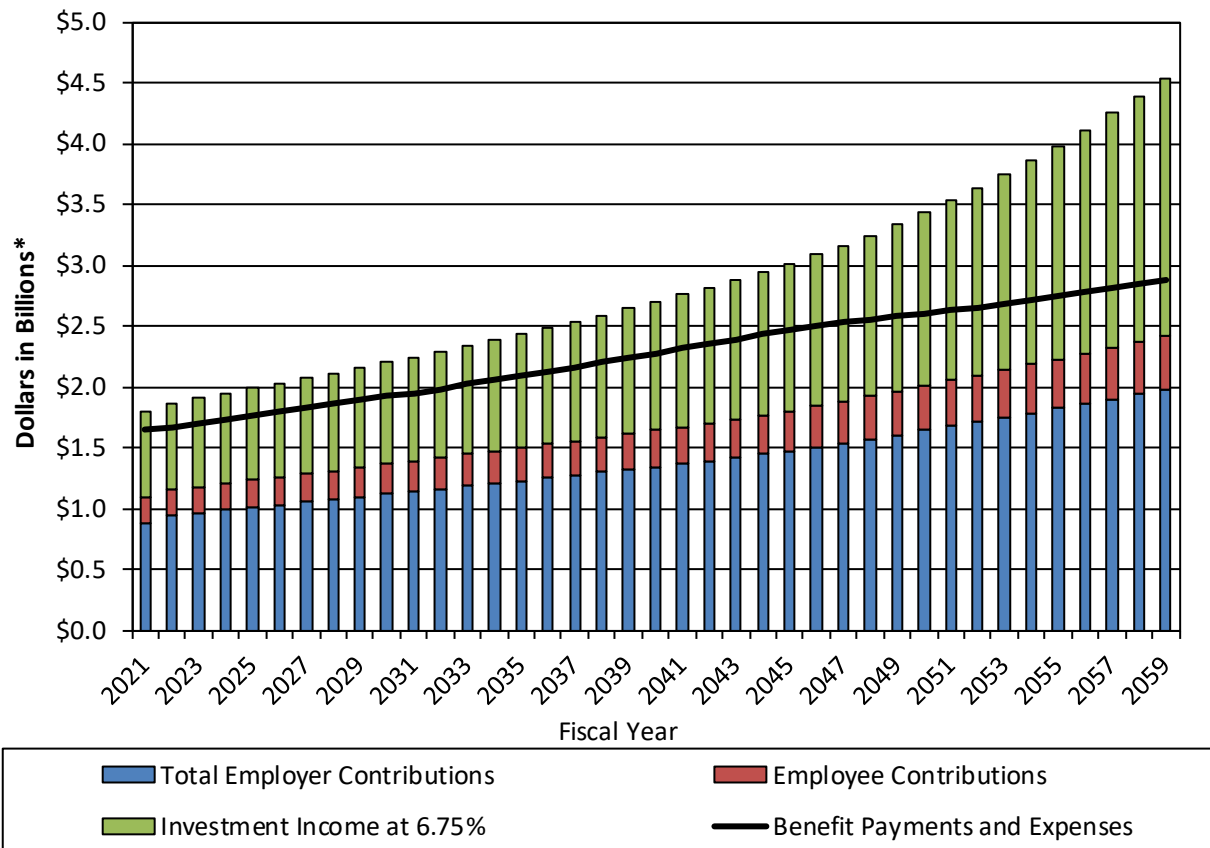
The statutory funding policy required for CTPF provides for level percent of pay funding that produces a funding target of 90 percent by 2059, assuming an open group projection. The following graph shows the projected funded ratio. A key observation is that the funded ratio does not grow markedly until after 2039. That is, a majority of the funding occurs between 2040 and 2059. This illustrates how significantly the current funding policy defers or back-loads contributions into the future.



Recommendations and Future Considerations

The following graph compares the projected benefits and expenses against employer contributions, employee contributions and investment income. From 2021 to 2059, the percentage of investment income needed to pay ongoing benefits decreases from 78.7 percent to 22.0 percent. This implies that a lower level of investment income is projected to be available for potential asset growth in the beginning of the projection period.

Comparison of Cash Flows



**Future dollar amounts are based on assumed inflationary increases.*

We are concerned about potential cash flow problems for CTPF. This is because the assets in the plan (\$10,937.1 million on a market value basis) are not sufficient to cover current retiree liabilities (\$17,300.9 million) and the ratio of market value of assets to retiree benefit payments and expenses is approximately 7.1. This means that approximately seven to eight years of retiree benefit payments can be paid from current assets; the ability to make such payments beyond that period is heavily dependent upon future employer contributions and future investment return.

Recommendations and Future Considerations

The calculations in this report were prepared based on the methods required by the statutory funding policy. GRS does not endorse this funding policy because the statutory funding policy defers funding for these benefits into the future and places a higher burden on future generations of taxpayers.

We recommend the following changes:

1. Implementing a funding policy that contributes normal cost plus closed period amortization as a level percentage of capped payroll amortization of the unfunded liability. (A policy which recognizes the unfunded liability at the valuation date and not a projected liability in the year 2059.)
2. Changing the actuarial cost method for calculating liabilities from the Projected Unit Credit to the Entry Age Normal method.

Change Funding Policy to a More Actuarially Sound Funding Method

We recommend a funding policy that contributes normal cost plus closed period amortization as a level percentage of capped payroll for paying off the current unfunded accrued liability (i.e., the amortization period declines by one year with each actuarial valuation) such that the funded ratio is projected to be 100 percent funded in 30 years or sooner. A 30-year closed amortization period (at the actuarial valuation as of June 30, 2013) methodology pays off the unfunded accrued liability in full by the end of the 30-year period in 2043. The fiscal year 2021 contribution would be \$1,219.9 million under this funding policy. The current statutory contribution does not comply with this recommendation. Underfunding the Fund creates the risk that ultimately benefit obligations cannot be met from the trust, and will require a greater amount of funding from other City and State resources. In addition, continually underfunding the Fund also creates more of a funding need from contributions and less is available from investment return – thereby creating a more expensive plan.

Change the Actuarial Cost Method to the Entry Age Normal Method

The current actuarial cost method is the Projected Unit Credit method, which is required by statute. The Projected Unit Credit method recognizes costs such that the normal cost for an individual member increases as a percentage of payroll throughout the member's career. The Entry Age Normal cost method is the most commonly used method in the public sector. It is also the method required to be used for financial reporting under GASB Statement Nos. 67 and 68. The Entry Age Normal method recognizes costs as a level percentage of payroll over a member's career. We recommend a change to the Entry Age Normal method.

We recognize that the State Statute governs the funding policy of the Fund. The purpose of these comments is to highlight the difference between the statutory appropriation methodology and an actuarially sound funding policy and to highlight the risks and additional costs of continuing to underfund the Fund.



Recommendations and Future Considerations

Future Considerations

Changes (such as the addition of a new benefit tier and delaying the 90 percent funding target year from 2045 to 2059) have had the effect of reducing the statutory contribution amounts that would have otherwise been made. However, recent changes in the investment return assumption and other demographic assumption changes to more closely align the actuarial assumptions with current market expectations have increased the contribution amounts that would otherwise have been made. Assuming the statutory contributions are received (and the actuarial assumptions are met including a 6.75 percent investment rate of return, each year through 2059) CTPF is currently projected to have contributions sufficient to increase the funded ratio from the current level of 46.7 percent to 90.0 percent by 2059.

This is a severely underfunded plan and the ability of the plan to reach 90 percent funding by 2059 is heavily dependent on the State and the Board of Education contributing the statutory contributions each and every year until 2059. Actuarial standards do not require the actuary to evaluate the ability of the plan sponsor or other contributing entity to make required contributions to the plan when due. Such an evaluation was not within the scope of this project and is not within the actuary's domain of expertise. Consequently, we are not able to assess the ability of the State and/or the Board of Education to make contributions when due.

Number of Projected Future Active Members

The total required employer contribution is based on performing an open group projection through the year 2059. The projection is based on assuming that new active members are hired to replace the current members who leave active membership (through termination, retirement, death or disability). As shown in Table 12 on page 42, the number of active members decreased by about 10 percent between 2008 and 2017, which is an average annualized decrease of about 1.0 percent. The number of actives increased by less than 1.0 percent in 2018, by 1.2 percent in 2019, and by 2.7 percent in 2020.

Currently, the actuarial valuation assumes that the total number of active members in the future will be equal to the number active in the current actuarial valuation. Although there is only three years of experience where the number of active members has increased, we believe that it is reasonable to maintain the current level future active member population assumption, but continue to annually monitor the number of active members in the coming years. Given the decrease in the number of active members between 2008 and 2017, if future valuations show additional declines in the active population, the Board may want to consider an update to the population projection assumption to include a decreasing population in the near term before reaching an equilibrium number of active members long term.

Recommendations and Future Considerations

Actuarial Standards of Practice (ASOP) No. 4 Disclosures

General Implications of Contribution Allocation Procedure or Funding Policy on Future Expected Plan Contributions and Funded Status

Given the plan's contribution allocation procedure, if all actuarial assumptions are met (including the assumption of the plan earning 6.75 percent on the actuarial value of assets), it is expected that:

1. The combined State and BOE contribution rate will be level as a percentage of payroll through 2059 (after all deferred asset gains and losses are fully recognized);
2. The unfunded liability will increase in dollar amount through 2038 before it begins to decrease;
3. The unfunded actuarial accrued liabilities will never be fully amortized; and
4. The funded status of the plan will increase gradually towards a 90 percent funded ratio in 2059.

Limitations of Funded Status Measurements

Unless otherwise indicated, a funded status measurement presented in this report is based upon the actuarial accrued liability and the actuarial value of assets. Unless otherwise indicated, with regard to any funded status measurements presented in this report:

1. The measurement is inappropriate for assessing the sufficiency of plan assets to cover the estimated cost of settling the plan's benefit obligations; in other words, of transferring the obligations to a unrelated third party in an arm's length market value type transaction.
2. The measurement is dependent upon the actuarial cost method which, in combination with the plan's amortization policy, affects the timing and amounts of future contributions. The amounts of future contributions will most certainly differ from those assumed in this report due to future actual experience differing from assumed experience based upon the actuarial assumptions.
3. A funded status measurement in this report of 100 percent is not synonymous with no required future contributions. If the funded status were 100 percent, the plan would still require future normal cost contributions (i.e., contributions to cover the cost of the active membership accruing an additional year of service credit).
4. The measurement would produce a different result if the market value of assets were used instead of the actuarial value of assets.

Limitation of Project Scope

Actuarial standards do not require the actuary to evaluate the ability of the plan sponsor or other contributing entity to make required contributions to the plan when due. Such an evaluation was not within the scope of this project and is not within the actuary's domain of expertise. Consequently, the actuary performed no such evaluation.



Risks Associated with Measuring the Accrued Liability and Total Required Employer Contribution

The determination of the accrued liability and the total required employer contribution requires the use of actuarial assumptions regarding future economic and demographic experience. Risk measures, as illustrated in this report, are intended to aid in the understanding of the effects of future experience differing from the actuarial assumptions used in the course of the actuarial valuation. Risk measures may also help with illustrating the potential volatility in the accrued liability and the total required employer contribution that result from the differences between actual experience and the actuarial assumptions.

Future actuarial measurements may differ significantly from the current measurements presented in this report due to such factors as the following: plan experience differing from that anticipated by the economic or demographic assumptions; changes in economic or demographic assumptions due to changing conditions; increases or decreases expected as part of the natural operation of the methodology used for these measurements (such as the end of an amortization period, or additional cost or contribution requirements based on the Fund's funded status); and changes in plan provisions or applicable law. The scope of an actuarial valuation does not include an analysis of the potential range of such future measurements.

Examples of risk that may reasonably be anticipated to significantly affect the Fund's future financial condition include:

1. Investment risk – actual investment returns may differ from the expected returns;
2. Asset/Liability mismatch – changes in asset values may not match changes in liabilities, thereby altering the gap between the accrued liability and assets and consequently altering the funded status and contribution requirements;
3. Contribution risk – actual contributions may differ from expected future contributions. For example, actual contributions may not be made in accordance with the Fund's funding policy or material changes may occur in the anticipated number of covered employees, covered payroll or other relevant contribution base;
4. Salary and Payroll risk – actual salaries and total payroll may differ from expected, resulting in actual future accrued liability and contributions differing from expected;
5. Longevity risk – members may live longer or shorter than expected and receive pensions for a period of time other than assumed; and
6. Other demographic risks – members may terminate, retire or become disabled at times or with benefits other than assumed resulting in actual future accrued liability and contributions differing from expected.

The effects of certain trends in experience can generally be anticipated. For example if the investment return since the most recent actuarial valuation is less (or more) than the assumed rate, the cost of the plan can be expected to increase (or decrease). Likewise if longevity is improving (or worsening), increases (or decreases) in cost can be anticipated.

The statutory contribution for fiscal year 2022 shown on page 8 should be considered as the minimum contribution that complies with the funding policy governed by State statute (Section 17-129(b)(vi) of the Illinois Pension Code). The timely receipt of the statutory contribution is critical to support the financial



Risks Associated with Measuring the Accrued Liability and Total Required Employer Contribution

health of the Fund. Users of this report should be aware that contributions made at the statutorily determined amount do not necessarily guarantee benefit security.

Plan Maturity Measures

Risks facing a pension plan evolve over time. A young plan with virtually no investments and paying few benefits may experience little investment risk. An older plan with a large number of members in pay status and a significant trust may be much more exposed to investment risk. Generally accepted plan maturity measures include the following:

	2020	2019
Ratio of the Market Value of Assets to Payroll	4.84	5.02
Ratio of Actuarial Accrued Liability to Payroll	10.64	10.58
Ratio of Actives to Retirees and Beneficiaries	1.07	1.03
Ratio of Net Cash Flow to Market Value of Assets	-4.94%	-5.25%

Ratios exclude inactive members not receiving benefits.

Ratio of Market Value of Assets to Payroll

The relationship between assets and payroll is a useful indicator of the potential volatility of contributions. For example, if the market value of assets is 5.0 times the payroll, a return on assets 5 percent different than assumed would equal 25 percent of payroll. A higher (lower) or increasing (decreasing) level of this maturity measure generally indicates a higher (lower) or increasing (decreasing) volatility in plan sponsor contributions as a percentage of payroll.

Ratio of Actuarial Accrued Liability to Payroll

The relationship between actuarial accrued liability and payroll is a useful indicator of the potential volatility of contributions for a fully funded plan. A funding policy that targets a funded ratio of 100 percent is expected to result in the ratio of assets to payroll and the ratio of liability to payroll converging over time.

The ratio of liability to payroll may also be used as a measure of sensitivity of the liability itself. For example, if the actuarial accrued liability is 11 times the payroll, a change in liability 2 percent other than assumed would equal 22 percent of payroll. A higher (lower) or increasing (decreasing) level of this maturity measure generally indicates a higher (lower) or increasing (decreasing) volatility in liability (and also plan sponsor contributions) as a percentage of payroll.



Risks Associated with Measuring the Accrued Liability and Total Required Employer Contribution

Ratio of Actives to Retirees and Beneficiaries

A young plan with many active members and few retirees will have a high ratio of active to retirees. A mature open plan may have close to the same number of actives to retirees resulting in a ratio near 1.0. A super-mature or closed plan may have significantly more retirees than actives resulting in a ratio below 1.0.

Ratio of Net Cash Flow to Market Value of Assets

A positive net cash flow means contributions exceed benefits and expenses. A negative cash flow means existing funds are being used to make payments. A certain amount of negative net cash flow is generally expected to occur when benefits are prefunded through a qualified trust. Large negative net cash flows as a percent of assets may indicate a super-mature plan or a need for additional contributions.

Additional Risk Assessment

Additional risk assessment is outside the scope of the annual actuarial valuation. Additional assessment may include scenario tests, sensitivity tests, stochastic modeling, stress tests and a comparison of the present value of accrued benefits at low-risk discount rates with the actuarial accrued liability. At the Board's request, we conducted additional risk assessment of investment and contribution risk through stress testing the investment return assumption and future active population growth.

SECTION C

ACTUARIAL DETERMINATIONS

Table 1

Results of Actuarial Valuation as of June 30, 2020

1. Number of Members		
a. Active ^a		30,091
b. Inactive:		
i. Eligible for deferred vested pension benefits		10,024
ii. Eligible for return of contributions only		21,260
c. Current Benefit Recipients:		
i. Retirement annuities		24,293
ii. Survivor annuities		3,262
iii. Disability annuities		460
d. Total		89,390
2. Covered Payroll		
a. As of the Actuarial Valuation Date	\$	2,262,053,256
b. Projected Capped Payroll for Fiscal Year 2021		2,267,288,286
c. Projected Capped Payroll for Fiscal Year 2022		2,325,129,719
3. Annualized Benefit Payments Currently Being Made		
a. Retirement annuities	\$	1,388,627,977
b. Survivor annuities		81,549,497
c. Disability annuities		18,571,456
d. Total	\$	1,488,748,930
4. Actuarial Accrued Liability—Annuitants		
a. Current Benefit Recipients:		
i. Retirement annuities	\$	16,360,652,738
ii. Survivor annuities		730,552,004
iii. Disability annuities		209,701,712
b. Total	\$	17,300,906,454
5. Actuarial Accrued Liability—Inactive Members		
a. Eligible for Deferred Vested Pension Benefits	\$	470,202,661
b. Eligible for Return of Contributions Only		95,083,036
c. Total	\$	565,285,697

^a Active count excludes 106 members expected to be hired to replace retirements and terminations that occurred in June 2020.



Table 1 (Continued)

Results of Actuarial Valuation as of June 30, 2020

	Normal Cost	Actuarial Accrued Liability
6. Active Members		
a. Retirement Benefits	\$ 288,007,934	\$ 5,280,438,137
b. Withdrawal	75,911,356	753,862,767
c. Death Benefits	5,149,501	73,822,566
d. Disability	7,340,931	99,166,986
e. Administrative Expenses	23,319,842	-
f. Total	\$ 399,729,564	\$ 6,207,290,456
7. Total Actuarial Accrued Liability (4. + 5. + 6.)		\$ 24,073,482,607
8. Market Value of Assets (MVA)		\$ 10,937,062,021
9. Unfunded Actuarial Accrued Liability Based on MVA (7. – 8.)		\$ 13,136,420,586
10. Funded Percentage Based on MVA (8. ÷ 7.) ^a		45.43%
11. Actuarial Value of Assets (AVA)		\$ 11,240,208,045
12. Unfunded Actuarial Accrued Liability Based on AVA (7. – 11.)		\$ 12,833,274,562
13. Funded Percentage Based on AVA (11. ÷ 7.) ^a		46.69%
14. Total Normal Cost	\$ 399,729,564	
15. Expected Employee Contributions	\$ 204,055,946	
16. Annual Employer Normal Cost (% of Projected Capped Payroll for Fiscal Year 2021)	\$ 195,673,618 8.63%	
17. Health Insurance Subsidy	\$ 65,000,000	
18. Annual Employer Normal Cost, including Health Insurance Reimbursement (16. + 17.) % of Projected Capped Payroll for Fiscal Year 2021 ^b	\$ 260,673,618 11.50%	

^a The funded status measure is appropriate for assessing the need for future contributions. The funded status is not appropriate for assessing the sufficiency of plan assets to cover the estimated cost of settling the plan's benefit obligations.

^b Used for calculation of the ADC.



Table 2
Components of Actuarial Accrued Liability and Normal Cost by Tier

Actuarial Valuation Results	Tier 1		Tier 2 ^a		Total	
1. Count	18,118		11,973		30,091	
2. Covered Payroll						
a. As of the Valuation Date	\$	1,589,735,140		\$672,318,116	\$	2,262,053,256
b. Projected Capped Payroll for Fiscal Year 2021		1,589,735,140		677,553,146		2,267,288,286
c. Projected Capped Payroll for Fiscal Year 2022		1,554,881,677		770,248,042		2,325,129,719
3. Actuarial Accrued Liability						
a. Retirement Benefits	\$	5,187,619,984	\$	92,818,153	\$	5,280,438,137
b. Withdrawal		643,691,979		110,170,788		753,862,767
c. Death Benefits		68,498,453		5,324,113		73,822,566
d. Disability		89,770,423		9,396,563		99,166,986
e. Total	\$	5,989,580,839	\$	217,709,617	\$	6,207,290,456
4. Normal Cost	Amount	% of Payroll	Amount	% of Payroll	Amount	% of Payroll
a. Retirement Benefits	\$ 268,702,361	16.90%	\$ 19,305,573	2.85%	\$ 288,007,934	12.70%
b. Withdrawal	47,944,261	3.02%	27,967,095	4.13%	75,911,356	3.35%
c. Death Benefits	4,000,352	0.25%	1,149,149	0.17%	5,149,501	0.24%
d. Disability	5,457,272	0.34%	1,883,659	0.28%	7,340,931	0.32%
e. Administrative Expenses	16,350,974	1.03%	6,968,868	1.03%	23,319,842	1.03%
f. Total	\$ 342,455,220	21.54%	\$ 57,274,344	8.45%	\$ 399,729,564	17.63%
5. Expected Employee Contributions	\$ 143,076,163	9.00%	\$ 60,979,783	9.00%	\$ 204,055,946	9.00%
6. Annual Employer Normal Cost	\$ 199,379,057	12.54%	\$ (3,705,439)	-0.55%	\$ 195,673,618	8.63%

^aThe actuarial accrued liability, normal cost, projected capped payroll and expected employee contributions include the results for 106 members expected to be hired to replace retirements and terminations that occurred in June 2020. Active count excludes 106 members expected to be hired to replace retirements and terminations that occurred in June 2020.

Actuarial Accrued Liability and Normal Cost are determined under the Projected Unit Credit actuarial cost method.

Normal Cost rates as a percent of pay under the Projected Unit Credit actuarial cost method increase over a member's career.



Table 3

Analysis of Change in Unfunded Actuarial Accrued Liability

In addition to the expected change in the unfunded accrued actuarial liability, changes in membership demographics and fund assets have affected the actuarial valuation results. The increase in the unfunded actuarial accrued liability (UAAL) of \$602,922,889 was due to the following:

1. Unfunded Actuarial Accrued Liability (UAAL) at 06/30/2019	\$ 12,230,351,673
2. Contributions	
a. Contributions due (Normal Cost plus interest on the UAAL)	
i interest on item 1.	\$ 856,124,617
ii members contributions	196,097,115
iii employer normal cost (middle of year)	251,897,722
iv interest on ii and iii	15,414,626
v total due	<u>\$ 1,319,534,080</u>
b. Contributions paid (Actual)	
i member contributions	\$ 196,097,115
ii employer	854,500,000
iii interest on i and ii ^a	21,612,937
iv total paid	<u>\$ 1,072,210,052</u>
c. Expected increase in Unfunded Actuarial Accrued Liability	\$ 247,324,028
3. Expected Unfunded Actuarial Accrued Liability at 06/30/2020	\$ 12,477,675,701
4. (Gains)/Losses	
a. investment income	\$ (22,146,029)
b. retiree health insurance cash flows	(14,032,804)
c. salary increases	(118,074,777)
d. demographic	(55,354,066)
e. total	<u>\$ (209,607,676)</u>
5. Plan Provision Changes	\$ -
6. Assumption Changes	\$ 565,206,537
7. Total Change in UAAL	\$ 602,922,889
8. UAAL at 06/30/2020	\$ 12,833,274,562

^aInterest on employer contributions is estimated based on a weighted timing of middle of year, 8/12^{ths} of a year, and end of year.



Table 4

Analysis of Financial (Gains) and Losses in Unfunded Actuarial Accrued Liability for Fiscal Year Ending June 30, 2020

	Activity	(Gain)/Loss	% of 06/30/2019 AAL
1.	Actuarial (Gain)/Loss		
	a. Retirements	\$ 6,878,558	0.03%
	b. Incidence of Disability	(1,131,160)	0.00%
	c. In-Service Mortality	(70,494)	0.00%
	d. Retiree Mortality	(55,588,659)	-0.24%
	e. Salary Increases	(118,074,777)	-0.51%
	f. Terminations	(19,358,722)	-0.08%
	g. Investment Return	(22,146,029)	-0.10%
	h. Retiree Health Insurance Cash Flows	(14,032,804)	-0.06%
	i. New Entrant Liability	45,481,403	0.20%
	j. Other	(31,564,992)	-0.14%
	k. Total Actuarial (Gain)/Loss	\$ (209,607,676)	-0.90%
2.	Plan Provision Changes	\$ -	0.00%
3.	Assumption Changes	\$ 565,206,537	2.43%
4.	Contribution (Excess)/Shortfall ^a	\$ 247,324,028	1.06%
5.	Total Financial (Gain)/Loss	\$ 602,922,889	2.59%

^a Represents the increase in the Unfunded Actuarial Accrued Liability due to actual contributions being less than the Normal Cost plus interest on the beginning of year Unfunded Actuarial Accrued Liability.

Graph 1

Financial (Gains) and Losses in Unfunded Actuarial Accrued Liability for Fiscal Year Ending June 30, 2020, by Source

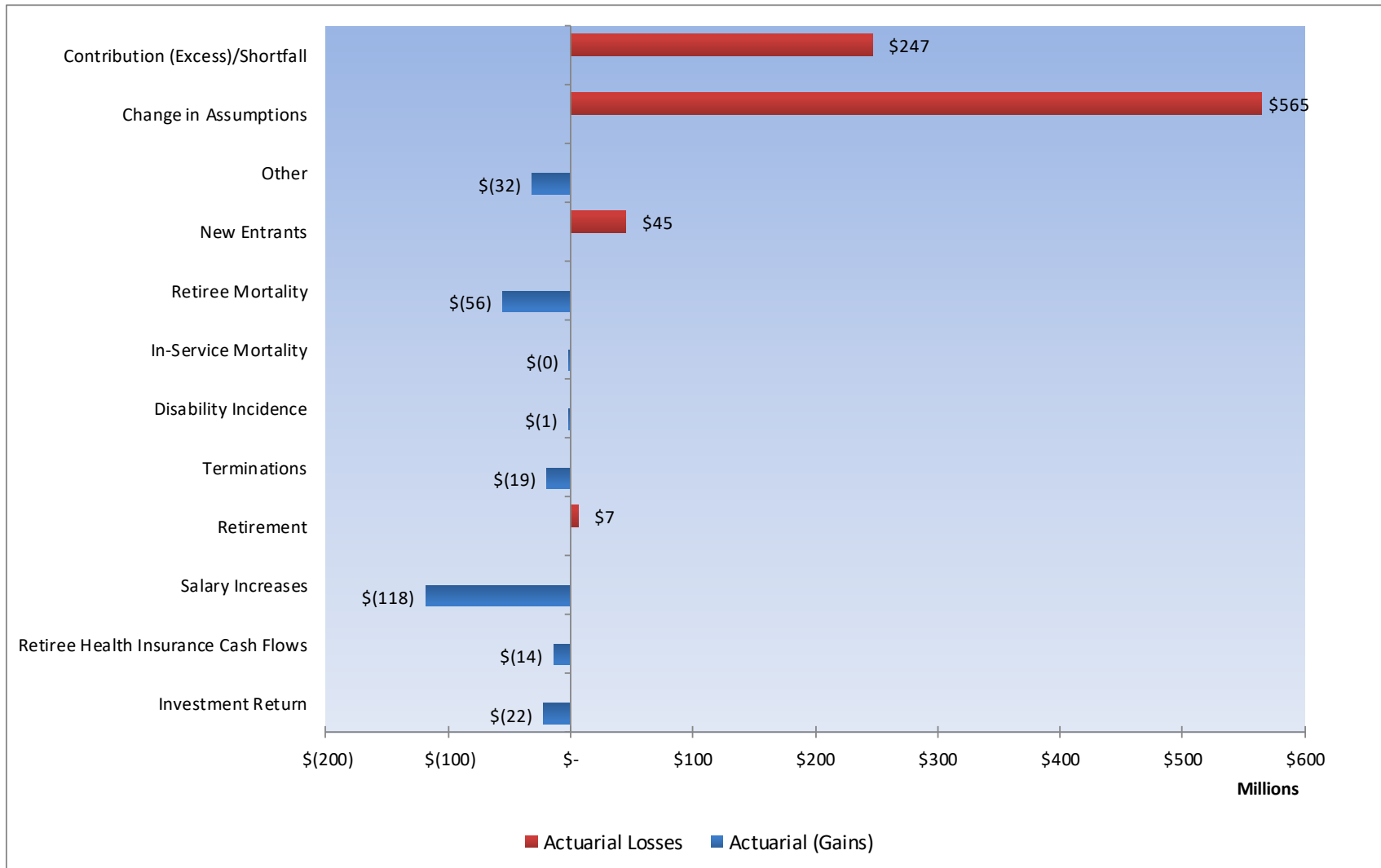


Table 5a
Historical Financial (Gains) and Losses in Unfunded Actuarial Accrued Liability

	Amount of (Gain) or Loss					Total Five-year Change
	FY 2016	FY 2017	FY 2018	FY 2019	FY 2020	
Investment Return (AVA Basis)	\$ (81,129,490)	\$ (80,937,857)	\$ 131,839,730	\$ 98,317,079	\$ (22,146,029)	\$ 45,943,433
Retiree Health Insurance Cash Flows	1,673,226	(17,178,251)	1,381,154	(6,608,283)	(14,032,804)	(34,764,958)
Salary Increases	(264,801,612)	(180,217,505)	6,927,266	(62,859,630)	(118,074,777)	(619,026,258)
Retirements	47,235,684	32,846,858	27,778,569	1,996,977	6,878,558	116,736,646
Terminations	(6,458,419)	(7,478,652)	(39,625,292)	(46,251,276)	(19,358,722)	(119,172,361)
Disability Incidence	-	(685,126)	(1,010,548)	(966,542)	(1,131,160)	(3,793,376)
In-Service Mortality	-	657,829	1,253,670	123,792	(70,494)	1,964,797
Retiree Mortality	15,811,894	14,544,206	70,809,472	5,854,235	(55,588,659)	51,431,148
New Entrants	-	32,961,135	38,467,726	44,926,283	45,481,403	161,836,547
Other ^a	57,480,517	(54,889,992)	(28,017,087)	(22,939,305)	(31,564,992)	(79,930,859)
Total Actuarial (Gain)/Loss	\$ (230,188,200)	\$ (260,377,355)	\$ 209,804,660	\$ 11,593,330	\$ (209,607,676)	\$ (478,775,241)
(Gain)/Loss as a % of BOY AAL	(1.2)%	(1.3)%	1.0%	0.1%	(0.9)%	
Total Non-Investment (Gain)/Loss	\$ (149,058,710)	\$ (179,439,498)	\$ 77,964,930	\$ (86,723,749)	\$ (187,461,647)	\$ (524,718,674)
(Gain)/Loss as a % of BOY AAL	(0.7)%	(0.9)%	0.4%	(0.4)%	(0.8)%	
(Gain)/Loss Due to Plan Provisions Changes	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
(Gain)/Loss as a % of BOY AAL	-	-	-	-	-	
(Gain)/Loss Due to Assumption Changes	\$ -	\$ 1,074,523,844	\$ 621,772,494	\$ -	\$ 565,206,537	\$ 2,261,502,875
(Gain)/Loss as a % of BOY AAL	0.0%	5.2%	2.8%	0.0%	2.4%	
(Gain)/Loss Due to Contribution (Excess)/Shortfall	\$ 260,150,252	\$ 220,857,399	\$ 233,351,269	\$ 264,851,308	\$ 247,324,028	\$ 1,226,534,256
(Gain)/Loss as a % of BOY AAL	1.3%	1.1%	1.1%	1.2%	1.1%	
Total Financial (Gain)/Loss	\$ 29,962,052	\$ 1,035,003,888	\$ 1,064,928,423	\$ 276,444,638	\$ 602,922,889	\$ 3,009,261,890
(Gain)/Loss as a % of BOY AAL	0.2%	5.1%	4.9%	1.2%	2.6%	
BOY Actuarial Accrued Liability (AAL)	\$ 19,951,289,974	\$ 20,484,951,277	\$ 21,822,010,297	\$ 22,922,992,558	\$ 23,252,163,307	

^a Includes other experience such as deviations between actual and expected benefit payments and unexpected changes in service.

Results prior to fiscal year 2017 were obtained from the prior Actuary's actuarial valuation reports.



Table 5b
Historical Normal Cost Amounts (\$ and % of Pay)

Actuarial Valuation Date: Required Employer Contributions for Fiscal Year Ending:	June 30, 2019 June 30, 2020		June 30, 2020 June 30, 2021		Projected from June 30, 2020 June 30, 2022	
1. Projected Capped Payroll for the Fiscal Year	\$ 2,203,055,500		\$ 2,267,288,286		\$ 2,325,129,719	
2. Normal Cost	Amount	% of Payroll	Amount	% of Payroll	Amount	% of Payroll
a. Retirement Benefits	\$ 271,774,168	12.34%	\$ 288,007,934	12.70%	\$ 296,255,150	12.74%
b. Withdrawal	71,756,704	3.26%	75,911,356	3.35%	75,552,759	3.25%
c. Death Benefits	4,913,776	0.22%	5,149,501	0.23%	5,176,130	0.22%
d. Disability	6,949,723	0.32%	7,340,931	0.32%	7,464,590	0.32%
e. Administrative Expenses	27,600,466	1.25%	23,319,842	1.03%	24,661,000	1.06%
f. Total	\$ 382,994,837	17.38%	\$ 399,729,564	17.63%	\$ 409,109,630	17.60%
3. Expected Employee Contributions	\$ 198,274,995	9.00%	\$ 204,055,946	9.00%	\$ 209,261,675	9.00%
4. Annual Employer Normal Cost	\$ 184,719,842	8.38%	\$ 195,673,618	8.63%	\$ 199,847,955	8.60%

SECTION D

ACTUARIAL PROJECTIONS

Table 6

Baseline Projections — Employer Contributions Determined under Public Act 90-0655, Public Act 91-0357, Public Act 96-0889 and Public Act 100-0465 Investment Return of 6.75% Each Year (\$ in Millions)

Fiscal Year	Actuarial Accrued Liability	Actuarial Value of Assets	Unfunded Liability	Funded Ratio	Pensionable Payroll	Annual Normal Cost				Total Employer Contributions						Total Expenses
						Total	Employee Contribution	Normal Cost	Percent of Pay	Required Employer Contribution	Additional State Contributions	State Normal Cost Contributions	Additional Board of Education Contributions	Required Board of Education Contributions	Percent of Pay	
2021	\$24,479.0	\$11,307.1	\$13,172.0	46.19%	\$2,267.3	\$464.7	\$204.1	\$260.6	11.49%	\$885.9	\$12.3	\$254.6	\$13.1	\$605.9	39.07%	\$1,645.0
2022	24,891.0	11,378.0	13,513.0	45.71%	2,325.1	474.1	209.3	264.8	11.39%	944.7	12.6	264.8	13.5	653.7	40.63%	1,674.6
2023	25,310.1	11,504.3	13,805.8	45.45%	2,379.8	483.2	214.2	269.0	11.30%	966.9	12.9	269.0	13.8	671.1	40.63%	1,703.7
2024	25,735.9	11,710.8	14,025.1	45.50%	2,434.2	491.7	219.1	272.6	11.20%	989.0	13.2	272.6	14.1	689.1	40.63%	1,733.1
2025	26,166.1	11,931.2	14,234.9	45.60%	2,488.0	499.3	223.9	275.4	11.07%	1,010.9	13.5	275.4	14.4	707.5	40.63%	1,764.3
2026	26,600.7	12,164.9	14,435.8	45.73%	2,541.5	506.6	228.7	277.9	10.93%	1,032.6	13.8	277.9	14.7	726.1	40.63%	1,795.4
2027	27,039.1	12,410.5	14,628.6	45.90%	2,594.8	513.3	233.5	279.8	10.78%	1,054.2	14.1	279.8	15.0	745.3	40.63%	1,826.8
2028	27,479.9	12,666.3	14,813.6	46.09%	2,647.7	519.2	238.3	280.9	10.61%	1,075.7	14.4	280.9	15.4	765.0	40.63%	1,859.1
2029	27,922.1	12,932.1	14,990.0	46.31%	2,700.1	524.5	243.0	281.5	10.43%	1,097.0	14.7	281.5	15.7	785.2	40.63%	1,891.8
2030	28,362.3	13,206.0	15,156.3	46.56%	2,753.1	529.1	247.8	281.3	10.22%	1,118.6	15.0	281.3	16.0	806.3	40.63%	1,927.3
2031	28,814.0	13,503.0	15,311.0	46.86%	2,807.2	533.3	252.6	280.7	10.00%	1,140.5	15.3	280.7	16.3	828.3	40.63%	1,949.1
2032	29,263.7	13,811.1	15,452.6	47.20%	2,861.3	537.0	257.5	279.5	9.77%	1,162.5	15.6	279.5	16.6	850.9	40.63%	1,984.1
2033	29,709.6	14,129.9	15,579.7	47.56%	2,915.8	540.1	262.4	277.7	9.52%	1,184.7	15.9	277.7	16.9	874.2	40.63%	2,020.4
2034	30,150.3	14,460.2	15,690.1	47.96%	2,971.2	542.7	267.4	275.3	9.27%	1,207.2	16.2	275.3	17.2	898.5	40.63%	2,057.1
2035	30,584.2	14,802.1	15,782.1	48.40%	3,026.9	544.7	272.4	272.3	9.00%	1,229.8	16.5	272.3	17.6	923.5	40.63%	2,094.5
2036	31,010.0	15,158.3	15,851.7	48.88%	3,083.0	544.2	277.5	266.7	8.65%	1,252.6	16.8	266.7	17.9	951.2	40.63%	2,130.2
2037	31,426.1	15,529.7	15,896.4	49.42%	3,139.3	543.0	282.5	260.4	8.29%	1,275.5	17.1	260.4	18.2	979.8	40.63%	2,166.1
2038	31,831.2	15,917.3	15,913.9	50.01%	3,196.2	541.2	287.7	253.5	7.93%	1,298.6	17.4	253.5	18.5	1,009.1	40.63%	2,202.2
2039	32,223.2	16,321.7	15,901.5	50.65%	3,254.0	539.0	292.9	246.2	7.57%	1,322.1	17.7	246.2	18.9	1,039.3	40.63%	2,239.2
2040	32,600.1	16,743.9	15,856.2	51.36%	3,313.2	536.6	298.2	238.5	7.20%	1,346.1	18.0	238.5	19.2	1,070.4	40.63%	2,277.1

Normal cost includes administrative expenses and health insurance subsidy. The health insurance subsidy is assumed to be \$65 million each year.

Total expenses shown include benefit payments, refunds, administrative expenses and health insurance subsidy.

Actuarial accrued liability and assets are measured at the end of the fiscal year. Normal Cost is measured at the middle of the fiscal year.

State contributions, benefit payments, refunds, administrative expenses and employee contributions are assumed to occur during the middle of the year.

55 percent of the Board of Education's dedicated property tax levy of 0.567 percent is assumed to be paid March 1, each fiscal year. The remaining Board of Education contributions are assumed to occur at the end of the year.

Total payroll is capped for members hired after December 31, 2010, as defined in Public Act 96-0889.

Table 6 (Continued)

Baseline Projections — Employer Contributions Determined under Public Act 90-0655, Public Act 91-0357, Public Act 96-0889 and Public Act 100-0465

Investment Return of 6.75% Each Year (\$ in Millions)

						Annual Normal Cost				Total Employer Contributions							
Fiscal Year	Actuarial Accrued Liability	Actuarial Value of Assets	Unfunded Liability	Funded Ratio	Pensionable Payroll	Employer				Required Employer Contribution	Additional State Contributions	State Normal Cost	Additional Board of Education Contributions	Required Board of Education Contributions	Percent of Pay	Total Expenses	
						Total	Employee Contribution	Normal Cost	Percent of Pay								
2041	\$32,959.7	\$17,185.5	\$15,774.2	52.14%	\$3,374.4	\$533.9	\$303.7	\$230.2	6.82%	\$1,371.0	\$18.4	\$230.2	\$19.6	\$1,102.9	40.63%	\$2,315.6	
2042	33,300.0	17,647.2	15,652.8	52.99%	3,436.1	530.9	309.3	221.6	6.45%	1,396.1	18.7	221.6	19.9	1,135.8	40.63%	2,354.8	
2043	33,621.1	18,132.3	15,488.8	53.93%	3,500.2	528.5	315.0	213.5	6.10%	1,422.1	19.0	213.5	20.3	1,169.2	40.63%	2,393.3	
2044	33,923.6	18,645.4	15,278.2	54.96%	3,567.7	527.1	321.1	206.0	5.77%	1,449.5	19.4	206.0	20.7	1,203.5	40.63%	2,430.8	
2045	34,209.8	19,191.6	15,018.2	56.10%	3,637.8	527.0	327.4	199.6	5.49%	1,478.0	19.8	199.6	21.1	1,237.6	40.63%	2,466.2	
2046	34,483.5	19,778.5	14,705.0	57.36%	3,712.4	528.6	334.1	194.5	5.24%	1,508.3	20.2	194.5	21.5	1,272.1	40.63%	2,498.7	
2047	34,747.2	20,412.5	14,334.7	58.75%	3,790.9	532.2	341.2	191.0	5.04%	1,540.2	20.6	191.0	22.0	1,306.6	40.63%	2,529.7	
2048	35,006.2	21,102.5	13,903.7	60.28%	3,873.2	537.6	348.6	189.0	4.88%	1,573.6	21.1	189.0	22.5	1,341.1	40.63%	2,557.0	
2049	35,264.2	21,856.6	13,407.6	61.98%	3,959.2	544.6	356.3	188.3	4.76%	1,608.6	21.5	188.3	23.0	1,375.8	40.63%	2,581.9	
2050	35,522.7	22,681.1	12,841.6	63.85%	4,047.5	552.6	364.3	188.3	4.65%	1,644.5	22.0	188.3	23.5	1,410.7	40.63%	2,606.2	
2051	35,782.6	23,581.0	12,201.6	65.90%	4,136.1	561.1	372.2	188.8	4.56%	1,680.4	22.5	188.8	24.0	1,445.1	40.63%	2,630.3	
2052	36,043.1	24,560.1	11,483.0	68.14%	4,224.4	570.0	380.2	189.8	4.49%	1,716.3	23.0	189.8	24.5	1,479.0	40.63%	2,655.6	
2053	36,303.4	25,622.4	10,681.0	70.58%	4,313.0	579.6	388.2	191.4	4.44%	1,752.3	23.5	191.4	25.0	1,512.4	40.63%	2,682.3	
2054	36,562.2	26,772.2	9,790.0	73.22%	4,402.8	589.7	396.2	193.5	4.39%	1,788.8	24.0	193.5	25.5	1,545.8	40.63%	2,711.0	
2055	36,816.0	28,012.0	8,804.0	76.09%	4,493.8	600.5	404.4	196.1	4.36%	1,825.8	24.4	196.1	26.1	1,579.2	40.63%	2,743.5	
2056	37,061.7	29,345.2	7,716.5	79.18%	4,585.9	611.8	412.7	199.1	4.34%	1,863.2	24.9	199.1	26.6	1,612.6	40.63%	2,779.2	
2057	37,298.4	30,777.6	6,520.8	82.52%	4,679.0	623.6	421.1	202.5	4.33%	1,901.0	25.5	202.5	27.1	1,645.9	40.63%	2,815.8	
2058	37,527.0	32,317.8	5,209.2	86.12%	4,772.9	635.8	429.6	206.2	4.32%	1,939.2	26.0	206.2	27.7	1,679.3	40.63%	2,851.3	
2059	37,750.0	33,975.7	3,774.3	90.00%	4,867.5	648.3	438.1	210.2	4.32%	1,977.6	26.5	210.2	28.2	1,712.6	40.63%	2,884.2	

Normal cost includes administrative expenses and health insurance subsidy. The health insurance subsidy is assumed to be \$65 million each year.

Total expenses shown include benefit payments, refunds, administrative expenses and health insurance subsidy.

Actuarial accrued liability and assets are measured at the end of the fiscal year. Normal Cost is measured at the middle of the fiscal year.

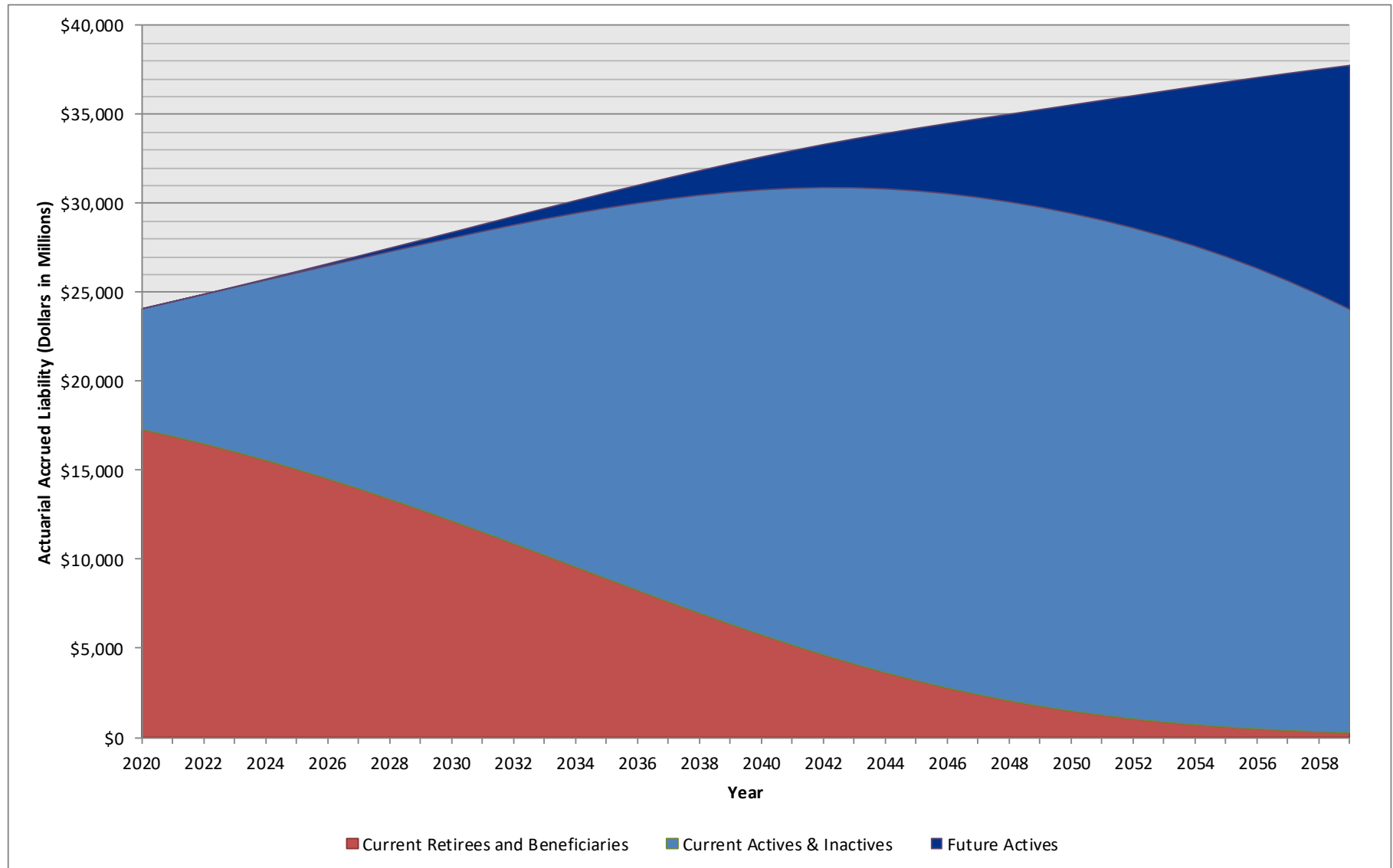
State contributions, benefit payments, refunds, administrative expenses and employee contributions are assumed to occur during the middle of the year.

55 percent of the Board of Education's dedicated property tax levy of 0.567 percent is assumed to be paid March 1, each fiscal year. The remaining Board of Education contributions are assumed to occur at the end of the year.

Total payroll is capped for members hired after December 31, 2010, as defined in Public Act 96-0889.



Graph 2
Projected Actuarial Accrued Liabilities
Actuarial Valuation as of June 30, 2020

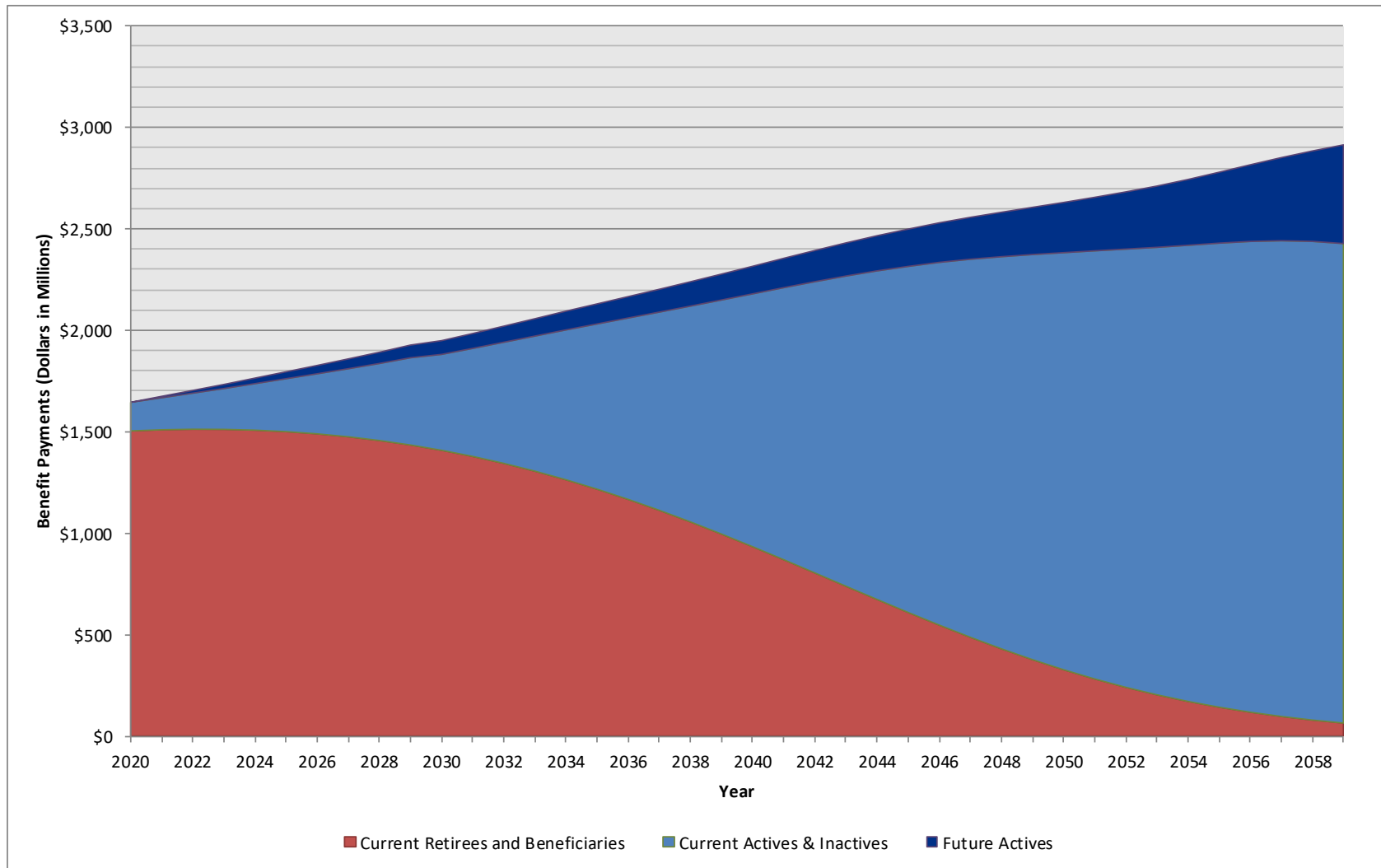


Graph 3

Projected Benefit Payments

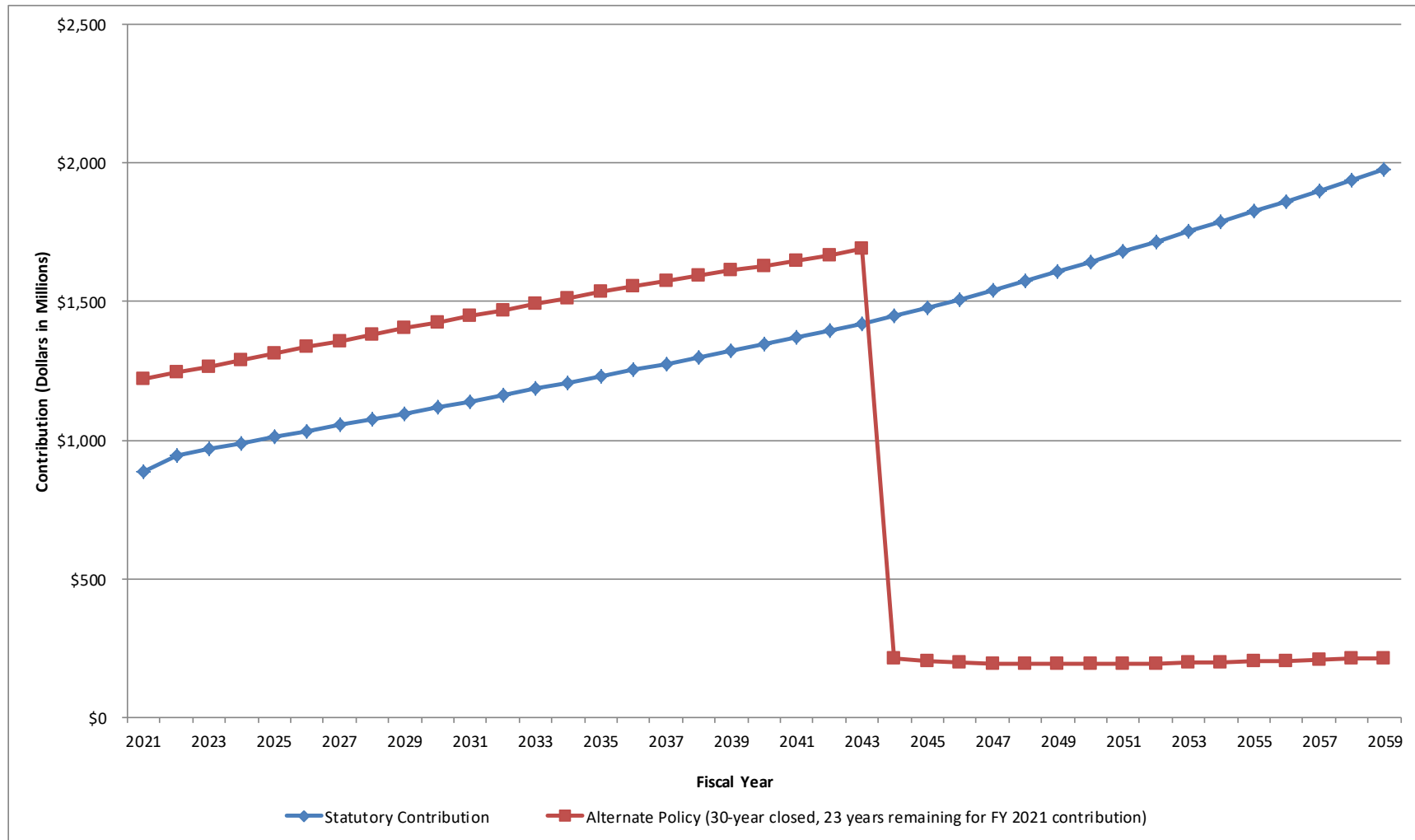
(Including Administrative Expenses and Health Insurance Subsidy)

Actuarial Valuation as of June 30, 2020



Graph 4

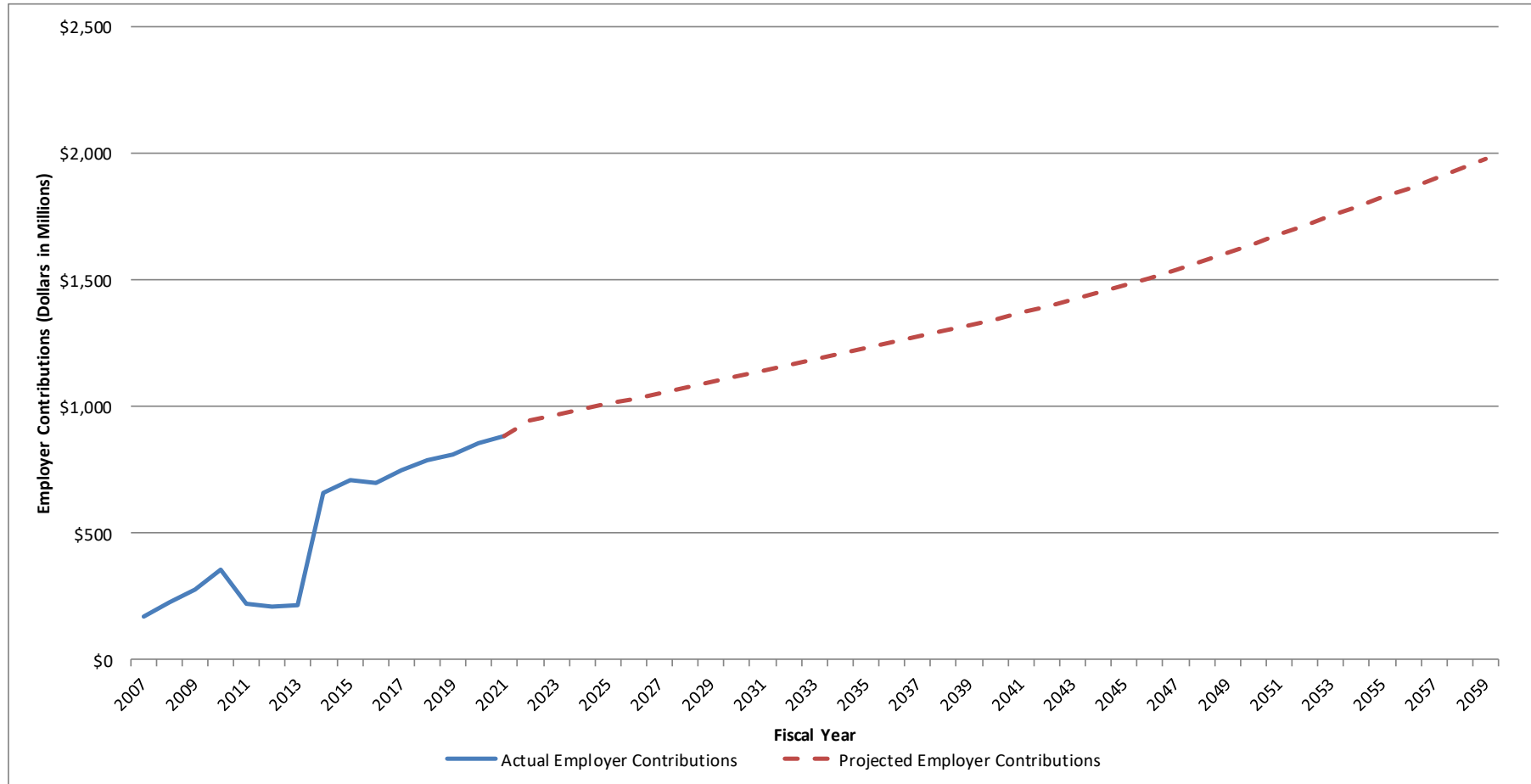
Projected Statutory Contributions vs. Contributions under Alternate Policy (Normal Cost Plus 30-Year Closed Period Level Percent of Pay Amortization) (23 Years Remaining in Amortization Period for FY 2021 Contribution)



Alternate funding policy of normal cost plus 30-year closed period amortization of the unfunded liability as a level percentage of capped payroll beginning in FY 2014 and 23 years remaining in FY 2021.

Graph 5

Actual and Projected Employer Contributions



SECTION E

FUND ASSETS

Table 7

Statement of Fiduciary Net Position

for Years Ended June 30, 2020, and 2019

	June 30, 2020			June 30, 2019		
	Pension Fund	Health Insurance Fund	Total Fund - Pension and Health Insurance	Pension Fund	Health Insurance Fund	Total Fund - Pension and Health Insurance
Assets:						
Cash	\$ 60,193,099	\$ -	\$ 60,193,099	\$ 121,996,481	\$ -	\$ 121,996,481
Prepaid expense	612,250	-	612,250	324,753	-	324,753
Receivables:	-	-	-	-	-	-
Minimum funding requirement (Employer)	232,392,721	-	232,392,721	186,608,225	-	186,608,225
Minimum funding requirement (State)	-	-	-	37,797,000	-	37,797,000
Employee	10,783,758	-	10,783,758	11,247,826	-	11,247,826
Accrued investment income	36,255,805	-	36,255,805	39,268,564	-	39,268,564
Due from brokers	81,201,107	-	81,201,107	162,389,723	-	162,389,723
Participating teachers' accounts for contributions	4,535,674	-	4,535,674	4,681,191	-	4,681,191
Other receivables	10,590,492	8,720,656	19,311,148	8,364,555	4,915,284	13,279,839
Total receivables	375,759,557	8,720,656	384,480,213	450,357,084	4,915,284	455,272,368
Investments, at fair value:						
U.S. government and agency fixed income	1,307,282,631	-	1,307,282,631	1,381,549,632	-	1,381,549,632
U.S. corporate fixed income	1,227,940,708	-	1,227,940,708	1,080,629,549	-	1,080,629,549
Foreign fixed income securities	49,326,494	-	49,326,494	60,378,150	-	60,378,150
U.S. equities	2,959,380,844	-	2,959,380,844	3,115,916,532	-	3,115,916,532
Foreign equities	3,033,910,935	-	3,033,910,935	3,249,511,598	-	3,249,511,598
Public REITs	115,436,398	-	115,436,398	157,753,245	-	157,753,245
Pooled short-term investment funds	479,732,946	-	479,732,946	359,473,153	-	359,473,153
Real estate	922,797,885	-	922,797,885	719,817,831	-	719,817,831
Infrastructure	203,049,254	-	203,049,254	210,950,505	-	210,950,505
Private equity	432,031,348	-	432,031,348	379,931,583	-	379,931,583
Total investments	10,730,889,443	-	10,730,889,443	10,715,911,778	-	10,715,911,778
Securities lending collateral	686,855,566	-	686,855,566	560,663,605	-	560,663,605
Capital assets, net of accumulated depreciation	1,922,247	-	1,922,247	705,360	-	705,360
Total assets	11,856,232,162	8,720,656	11,864,952,818	11,849,959,061	4,915,284	11,854,874,345
Liabilities:						
Benefits payable	5,164,409	8,696,400	13,860,809	4,510,886	4,902,177	9,413,063
Refunds payable	13,681,903	-	13,681,903	11,502,175	-	11,502,175
Accounts and administrative expenses payable	9,287,445	24,256	9,311,701	14,953,962	13,107	14,967,069
Employer required contribution payable	-	-	-	-	-	-
Securities lending collateral payable	686,655,357	-	686,655,357	559,602,193	-	559,602,193
Due to brokers	204,381,027	-	204,381,027	220,552,386	-	220,552,386
Total liabilities	919,170,141	8,720,656	927,890,797	811,121,602	4,915,284	816,036,886
Net Position Restricted for Pension Benefits	\$ 10,937,062,021	\$ -	\$ 10,937,062,021	\$ 11,038,837,459	\$ -	\$ 11,038,837,459



Table 8

Statement of Changes in Fiduciary Net Position for Years Ended June 30, 2020, and 2019

	June 30, 2020			June 30, 2019		
	Pension Fund	Health Insurance Fund	Total Fund - Pension and Health Insurance	Pension Fund	Health Insurance Fund	Total Fund - Pension and Health Insurance
Additions:						
Contributions:						
Employee	\$ 196,097,115	\$ -	\$ 196,097,115	\$ 190,565,220	\$ -	\$ 190,565,220
Minimum funding requirement (Employer)	597,151,000	-	597,151,000	569,701,000	-	569,701,000
Minimum funding requirement (State)	257,349,000	-	257,349,000	238,869,000	-	238,869,000
Allocation to health insurance fund	(51,962,540)	51,962,540	-	(59,089,369)	59,089,369	-
Total contributions	998,634,575	51,962,540	1,050,597,115	940,045,851	59,089,369	999,135,220
Investment income:						
Net appreciation (depreciation) in fair value	98,040,345	-	98,040,345	255,501,019	-	255,501,019
Interest	160,841,176	-	160,841,176	92,918,909	-	92,918,909
Dividends	221,153,927	-	221,153,927	199,555,823	-	199,555,823
Miscellaneous	1,245,405	-	1,245,405	1,253,766	-	1,253,766
Securities lending income, net	3,675,381	-	3,675,381	4,845,062	-	4,845,062
Less investment expense:						
Investment advisory and custodial fees	(46,143,884)	-	(46,143,884)	(40,498,179)	-	(40,498,179)
Net investment income	438,812,350	-	438,812,350	513,576,400	-	513,576,400
Interest on late required contribution payments	1,166,174	-	1,166,174	1,449,709	-	1,449,709
Miscellaneous	334,222	-	334,222	238,261	-	238,261
Total additions	1,438,947,321	51,962,540	1,490,909,861	1,455,310,221	59,089,369	1,514,399,590
Deductions:						
Pension benefits	1,499,920,081	-	1,499,920,081	1,467,513,811	-	1,467,513,811
Refunds	19,522,894	-	19,522,894	24,133,854	-	24,133,854
2.2 Legislative refunds	673,391	-	673,391	562,230	-	562,230
Refund of insurance premiums	-	51,433,976	51,433,976	-	58,611,532	58,611,532
Death benefits	2,759,158	-	2,759,158	3,406,487	-	3,406,487
Total benefits payments	1,522,875,524	51,433,976	1,574,309,500	1,495,616,382	58,611,532	1,554,227,914
Administrative and miscellaneous expenses	17,847,235	528,564	18,375,799	25,621,894	477,837	26,099,731
Total deductions	1,540,722,759	51,962,540	1,592,685,299	1,521,238,276	59,089,369	1,580,327,645
Net increase	(101,775,438)	-	(101,775,438)	(65,928,055)	-	(65,928,055)
Net Position Restricted for Pension Benefits						
Beginning of the Year	11,038,837,459	-	11,038,837,459	11,104,765,514	-	11,104,765,514
Transfer of residual assets to Pension Plan	-	-	-	-	-	-
End of year	\$ 10,937,062,021	\$ -	\$ 10,937,062,021	\$ 11,038,837,459	\$ -	\$ 11,038,837,459



Table 9
Development of the Actuarial Value of Assets

Year Ending June 30	2020	2021	2022	2023
Beginning of Year:				
(1) Market Value of Assets	\$ 11,038,837,459			
(2) Actuarial Value of Assets	11,021,811,634			
End of Year:				
(3) Market Value of Assets	10,937,062,021			
(4) Contributions and Disbursements				
(4a) Actual Employer & Misc. Contributions	856,000,396			
(4b) Employee Contributions	196,097,115			
(4c) Benefit Payouts & Refunds	(1,574,309,500)			
(4d) Administrative Expenses	(18,375,799)			
(4e) Net of Contributions and Disbursements	(540,587,788)			
(5) Total Investment Income				
=(3)-(1)-(4e)	438,812,350			
(6) Projected Rate of Return	7.00%			
(7) Projected Investment Income ^a	738,338,566			
(8) Investment Income in				
Excess of Projected Income	(299,526,216)			
(9) Excess Investment Income Recognized				
This Year (4-year recognition)				
(9a) From This Year	\$ (74,881,554)			
(9b) From One Year Ago	(54,664,804)	\$ (74,881,554)		
(9c) From Two Years Ago	35,555,686	(54,664,804)	\$ (74,881,554)	
(9d) From Three Years Ago	114,636,305	35,555,685	(54,664,803)	\$ (74,881,554)
(9e) Total Recognized Investment Gain	20,645,633	(93,990,673)	(129,546,357)	(74,881,554)
(10) Change in Actuarial Value of Assets				
=(4e)+(7)+(9e)	\$ 218,396,411			
End of Year:				
(3) Market Value of Assets	\$ 10,937,062,021			
(11) Actuarial Value of Assets	\$ 11,240,208,045			
(12) Difference between Market & Actuarial Values	\$ (303,146,024)			
(13) Estimated Actuarial Value Rate of Return	7.06%			
(14) Estimated Market Value Rate of Return	4.07%			
(15) Ratio of Actuarial Value to Market Value	102.77%			

^aProjected investment income is estimated based on the actuarial value of assets and weighted timing of middle of year, 8/12^{ths} of a year, and end of year for non-investment cash flows.



Graph 6
Historical Assets Values From June 30, 2011 — 2020

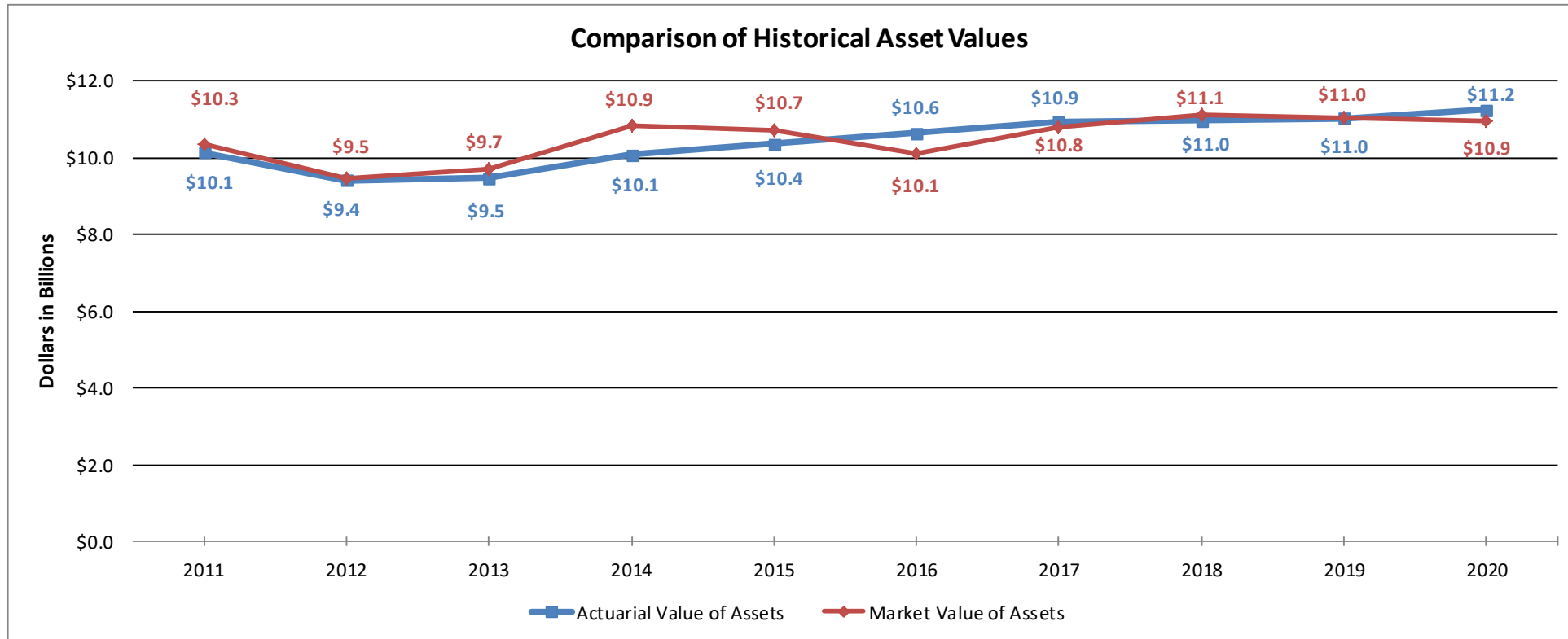
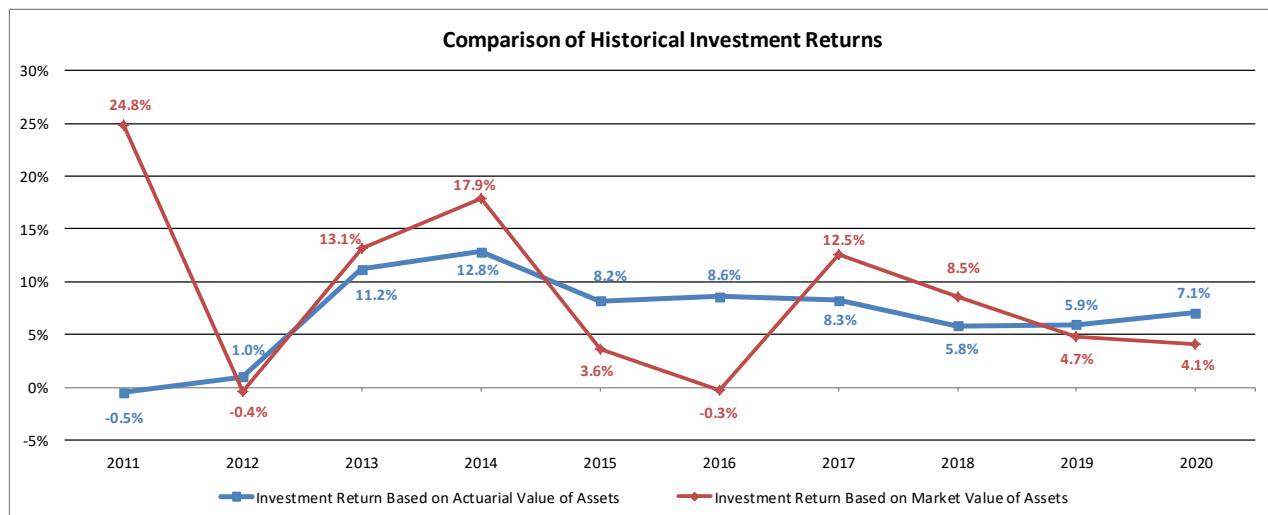


Table 10

Historical Investment Returns

Year Ended June 30,	Market Value Return	Actuarial Value Return
1996	16.3%	n/a
1997	19.8%	n/a
1998	18.2%	n/a
1999	10.7%	n/a
2000	9.5%	n/a
2001	(1.5%)	n/a
2002	(3.3%)	n/a
2003	4.0%	2.3%
2004	15.0%	3.2%
2005	10.8%	6.0%
2006	10.7%	9.6%
2007	17.7%	13.3%
2008	(5.3%)	7.9%
2009	(22.4%)	0.2%
2010	13.6%	(0.4%)
2011	24.8%	(0.5%)
2012	(0.4%)	1.0%
2013	13.1%	11.2%
2014	17.9%	12.8%
2015	3.6%	8.2%
2016	(0.3%)	8.6%
2017	12.5%	8.3%
2018	8.5%	5.8%
2019	4.7%	5.9%
2020	4.1%	7.1%
Average Returns		
Last 10 Years:	8.6%	6.8%
Last 25 Years:	7.6%	n/a

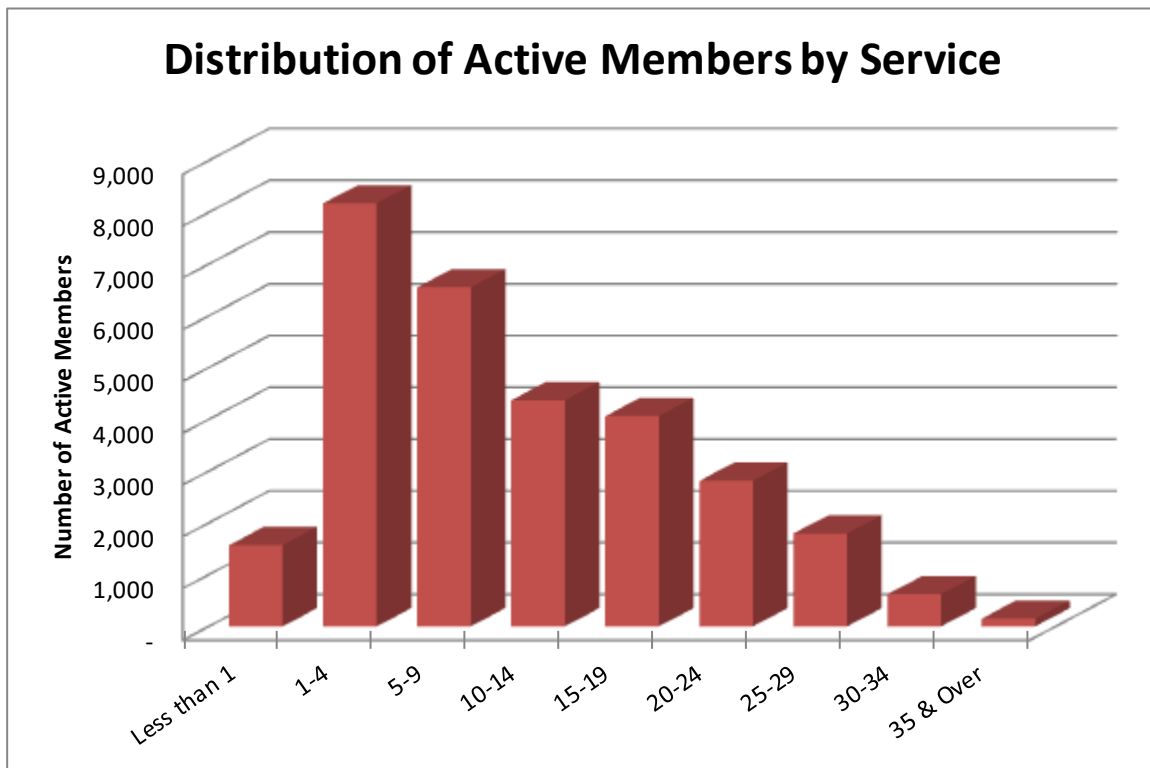
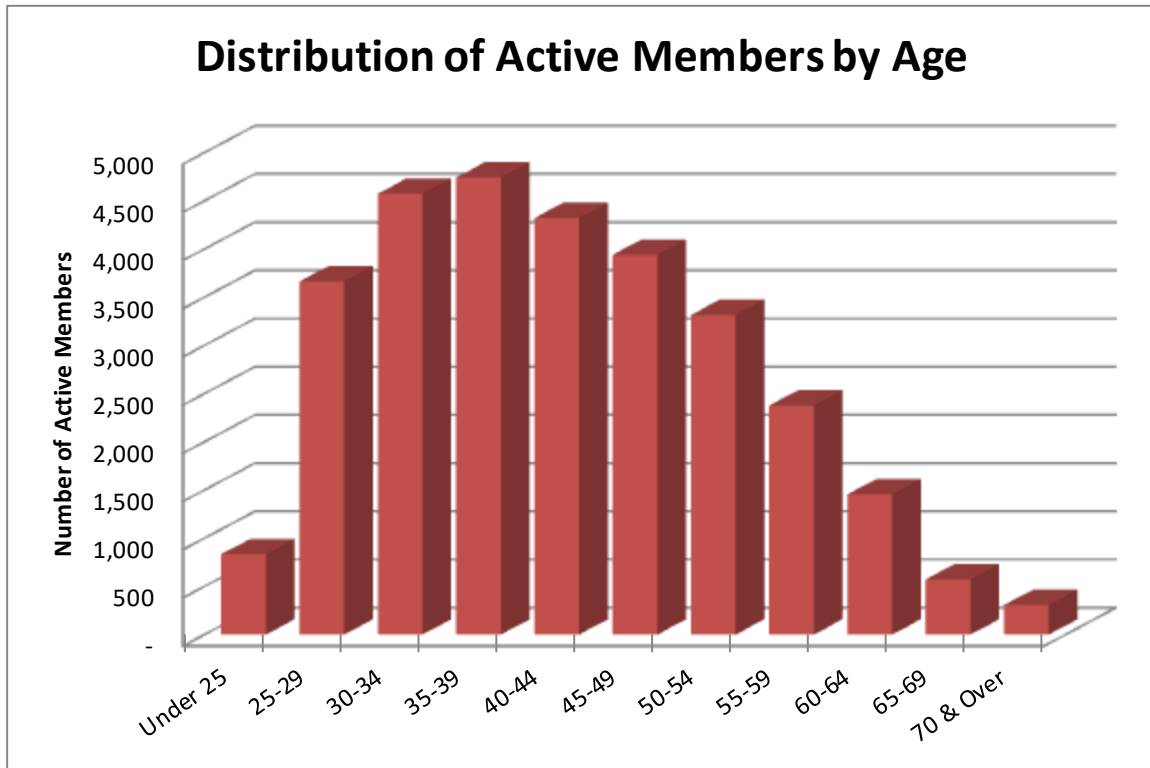


SECTION F

PARTICIPANT DATA

Graph 7

Active Members Classified by Age and Service



Graph 8

Retirees Classified by Benefit Amount and Age

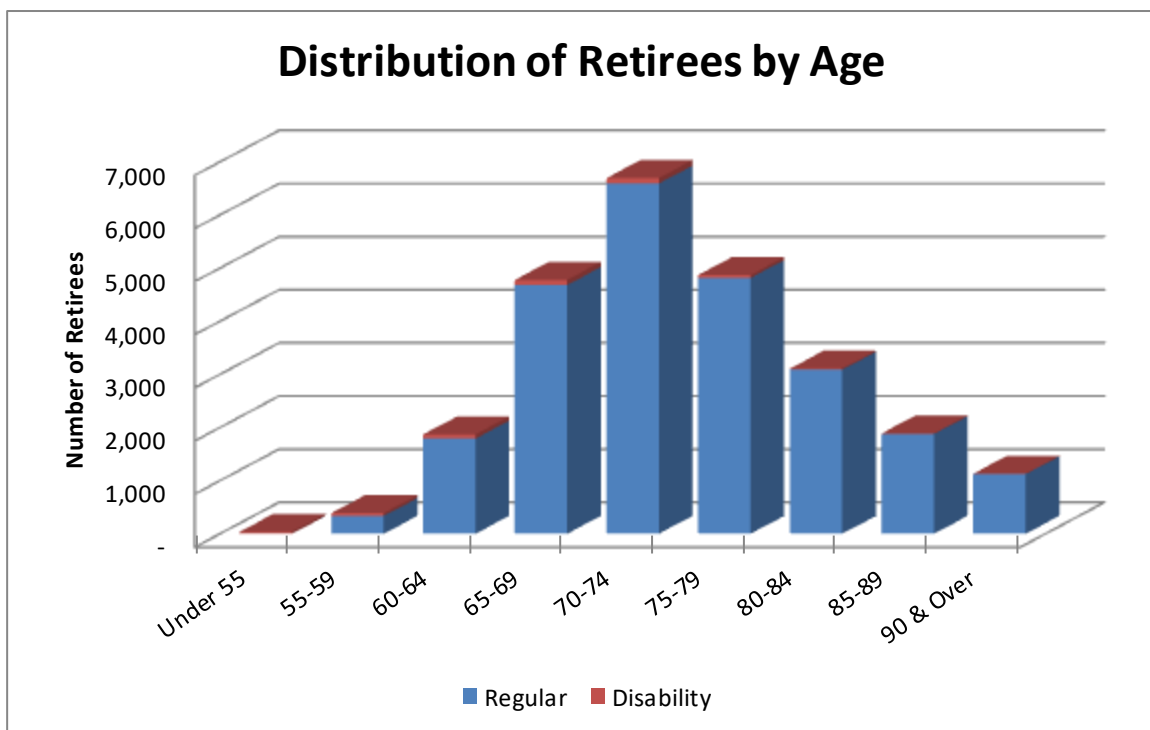
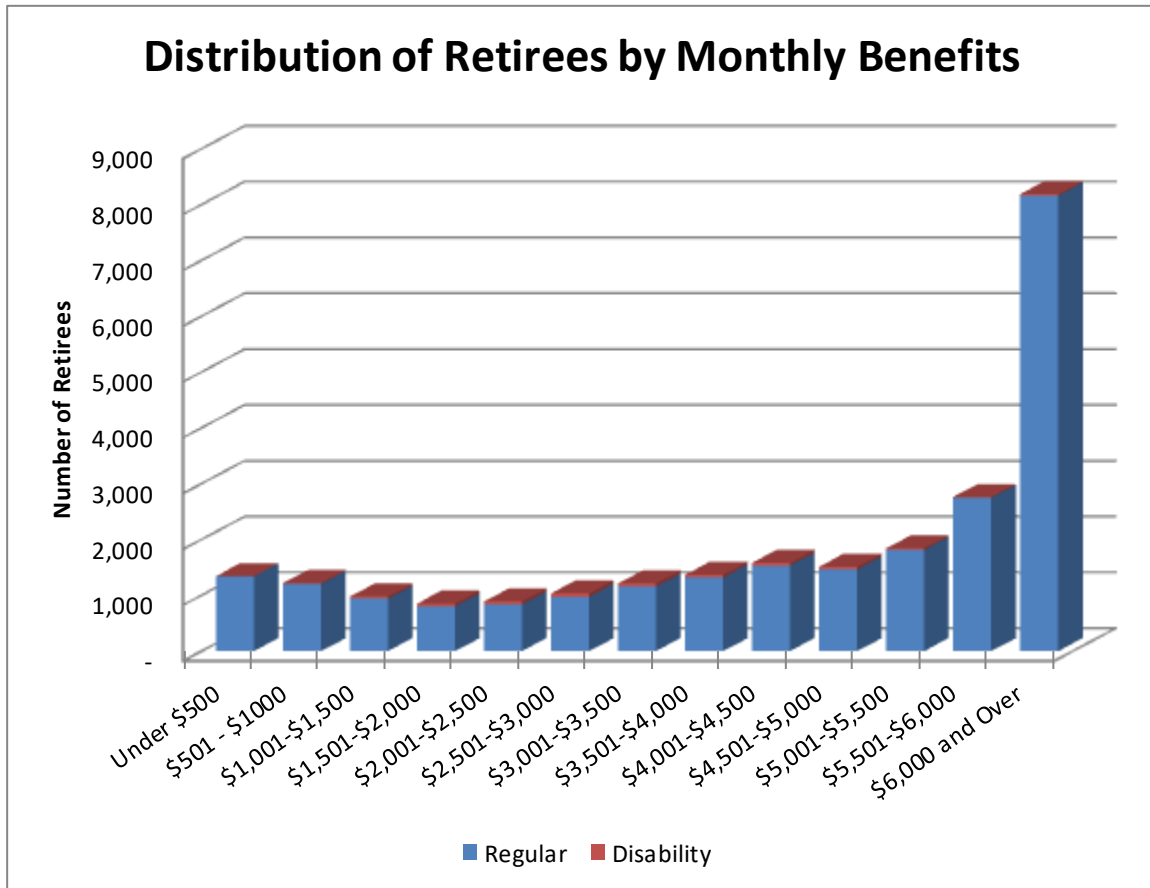


Table 11
Summary of Fund Membership

	Year Ended		Change from Prior Year
	June 30, 2020	June 30, 2019	
Active Members			
Number ^a	30,091	29,295	2.7%
Average Age	42.5	42.3	
Average Service	11.0	10.8	
Total Salary Supplied by Fund	\$2,249,491,403	\$2,179,054,844	3.2%
Average Annual Salary	\$74,756	\$74,383	0.5%
Total Active Vested Participants	16,266	16,655	-2.3%
Male Members	7,087	6,903	2.7%
Female Members	23,004	22,392	2.7%
Tier 1 Members	18,118	18,632	-2.8%
Tier 2 Members	11,973	10,663	12.3%
Inactive Vested Members^b			
Number	10,024	9,926	1.0%
Average Age	47.7	47.2	
Average Service (Excluding Reciprocal Service)	7.8	7.7	
Inactive Non-Vested Members			
Number	21,260	20,621	3.1%
Retirees			
Number	24,293	24,631	-1.4%
Average Age	75.0	74.6	
Average Annual Benefit	\$57,162	\$55,761	2.5%
Total Annual Benefit	\$1,388,627,977	\$1,373,459,588	1.1%
Disabled Retirees			
Number	460	469	-1.9%
Average Age	68.1	67.8	
Average Annual Benefit	\$40,373	\$38,717	4.3%
Total Annual Benefit	\$18,571,456	\$18,158,456	2.3%
Beneficiaries (Including Children)			
Number	3,262	3,217	1.4%
Average Age	77.4	77.0	
Average Annual Benefit	\$25,000	\$24,067	3.9%
Total Annual Benefit	\$81,549,497	\$77,424,498	5.3%
Total Members	89,390	88,159	1.4%

^a Active count excludes members expected to be hired to replace retirements and terminations that occurred in June.

^b Includes reciprocal service



Table 12
Member Population and Ratio of Non-Actives to Actives

Year Ended June 30,	Active Participants ^a	Vested Terminated Participants ^b	Retirees and Beneficiaries	Ratio of Non- Actives to Actives
2008	32,086	3,479	23,920	0.85
2009	31,905	3,056	24,218	0.85
2010	31,012	3,554	24,600	0.91
2011	30,133	4,253	25,199	0.98
2012	30,366	4,245	25,926	0.99
2013	30,969	4,502	27,440	1.03
2014	30,654	4,818	27,722	1.06
2015	29,706	5,464	28,114	1.13
2016	29,543	5,715	28,298	1.15
2017	28,855	6,062	28,439	1.20
2018	28,958	9,398	28,549	1.31
2019	29,295	9,926	28,317	1.31
2020	30,091	10,024	28,015	1.26

^a Active count excludes members expected to be hired to replace retirements and terminations that occurred in June.

^b Excludes non-vested terminated participants due a refund of member contributions.

Table 13
Total Lives and Annual Salaries of Active Members Classified by Age
and Years of Service as of June 30, 2020

Completed Years of Service																			
Attained Age	Under 1		1-4		5-9		10-14		15-19		20-24		25-29		30-34		35 & Over		Total
Under 20		-		-		-		-		-		-		-		-		-	-
	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$
20-24		156		677		-		-		-		-		-		-		-	833
	\$	2,376,431	\$	35,906,563	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$
25-29		323		2,691		649		-		-		-		-		-		-	3,663
	\$	4,589,847	\$	150,746,844	\$	42,303,935	\$	-	\$	-	\$	-	\$	-	\$	-	\$	-	\$
30-34		236		1,729		2,266		342		-		-		-		-		-	4,573
	\$	3,412,642	\$	97,822,577	\$	158,181,194	\$	29,739,209	\$	-	\$	-	\$	-	\$	-	\$	-	\$
35-39		170		986		1,586		1,573		426		-		-		-		-	4,741
	\$	2,359,830	\$	55,836,012	\$	114,125,608	\$	142,183,896	\$	41,117,115	\$	-	\$	-	\$	-	\$	-	\$
40-44		156		633		767		1,020		1,438		306		-		-		-	4,320
	\$	1,892,370	\$	35,691,764	\$	55,678,725	\$	93,347,487	\$	141,215,534	\$	32,450,924	\$	-	\$	-	\$	-	\$
45-49		137		466		503		598		917		1,052		266		-		-	3,939
	\$	1,533,556	\$	24,196,888	\$	36,443,780	\$	53,926,866	\$	88,991,095	\$	109,110,547	\$	28,883,429	\$	-	\$	-	\$
50-54		119		346		318		373		538		713		763		144		1	3,315
	\$	1,589,227	\$	17,253,169	\$	23,153,277	\$	32,422,669	\$	51,645,464	\$	71,886,017	\$	81,319,243	\$	15,745,610	\$	114,613	\$
55-59		92		288		218		221		367		415		455		286		33	2,375
	\$	749,446	\$	12,759,193	\$	14,388,296	\$	18,699,262	\$	34,491,437	\$	40,578,653	\$	46,606,743	\$	30,714,050	\$	3,839,176	\$
60-64		87		190		137		149		245		236		207		140		63	1,454
	\$	528,715	\$	6,998,768	\$	8,349,432	\$	10,772,079	\$	22,130,889	\$	23,361,674	\$	20,994,797	\$	14,643,946	\$	7,212,797	\$
65-69		50		96		70		56		88		64		73		43		32	572
	\$	295,436	\$	2,390,751	\$	2,932,760	\$	3,534,251	\$	7,811,800	\$	5,744,775	\$	7,690,902	\$	4,379,293	\$	3,361,603	\$
70 & Over		42		69		35		36		42		26		29		7		20	306
	\$	219,249	\$	1,005,805	\$	1,016,780	\$	1,586,758	\$	2,814,530	\$	2,161,429	\$	2,624,231	\$	708,048	\$	2,199,692	\$
Total		1,568		8,171		6,549		4,368		4,061		2,812		1,793		620		149	30,091
	\$	19,546,748	\$	440,608,335	\$	456,573,786	\$	386,212,477	\$	390,217,864	\$	285,294,019	\$	188,119,345	\$	66,190,948	\$	16,727,882	\$

Total lives and annual salaries exclude 106 members expected to be hired to replace retirements and terminations that occurred in June 2020.



Table 14
Reconciliation of Member Data as of June 30, 2020

	Actives	Deferred Vested	Expected Refunds	Retirees	Disabilities	Beneficiaries	Total
Totals as of the June 30, 2019 Actuarial Valuation	29,295	9,926	20,621	24,631	469	3,217	88,159
New Entrants	2,848	-	-	-	-	-	2,848
Rehires — Members with Service	653	(234)	(419)	-	-	-	-
Non-Vested Terminations	(1,531)	(15)	1,546	-	-	-	-
Vested Terminations	(602)	676	(74)	-	-	-	-
Service Retirements	(332)	(203)	-	539	(4)	-	-
Disabilities	(8)	(6)	-	(2)	16	-	-
Deaths	(33)	(17)	(34)	(887)	(21)	(180)	(1,172)
New Beneficiaries	-	-	-	-	-	223	223
Refunds and Benefit Terminations	(199)	(108)	(565)	-	-	-	(872)
Data Adjustments	-	5	185	12	-	2	204
Net Change	796	98	639	(338)	(9)	45	1,231
Totals as of the June 30, 2020 Actuarial Valuation	30,091	10,024	21,260	24,293	460	3,262	89,390

Active count excludes members expected to be hired to replace retirements and terminations that occurred in June.

Table 15
History of Retirees and Beneficiaries Added to Rolls during
Fiscal Year Ended June 30, 2020

Fiscal Year	Added to Rolls		Removed from Rolls		Rolls - End of Year		Average Annual Allowances	Increase in Avg. Annual Allowance
	No.	Annual Allowance	No.	Annual Allowance	No.	Annual Allowance		
2011	1,394	\$ 86,404,558	795	\$ 26,694,342	25,199	\$ 1,047,873,059	\$ 41,584	3.52%
2012	1,579	96,719,811	852	28,511,983	25,926	1,116,080,887	43,049	3.52%
2013	2,129	130,553,477	615	21,764,846	27,440	1,224,869,518	44,638	3.69%
2014	1,006	70,963,134	724	26,376,522	27,722	1,269,456,130	45,792	2.59%
2015	1,302	85,087,053	910	34,678,799	28,114	1,319,864,384	46,947	2.52%
2016	1,101	78,909,433	917	36,891,261	28,298	1,361,882,556	48,126	2.51%
2017	1,072	79,434,960	931	37,328,244	28,439	1,403,989,272	49,368	2.58%
2018	1,025	79,219,962	915	39,123,440	28,549	1,444,085,794	50,583	2.46%
2019	708	67,514,478	940	42,557,730	28,317	1,469,042,542	51,878	2.56%
2020	792	71,924,782	1,094	52,218,394	28,015	1,488,748,930	53,141	2.43%

Table 16
Number of Annuitants Classified by Benefit Type and Amount as of June 30, 2020

Amount of Monthly Benefit	Retirees		Disabled Retirees		Beneficiaries		Total	
	Male	Female	Male	Female	Male	Female	Male	Female
\$ 1 - 500	374	966	1	1	161	229	536	1,196
501 - 1000	345	861	3	9	174	326	522	1,196
1,001-1,500	294	648	11	25	147	180	452	853
1,501-2,000	229	573	6	42	168	195	403	810
2,001-2,500	207	629	13	39	144	235	364	903
2,501-3,000	230	744	8	49	186	300	424	1,093
3,001-3,500	245	913	11	42	148	318	404	1,273
3,501-4,000	286	1,033	10	37	26	101	322	1,171
4,001-4,500	290	1,242	14	33	16	80	320	1,355
4,501-5,000	296	1,166	9	39	13	40	318	1,245
5,001-5,500	420	1,394	5	20	8	31	433	1,445
5,501-6,000	645	2,100	4	9	5	18	654	2,127
6,001-6,500	718	2,395	1	8	3	6	722	2,409
6,501-7,000	471	1,508	1	1	-	3	472	1,512
7,001-7,500	218	662	2	1	-	-	220	663
7,501-8,000	155	277	1	-	-	-	156	277
8,001-8,500	122	221	1	2	-	-	123	223
8,501-9,000	88	193	-	-	-	-	88	193
9,001-9,500	79	184	1	-	-	-	80	184
Over \$9,500	293	579	1	-	-	1	294	580
Totals	6,005	18,288	103	357	1,199	2,063	7,307	20,708

Table 17
Initial Year of Retirement Analysis

	Years of Credited Service								Total
	0-4	5-9	10-14	15-19	20-24	25-29	30-34		
Fiscal Year 2016									
Avg Monthly Annuity	\$ 326	\$ 840	\$ 1,493	\$ 2,432	\$ 3,440	\$ 4,294	\$ 5,701	\$ 3,286	
Avg Monthly FAS	\$ 7,267	\$ 5,266	\$ 5,627	\$ 6,515	\$ 7,301	\$ 7,711	\$ 8,026	\$ 7,054	
Number of Retirees	61	92	77	113	184	123	202	852	
Average Age								63.1	
Fiscal Year 2017									
Avg Monthly Annuity	\$ 323	\$ 734	\$ 1,578	\$ 2,516	\$ 3,438	\$ 4,301	\$ 5,684	\$ 3,466	
Avg Monthly FAS	\$ 6,255	\$ 4,332	\$ 5,819	\$ 6,705	\$ 7,268	\$ 7,612	\$ 7,975	\$ 6,961	
Number of Retirees	38	80	83	99	167	129	219	815	
Average Age								63.7	
Fiscal Year 2018									
Avg Monthly Annuity	\$ 336	\$ 823	\$ 1,503	\$ 2,578	\$ 3,471	\$ 4,505	\$ 5,867	\$ 3,382	
Avg Monthly FAS	\$ 6,507	\$ 5,349	\$ 5,502	\$ 6,738	\$ 7,407	\$ 7,927	\$ 8,166	\$ 7,089	
Number of Retirees	39	92	81	88	175	122	171	768	
Average Age								63.2	
Fiscal Year 2019									
Avg Monthly Annuity	\$ 305	\$ 699	\$ 1,634	\$ 2,547	\$ 3,672	\$ 4,789	\$ 6,009	\$ 3,150	
Avg Monthly FAS	\$ 6,069	\$ 4,827	\$ 6,098	\$ 6,673	\$ 7,644	\$ 8,482	\$ 8,446	\$ 7,046	
Number of Retirees	42	64	64	62	77	62	94	465	
Average Age								63.7	
Fiscal Year 2020									
Avg Monthly Annuity	\$ 318	\$ 768	\$ 1,615	\$ 2,578	\$ 3,587	\$ 4,659	\$ 6,170	\$ 3,211	
Avg Monthly FAS	\$ 6,442	\$ 5,315	\$ 5,859	\$ 6,746	\$ 7,717	\$ 8,076	\$ 8,578	\$ 7,179	
Number of Retirees	49	76	51	72	114	98	91	551	
Average Age								63.5	

Table 18
Schedule of Active Member Payroll

Valuation Date June 30,	Number	Covered Payroll	Average Annual Payroll	% Increase in Avg. Payroll
2011	30,133	\$2,090,131,858	\$ 69,364	
2012	30,366	2,224,903,121	73,270	5.63%
2013	30,969	2,239,347,051	72,309	-1.31%
2014	30,654	2,233,280,995	72,854	0.75%
2015	29,706	2,273,551,432	76,535	5.05%
2016	29,543	2,281,268,890	77,219	0.89%
2017	28,855	2,030,175,116	70,358	-8.88%
2018	28,958	2,094,830,446	72,340	2.82%
2019	29,295	2,179,054,844	74,383	2.82%
2020	30,091	2,249,491,403	74,756	0.50%

Table 19
Solvency Test

Valuation Date June 30,	Total Actuarial Accrued Liability	(1) Active Member Contributions	(2) Retirees, Term Vested and Beneficiaries	(3) Active Members (Employer Financed Portion)	Actuarial Value of Assets	Portion (%) of Present Value Covered By Assets		
						(1)	(2)	(3)
2017	\$ 21,822,010,297	\$ 1,608,474,476	\$ 16,244,526,663	\$ 3,969,009,158	\$ 10,933,031,685	100%	57%	0%
2018	22,922,992,558	1,659,408,813	17,510,678,092	3,752,905,653	10,969,085,523	100%	53%	0%
2019	23,252,163,307	1,752,007,367	17,532,864,523	3,967,291,417	11,021,811,634	100%	53%	0%
2020	24,073,482,607	1,851,487,971	17,866,192,151	4,355,802,485	11,240,208,045	100%	53%	0%

10 fiscal years will be built prospectively.

Table 20
Participating Employers

Participating Employer	2020		2019	
	Covered Employees	% of Total CTPF Membership	Covered Employees	% of Total CTPF Membership
Chicago Public Schools	25,450	84.58%	24,805	84.67%
Noble Network of Charter Schools	738	2.45%	717	2.45%
UNO Charter School Network	690	2.29%	635	2.17%
Learn Charter School Network	264	0.88%	236	0.81%
KIPP Chicago Schools	209	0.69%	180	0.61%
Distinctive Schools	198	0.66%	245	0.84%
Charter Schools USA	174	0.58%	72	0.25%
Concept Schools	168	0.56%	178	0.61%
Civitas Schools	168	0.56%	176	0.60%
Youth Connection Charter Schools	132	0.44%	136	0.46%
Total, Largest 10 Employers	28,191	93.69%	27,380	93.46%
All Other Employers	1,900	6.31%	1,915	6.54%
Grand Total	30,091	100.00%	29,295	100.00%

SECTION G

ACTUARIAL METHODS AND ASSUMPTIONS

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Actuarial Cost Method as Mandated by 40 ILCS 5/17-129, Adopted August 31, 1991

The Projected Unit Credit normal cost method is used. Under this method, the projected pension at retirement age is first calculated and the present value at the individual member's current or attained age is determined. The normal cost for the member for the current year is equal to the actuarial present value divided by the member's projected service at retirement. The normal cost for the plan for the year is the sum of the individual normal costs.

The actuarial liability at any point in time is the present value of the projected pensions at that time less the present value of future normal costs.

For ancillary benefits for active members, in particular death and survivor benefits, termination benefits and the postretirement increases, the same procedure as outlined above is followed.

Estimated annual administrative expenses are added to the normal cost.

For actuarial valuation purposes, as well as projection purposes, an actuarial value of assets is used. The actuarial value of assets is projected to equal about 100 percent of the market value on and after June 30, 2025.



Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Actuarial Assumptions

Actuarial assumptions are set by the Board of Trustees. All actuarial assumptions are expectations of future experience and are not market measures. The rationale for the actuarial assumptions may be found in the 2018 Actuarial Experience Study issued on May 25, 2018 and the 2020 Actuarial Assumption Study issued on September 10, 2020.

Rate of Investment Return

6.75 percent per year, compounded annually, net of investment expenses. First effective with the actuarial valuation as of June 30, 2020.

Price Inflation (Increase in Consumer Price Index “CPI”)

2.25 percent per annum, compounded annually. First effective with the actuarial valuation as of June 30, 2020.

This assumption serves as the basis for the determination of annual increases in pension and the pensionable salary cap for Tier 2 members.

Cost of Living Adjustment “COLA”

The assumed rate is 3.00 percent per year for members hired before January 1, 2011, based on the benefit provision of 3.00 percent annual compound increases. The assumed rate is 1.125 percent for members hired on and after January 1, 2011, based on the benefit provision of increases equal to $\frac{1}{2}$ of the increase in CPI-U with a maximum increase of 3.00 percent. First effective with the actuarial valuation as of June 30, 2020.

Wage Inflation

2.75 percent per annum, compounded annually. First effective with the actuarial valuation as of June 30, 2020.

Calculation of the Actuarially Determined Contribution

The amortization factor used to calculate the ADC is based on the Fund’s assumed interest rate of 6.75 percent and an annualized assumed rate of increase in total capped payroll of 2.00 percent (which is consistent with the projected increase in total payroll from the projections used to calculate the statutory contribution requirements). The Employer Normal Cost and Amortization Payment are adjusted for expected contribution timing.



Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Total Payroll

Unless stated otherwise, total payroll includes employee contributions of 7.00 percent of salary picked up by the Board of Education for employees hired prior to January 1, 2017. All contributions are calculated based on total payroll.

Mortality

Applicable Group	Base Mortality Table	Male Scaling Factor	Female Scaling Factor
Pre-retirement Mortality	RP-2014 White Collar Employee, sex distinct	98%	113%
Post-retirement Disabled Mortality	RP-2014 Disabled Annuitant, sex distinct	103%	106%
Post-retirement Healthy Mortality	RP-2014 White Collar Healthy Annuitant, sex distinct	108%	94%

Future mortality improvements are reflected by projecting the base mortality tables back from the year 2014 to the year 2006 using the Society of Actuaries (SOA) MP-2014 (referred to as the RP-2006 base mortality tables) and projecting from 2006 using the MP-2017 projection scale. The assumptions are generational mortality tables and include a margin for improvement.

Age	Future Life Expectancy (years) in 2020				Future Life Expectancy (years) in 2030			
	Post-retirement Healthy		Post-retirement Disabled		Post-retirement Healthy		Post-retirement Disabled	
	Male	Female	Male	Female	Male	Female	Male	Female
35	51.15	54.41	34.29	40.67	52.13	55.34	35.90	42.14
40	45.97	49.22	30.58	36.23	46.93	50.13	32.03	37.60
45	40.88	44.09	27.28	32.13	41.81	44.98	28.60	33.41
50	35.88	38.99	24.11	28.17	36.79	39.87	25.34	29.37
55	31.00	33.95	21.01	24.41	31.88	34.81	22.16	25.53
60	26.25	29.04	18.00	20.93	27.09	29.86	19.03	21.93
65	21.69	24.32	15.16	17.58	22.46	25.09	16.03	18.44
70	17.37	19.82	12.45	14.27	18.06	20.53	13.18	15.02
75	13.38	15.57	9.86	11.16	13.99	16.22	10.48	11.84

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Termination

Service-based termination rates were used. Sample rates are as follows:

Termination			
Service (Beginning of Year)	Rate (%)	Service (Beginning of Year)	Rate (%)
0	30.00%	16	2.25%
1	16.00%	17	2.25%
2	13.00%	18	2.25%
3	12.00%	19	2.25%
4	9.00%	20	2.25%
5	9.00%	21	2.25%
6	8.00%	22	2.25%
7	6.00%	23	2.25%
8	5.00%	24	2.25%
9	5.00%	25	2.25%
10	4.00%	26	2.25%
11	3.00%	27	2.25%
12	3.00%	28	2.25%
13	3.00%	29	2.25%
14	3.00%	30	1.75%
15	3.00%	31+	1.75%

It is assumed that terminated employees will not be rehired. The rates apply only to employees who have not fulfilled the service requirement necessary for retirement at any given age.

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Salary Increases

Illustrative rates of increase per individual employee per year, compounded annually:

Age	Annual Increase
20	12.60%
25	7.50%
30	6.00%
35	5.25%
40	4.25%
45	3.50%
50	3.00%
55	2.75%
60	2.75%
65	2.75%
70	2.75%

The underlying salary increase assumption is based on a wage inflation assumption of 2.75 percent per year. First effective with the actuarial valuation as of June 30, 2020.

Disability

Disability rates, based on recent experience of the Fund, were applied to members with at least 10 years of service. All disabilities are assumed to be non-duty disabilities. Sample rates are as follows:

Age	Rate (%)
20	0.04%
25	0.04%
30	0.04%
35	0.05%
40	0.06%
45	0.08%
50	0.19%
55	0.24%
60	0.29%

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Retirement

Employees are assumed to retire in accordance with the rates shown below. The rates apply only to employees who have fulfilled the service requirement necessary for retirement at any given age.

Retirement Rates for Tier 1 Employees		
Age	<34 Years of Service Rate (%)	34+ Years of Service Rate (%)
55	5.00%	20.00%
56	5.00%	20.00%
57	5.00%	20.00%
58	5.00%	20.00%
59	7.00%	20.00%
60	9.00%	22.50%
61	11.00%	22.50%
62	12.00%	22.50%
63	13.00%	22.50%
64	14.00%	22.50%
65	15.00%	25.00%
66	16.00%	25.00%
67	17.00%	25.00%
68	18.00%	27.50%
69	19.00%	27.50%
70	20.00%	30.00%
71	20.00%	30.00%
72	20.00%	30.00%
73	20.00%	30.00%
74	20.00%	30.00%
75	100.00%	100.00%

Retirement Rates for Tier 2 Employees	
Age	Rate (%)
62	40.00%
63	25.00%
64	25.00%
65	30.00%
66	25.00%
67	30.00%
68	20.00%
69	20.00%
70	20.00%
71	20.00%
72	20.00%
73	20.00%
74	20.00%
75	100.00%

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Active Member Population as of the Actuarial Valuation Date

The Tier 2 active population as of the actuarial valuation date of June 30, 2020, was increased by 106 members in order to estimate the total expected number of active members that will be working and making contributions in the upcoming fiscal year. Members who retire at the end of the school year have June retirement dates and are already reflected as retirees in the data received as of June 30, but new active members to replace these members are not hired until August or September and are not included in the census data until the following fiscal year. These members are assumed to have a similar demographic profile as new entrants who have been hired in the last three years.

Population Projection

For purposes of determining annual appropriation as a percent of total covered payroll, the size of the active group is assumed to remain level at the number of actives as of the actuarial valuation date including expected new hires, or 30,197. New entrants are assumed to enter with an average age and an average pay as disclosed below. New entrants are assumed to have a similar demographic profile of recent new entrants to the Fund (as shown in the table below). The average increase in uncapped payroll for the projection period is 2.75 percent per year.

New Entrant Profile			
Age Group	No.	Salary	
Under 20			
20-24	1,103	\$	53,724,459
25-29	1,644		82,971,466
30-34	859		44,702,997
35-39	529		27,506,887
40-44	344		17,591,104
45-49	295		14,300,014
50-54	221		10,269,141
55-59	165		6,304,777
60-64	103		2,830,616
65-69	13		351,188
70 & Over			
Total	5,276	\$	260,552,649
Avg. Salary		\$	49,385
Avg. Age			32.69
Percent Female			76%

Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Assets

The asset values used for the actuarial valuation were based on asset information contained in the financial statements for the year ended June 30, 2020, as prepared by the Fund. The actuarial value of assets was determined by smoothing unexpected gains or losses over a period of four years. The investment gain or loss for a year is calculated as the total investment income on the market value of assets, minus expected investment return on the prior actuarial value of assets. The final actuarial value is equal to the expected actuarial value plus (or minus) 25 percent of the calculated gain (or loss) in the prior four years.

Expenses

Administrative expenses included in the normal cost for fiscal year 2021 are the budgeted administrative expense amount of \$23,319,842, as provided by Staff. Future administrative expenses are assumed to increase by 5.75 percent per year for 14 years and then increase at a rate consistent with the increase in projected capped payroll thereafter.

Marriage Assumption

75.0 percent of active male participants and 65.0 percent of active female participants are assumed to be married. Actual marital status at benefit commencement is used for retirees.

Spouse's Age

The female spouse is assumed to be two years younger than the male spouse.

Total Service at Retirement

A teacher's total service credit at retirement is assumed to be 103.3 percent of the teacher's regular period of service at retirement.

Valuation of Inactive Members Eligible for Deferred Vested Pension Benefits

Benefits for inactive deferred vested members were determined by projecting the accumulated contribution balance to retirement (age 62) with interest at the assumed investment rate of return, converted to an annuity, and then loaded by 35 percent.

Assumption for Missing Data

Members whose gender was not provided are assumed to be female.

Benefit Option

Retirees whose record includes a spouse date of birth are assumed to have the automatic 50% Joint and Survivor benefit. All other retirees are assumed to have a straight life benefit.



Actuarial Methods and Assumptions (Most Adopted Effective with the June 30, 2018 Actuarial Valuation)

Contribution Timing

Projected employer contributions are assumed to occur based on the following timing:

1. Additional Board of Education Contribution (0.58 percent of pay) — June 30th (End of Year)
2. Additional State Contribution (0.544 percent of pay) — Monthly (Middle of Year)
3. State Normal Cost Contribution — Monthly (Middle of Year)
4. Board of Education Early Payment of Special Tax Levy — March 1st, annually
 - a. 55 percent of prior year's tax levy is assumed to occur each March 1st
 - i. The March 1st payment amount as provided by Staff is \$263,523,630 for fiscal year 2020.
5. Remaining Board of Education Contribution — June 30th (End of Year)

Decrement Timing

All decrements are assumed to occur during the middle of the year.

Decrement Relativity

Decrement rates are used directly from the experience study, without adjustment for multiple decrement table effects.

Decrement Operation

Turnover decrements do not operate after a member reaches retirement eligibility. Disability decrements do not operate after a member reaches normal retirement eligibility.

Eligibility Testing

Eligibility for benefits is determined based upon the age nearest birthday and service on the date the decrement is assumed to occur.

Assumptions as a Result of Public Act 96-0889

Members hired on or after January 1, 2011, are assumed to make contributions on salary up to the final average compensation cap in a given year.

State contributions, expressed as a percentage of pay, are calculated based upon capped pay.

Capped (pensionable) pay was \$115,929 for fiscal year 2020 and increases at ½ the annual increase in the Consumer Price Index-U thereafter.

The annual increase in the Consumer Price Index-U is assumed to be 2.25 percent for all years. First effective with the actuarial valuation as of June 30, 2020.



Projection Methodology and Appropriation Requirements under P.A. 90-0655, P.A. 91-0357, P.A. 96-0889, P.A. 99-0521 and P.A. 100-0465

Employer Contributions under P.A. 96-0889

The following is an excerpt from the Illinois Compiled statutes 40 ILCS 5/17-129 (b)(iv)-(b)(vii):

(iv) For fiscal years 2014 through 2059, the minimum contribution to the Fund to be made by the Board of Education in each fiscal year shall be an amount determined by the Fund to be sufficient to bring the total assets of the Fund up to 90% of the total actuarial liabilities of the Fund by the end of fiscal year 2059. In making these determinations, the required Board of Education contribution shall be calculated each year as a level percentage of the applicable employee payrolls over the years remaining to and including fiscal year 2059 and shall be determined under the Projected Unit Credit actuarial cost method.

(v) Beginning in fiscal year 2060, the minimum Board of Education contribution for each fiscal year shall be the amount needed to maintain the total assets of the Fund at 90% of the total actuarial liabilities of the Fund.

(vi) Notwithstanding any other provision of this subsection (b), for any fiscal year, the contribution to the Fund from the Board of Education shall not be required to be in excess of the amount calculated as needed to maintain the assets (or cause the assets to be) at the 90% level by the end of the fiscal year.

(vii) Any contribution by the State to or for the benefit of the Fund, including, without limitation, as referred to under Section 17-127, shall be a credit against any contribution required to be made by the Board of Education under this subsection (b).

Additional State and Employer Contributions under P.A. 90-0655 and P.A. 91-0357

The following is an excerpt from the Illinois Compiled statutes 40 ILCS 5/17-127 (b):

(b) The General Assembly finds that for many years the State has contributed to the Fund an annual amount that is between 20% and 30% of the amount of the annual State contribution to the Article 16 retirement system, and the General Assembly declares that it is its goal and intention to continue this level of contribution to the Fund in the future. Beginning in State fiscal year 1999, the State shall include in its annual contribution to the Fund an additional amount equal to 0.544% of the Fund's total teacher payroll; except that this additional contribution need not be made in a fiscal year if the Board has certified in the previous fiscal year that the Fund is at least 90% funded, based on actuarial determinations. These additional State contributions are intended to offset a portion of the cost to the Fund of the increases in retirement benefits resulting from this amendatory Act of 1998.

The following is an excerpt from the Illinois Compiled statutes 40 ILCS 5/17-127.2 (1)-(2):

Sec. 17-127.2. Additional contributions by employer of teachers. Beginning July 1, 1998, the employer of a teacher shall pay to the Fund an employer contribution computed as follows:

(1) Beginning July 1, 1998 through June 30, 1999, the employer contribution shall be equal to 0.3% of each teacher's salary.



Projection Methodology and Appropriation Requirements under P.A. 90-0655, P.A. 91-0357, P.A. 96-0889, P.A. 99-0521 and P.A. 100-0465

(2) Beginning July 1, 1999 and thereafter, the employer contribution shall be equal to 0.58% of each teacher's salary. The employer may pay these employer contributions out of any source of funding available for that purpose and shall forward the contributions to the Fund on the schedule established for the payment of member contributions. These employer contributions need not be made in a fiscal year if the Board has certified in the previous fiscal year that the Fund is at least 90% funded, based on actuarial determinations. These employer contributions are intended to offset a portion of the cost to the Fund of the increases in retirement benefits resulting from Public Act 90-582.

Board of Education Dedicated Property Tax Levy under P.A. 99-0521 as Amended by P.A. 100-0465

The following is an excerpt from the Illinois Compiled statutes 105 ILCS 5/34-53:

Beginning on the effective date of this amendatory Act of the 99th General Assembly, for the purpose of making an employer contribution to the Public School Teachers' Pension and Retirement Fund of Chicago, the board may levy annually for taxable years prior to 2017, upon all taxable property located within the district, a tax at a rate not to exceed 0.383%. Beginning with the 2017 taxable year, for the purpose of making an employer contribution to the Public School Teachers' Pension and Retirement Fund of Chicago, the board may levy annually, upon all taxable property within the district, a tax at a rate of 0.567%. The proceeds from this additional tax shall be paid, as soon as possible after collection, directly to the Public School Teachers' Pension and Retirement Fund of Chicago and not to the Board of Education.

State Contributions under P.A. 100-0465

The following is an excerpt from the Illinois Compiled statutes 40 ILCS 5/17-127 (d)(1)-(d)(2):

(d) In addition to any other contribution required under this Article, including the contribution required under subsection (c), the State shall contribute to the Fund the following amounts:

(1) For State fiscal year 2018, the State shall contribute \$221,300,000 for the employer normal cost for fiscal year 2018 and the amount allowed under paragraph (3) of Section 17-142.1 of this Code to defray health insurance costs. Funds to this paragraph (1) shall come from funds appropriated for Evidence-Based Funding pursuant to Section 18-8.15 of the School Code.

(2) Beginning in State fiscal year 2019, the State shall contribute for each fiscal year an amount to be determined by the Fund, equal to the employer normal cost for that fiscal year, plus the amount allowed pursuant to paragraph (3) of Section 17-142.1 to defray health insurance costs.

We calculated the required contribution based on the above legislation; the results are shown in the summary section of this report.



SECTION H

SUMMARY OF PLAN PROVISIONS

Summary of Plan Provisions (as of June 30, 2020)

It should be noted that the purpose of this section is to describe the benefit structures of CTPF for which actuarial values have been generated. Many portions of the defined plans are described in a manner which may not be legally complete or precise.

It is not our intent to provide an exhaustive description of all benefits provided under CTPF or the policies and procedures utilized by CTPF staff. A more precise description of the provisions of CTPF can be found in Illinois Compiled Statutes (ILCS) Chapter 40, Articles 1, 17 and 20. In all situations, the plan provisions described in the Statutes govern.



Summary of Plan Provisions (as of June 30, 2020)

Purpose

The Public School Teachers' Pension and Retirement Fund of Chicago, established in 1895 by the Illinois state legislature, is a defined benefit public employee retirement fund that provides retirement, survivor and disability benefits to certain teachers and employees of the Chicago Public Schools.

Administration

Responsibility for the operation of the Fund and the direction of its policies is vested in a Board of Trustees of 12 members. The 12-member Board of Trustees is comprised of six members elected by the teacher contributors, three members elected by the annuitants, one member elected by the principal contributors and two members are appointed by the Board of Education. The administration of the detailed affairs of the Fund is the responsibility of the Executive Director who is appointed by the Board of Trustees. Administrative policies and procedures are designed to ensure an accurate accounting of funds of CTPF and prompt payment of claims for benefits within the applicable statute.

Membership

Any teacher and certain other employees of the Chicago Public Schools, approved charter schools and the Chicago Teachers' Pension Fund are participants of CTPF. Members hired prior to January 1, 2011, participate under the Tier 1 benefit structure. Members hired on and after January 1, 2011, participate under the Tier 2 benefit structure.

Membership Service

Membership service includes all service rendered while a member of the Fund for which credit is allowable. Contributors to the Fund cannot earn more than one year of service credit per fiscal year. Validated service within a fiscal year is determined on a schedule of 170 days.

Member Contributions

Members are required to contribute a percentage of salary as their share of meeting the cost of the various benefits. The total contribution rate of 9.0 percent of salary consists of 7.5 percent towards the retirement pension, 1.0 percent towards the survivor pension and 0.5 percent towards the post-retirement increase.

As of September 1981, the Board of Education has been paying 7.0 percent of the required teacher contributions for Chicago public school teachers. Charter school contributions may be contributed at various rates by the employers and teachers.

As a result of the collective bargaining agreement between the Board of Education of the City of Chicago and the Chicago Teachers Union, Local No. 1, American Federation of Teachers, AFL-CIO, which became effective December 7, 2016, teachers hired on and after January 1, 2017, will no longer receive the pension pick-up of 7.0 percent from the Board of Education.



Summary of Plan Provisions (as of June 30, 2020)

Retirement Pension

Qualification of Member

A member is eligible for a retirement pension after (1) completing 20 years of validated service, with the pension payable at age 55 or older, or (2) after completing five years of service with the pension payable at age 62 or older.

Amount of Pension

The pension is based on the member's final average salary and the number of years of service credit that has been established.

Final Average Salary is the average of the highest rates of salary for any four consecutive years of validated service within the last 10 years.

For service earned before July 1, 1998, the amount of the service retirement pension is 1.67 percent of final average salary for the first 10 years, 1.90 percent for each of the next 10 years, 2.10 percent for each of the following 10 years and 2.30 percent for each year above 30. For service earned after June 30, 1998, the amount of the service retirement pension is 2.2 percent of final average salary for each year of service.

Service earned before July 1, 1998, can be upgraded to the 2.2 percent formula through the payment of additional employee contributions of 1 percent of the teacher's highest salary within the last four years for each year of prior service, up to a maximum of 20 percent, which upgrades all service years. The number of years for which contributions are required is reduced by one for each three full years of service after June 30, 1998. No contribution is required if the member has at least 30 years of service.

The maximum pension payable is 75 percent of final average salary or \$1,500 per month, whichever is greater.

Annual Increases in Pension

Postretirement increases of 3.0 percent of the current pension (i.e., increases are compounded) are granted to members effective each January 1, occurring on or after the first anniversary of the pension or the 61st birthday, whichever is later.

Reductions

Except for retirement after 34 years of service, the retirement pension is reduced by $\frac{1}{2}$ of 1.0 percent for each month the member is under age 60.



Summary of Plan Provisions (as of June 30, 2020)

Survivors Annuity

Qualification of Survivor

A surviving spouse or unmarried minor children is entitled to a pension upon the death of a member while in service or in retirement. Survivor's pensions are conditioned upon marriage having been in effect at least one year prior to death.

Amount of Pension

The minimum survivor's pension upon death of an active or retired member is 50 percent of the deceased member's pension at the date of death. If the surviving spouse is under age 50, and no unmarried minor children under age 18 survive, payment of the survivor's pension is deferred until age 50.

Annual Increases in Pension

Survivor's pensions are subject to annual increases of 3.0 percent per year based on the current amount of pension starting the later of when the member would have attained age 61 and receipt of one year's pension payments.

Death Benefits

Amount and Duration of Payment

Upon the death of a member in service, a refund equal to the total contributions less contributions for survivor's pensions is payable without interest to a designated beneficiary or the estate of the member. The death benefit payable is the lesser of \$10,000 and salary earned for the most recent six months.

Upon death of a member after retirement, the death benefit consists of the excess, if any, of the total contributions over the total pension payments paid to the member or his/her beneficiary. Furthermore, the death benefit is the lesser of \$10,000 and the most recent salary earned for a six-month period less 20% of the death benefit for each year that the member has been on pension, to a minimum of \$5,000.

Non-Duty Disability Benefits

Qualification and Amount of Payment

A disability retirement pension is payable in the event of total or permanent disability with 10 or more years of service, irrespective of age. The benefit is the unreduced service retirement pension. However, if the participant has 20 or more and less than 25 years of service and is under age 55, the benefit is reduced by $\frac{1}{2}$ of 1.0 percent for each month that the age of the member is below age 55 down to a minimum age of 50. If total service is 20 years or more and the member has attained age 55, or after 25 years of service, regardless of age, the retirement pension is payable without reduction.



Summary of Plan Provisions (as of June 30, 2020)

Annual Increases in Annuity

Postretirement increases of 3.0 percent of the current pension (i.e., increases are compounded) are granted to members effective each January 1, occurring on or after the first anniversary of the pension or the 61st birthday, whichever is later.

Duty Disability Benefits

Qualification and Amount of Payment

A disability retirement pension is payable in the event of total or permanent disability from an injury that occurred while working. The disability benefit provided is 75 percent of final average salary until attainment of age 65. At age 65, the disabled retiree shall receive a service retirement pension, which includes service earned while disabled.

Annual Increases in Annuity

Postretirement increases of 3.0 percent of the current pension (i.e., increases are compounded) are granted to members effective each January 1, occurring on or after the first anniversary of the pension or the 61st birthday, whichever is later.

Refunds

Upon termination of employment, a member may obtain a refund of his/her total contributions and those contributions made on his/her behalf, without interest.

A member who is unmarried at the date of retirement is entitled to a refund of the full amount contributed for the survivor's pension, without interest.

Reversionary Pension

A member can provide a reversionary pension for a surviving beneficiary by having his/her current pension reduced. If the beneficiary survives the date of the member's retirement, but does not survive the retired member, the member's pension shall be restored to the full amount of pension in place prior to choosing the reversionary pension.

Health Insurance Subsidy

The board may pay each recipient of a retirement, disability or survivor's pension an amount to be determined by the board, which shall represent partial reimbursement for the cost of the recipient's health insurance coverage, with the total amount of payment not to exceed \$65,000,000, or 75% of the total cost of health insurance coverage in any year.



Summary of Plan Provisions (as of June 30, 2020)

Retirement Systems Reciprocal Act

The Fund complies with the Retirement Systems Reciprocal Act (Chapter 40 Act 5 Article 20 of the Illinois Compiled Statutes) to provide reciprocal benefits if a member has service credit for other public employment in Illinois.

Provisions Applicable to Members Hired on or after January 1, 2011, as a result of Public Act 96-0889

Final Average Compensation

Based on last eight years of service and may not exceed \$106,800, as automatically increased by the lesser of 3 percent or one-half of the annual increase in the consumer price index-u during the preceding 12-month calendar year.

Retirement Eligibility

Normal retirement – 67 years old with 10 years of service.

Early Retirement – 62 years old with 10 years of service with a 6.0 percent per year reduction in benefit for each year age is under 67.

Annual Increases in Annuity

Annual increases begin at the later of the first anniversary of retirement or age 67. The annual increases are equal to the lesser of 3.0 percent or one-half of the annual increase in the consumer price index-u during the preceding 12-month calendar year and are not compounded.

Survivor Benefits

Benefit equal to 66.67 percent of the earned retirement benefit at death. Survivor benefits are increased by the lesser of 3.0 percent or one-half of the annual increase in the consumer price index-u during the preceding 12-month calendar year and are not compounded.



Summary of Plan Provisions (as of June 30, 2020)

Salary and COLA Development for Members Hired on or After January 1, 2011

Year Ending	CPI-U	1/2 CPI-U	COLA	Maximum Annual Pensionable Earnings
2011			3.00%	\$106,800.00
2012	3.90%	1.95%	1.95%	\$108,882.60
2013	2.00%	1.00%	1.00%	\$109,971.43
2014	1.20%	0.60%	0.60%	\$110,631.26
2015	1.70%	0.85%	0.85%	\$111,571.63
2016	0.00%	0.00%	0.00%	\$111,571.63
2017	1.50%	0.75%	0.75%	\$112,408.42
2018	2.20%	1.10%	1.10%	\$113,644.91
2019	2.30%	1.15%	1.15%	\$114,951.83
2020	1.70%	0.85%	0.85%	\$115,928.92

SECTION I

GLOSSARY OF TERMS

Glossary of Terms

<i>Actuarial Accrued Liability ("AAL")</i>	The difference between the Actuarial Present Value of Future Benefits, and the Actuarial Present Value of Future Normal Costs.
<i>Actuarial Assumptions</i>	Assumptions about future plan experience that affect costs or liabilities, such as: mortality, withdrawal, disablement and retirement; future increases in salary; future rates of investment earnings; future investment and administrative expenses; characteristics of members not specified in the data, such as marital status; characteristics of future members; future elections made by members; and other items.
<i>Actuarial Cost Method</i>	A procedure for allocating the Actuarial Present Value of Future Benefits between the Actuarial Present Value of future Normal Costs and the Actuarial Accrued Liability.
<i>Actuarial Equivalent</i>	Of equal Actuarial Present Value, determined as of a given date and based on a given set of Actuarial Assumptions.
<i>Actuarial Present Value ("APV")</i>	The amount of funds required to provide a payment or series of payments in the future. It is determined by discounting the future payments with an assumed interest rate and with the assumed probability each payment will be made.
<i>Actuarial Present Value of Future Benefits ("APVFB")</i>	The Actuarial Present Value of amounts which are expected to be paid at various future times to active members, retired members, beneficiaries receiving benefits and inactive, nonretired members entitled to either a refund or a future retirement benefit. Expressed another way, it is the value that would have to be invested on the valuation date so that the amount invested plus investment earnings would provide sufficient assets to pay all projected benefits and expenses when due.
<i>Actuarial Valuation</i>	The determination, as of a valuation date, of the Normal Cost, Actuarial Accrued Liability, Actuarial Value of Assets and related Actuarial Present Values for a plan. An Actuarial Valuation for a governmental retirement system typically also includes calculations of items needed for compliance with GASB No. 67, such as the Funded Ratio and the Actuarially Determined Contribution ("ADC").
<i>Actuarial Value of Assets ("AVA")</i>	The value of the assets as of a given date, used by the actuary for valuation purposes. This may be the market or fair value of plan assets or a smoothed value in order to reduce the year-to-year volatility of calculated results, such as the funded ratio or contribution requirement.

Glossary of Terms

<i>Actuarially Determined Contribution ("ADC")</i>	The employer's periodic required contributions, expressed as a dollar amount or a percentage of covered plan compensation. The ADC consists of the Employer Normal Cost and Amortization Payment.
<i>Amortization Method</i>	A method for determining the Amortization Payment. The most common methods used are level dollar and level percentage of payroll. Under the Level Dollar method, the Amortization Payment is one of a stream of payments, all equal, whose Actuarial Present Value is equal to the UAAL. Under the Level Percentage of Pay method, the Amortization payment is one of a stream of increasing payments, whose Actuarial Present Value is equal to the UAAL. Under the Level Percentage of Pay method, the stream of payments increases at the rate at which total covered payroll of all active members is assumed to increase.
<i>Amortization Payment</i>	That portion of the plan contribution or ADC which is designed to pay interest on and to amortize the Unfunded Actuarial Accrued Liability.
<i>Amortization Period</i>	The period used in calculating the Amortization Payment.
<i>Closed Amortization Period</i>	A specific number of years that is reduced by one each year, and declines to zero with the passage of time. For example if the amortization period is initially set at 30 years, it is 29 years at the end of one year, 28 years at the end of two years, etc.
<i>Employer Normal Cost</i>	The portion of the Normal Cost to be paid by the employer. This is equal to the Normal Cost less expected member contributions.
<i>Equivalent Single Amortization Period</i>	For plans that do not establish separate amortization bases (separate components of the UAAL), this is the same as the Amortization Period. For plans that do establish separate amortization bases, this is the period over which the UAAL would be amortized if all amortization bases were combined upon the current UAAL payment.
<i>Experience Gain/Loss</i>	A measure of the difference between actual experience and that expected based upon a set of Actuarial Assumptions, during the period between two actuarial valuations. To the extent that actual experience differs from that assumed, Unfunded Actuarial Accrued Liabilities emerge which may be larger or smaller than projected. Gains are due to favorable experience; e.g., the assets earn more than projected, salaries do not increase as fast as assumed, members retire later than assumed, etc. Favorable experience means actual results produce actuarial liabilities not as large as projected by the actuarial assumptions. On the other hand, losses are the result of unfavorable experience; i.e., actual results that produce Unfunded Actuarial Accrued Liabilities which are larger than projected.

Glossary of Terms

<i>Funded Ratio</i>	The ratio of the Actuarial Value of Assets to the Actuarial Accrued Liability.
<i>GASB</i>	Governmental Accounting Standards Board.
<i>GASB Statement No. 67 and GASB Statement No. 68</i>	These are the governmental accounting standards that set the accounting rules for public retirement systems and the employers that sponsor or contribute to them. Statement No. 68, which replaced Statement No. 27 effective with the fiscal year ending June 30, 2015, sets the accounting rules for the employers that sponsor or contribute to public retirement systems. Statement No. 67, which replaced Statement No. 25 effective with fiscal year ending June 30, 2014, sets the rules for the systems themselves.
<i>Normal Cost</i>	The annual cost assigned, under the Actuarial Cost Method, to the current plan year.
<i>Open Amortization Period</i>	An open amortization period is one which is used to determine the Amortization Payment but which does not change over time. In other words, if the initial period is set as 30 years, the same 30-year period is used in determining the Amortization Period each year. In theory, if an Open Amortization Period is used to amortize the Unfunded Actuarial Accrued Liability, the UAAL will never completely disappear, but will become smaller each year, either as a dollar amount or in relation to covered payroll.
<i>Unfunded Actuarial Accrued Liability</i>	The difference between the Actuarial Accrued Liability and Actuarial Value of Assets.
<i>Valuation Date</i>	The date as of which the Actuarial Present Value of Future Benefits are determined. The benefits expected to be paid in the future are discounted to this date.

SECTION J

ADDITIONAL PROJECTION DETAILS

Table 21

Additional Projection Details — Actuarial Accrued Liability (\$ in Millions)

Fiscal Year Ending	Current Inactives		Actives			Grand Totals		
	Retirees & Beneficiaries	Deferreds	Tier 1	Current Tier 2	Future Tier 2	Current Retirees, Beneficiaries & Deferreds	Actives	Total
2020	\$17,300.91	\$565.29	\$5,989.58	\$217.71	\$0.00	\$17,866.19	\$6,207.29	\$24,073.48
2021	16,914.95	587.55	6,713.01	263.50	0.00	17,502.50	6,976.51	24,479.01
2022	16,497.02	610.66	7,463.54	311.79	8.01	17,107.68	7,783.34	24,891.03
2023	16,048.21	634.57	8,240.34	364.10	22.92	16,682.78	8,627.36	25,310.14
2024	15,569.71	659.10	9,040.87	422.03	44.20	16,228.81	9,507.09	25,735.90
2025	15,062.93	684.12	9,861.42	486.27	71.37	15,747.06	10,419.06	26,166.12
2026	14,529.54	709.54	10,698.66	557.86	105.09	15,239.07	11,361.62	26,600.69
2027	13,971.37	735.32	11,549.25	637.84	145.33	14,706.69	12,332.42	27,039.11
2028	13,390.52	761.31	12,408.81	726.74	192.55	14,151.84	13,328.11	27,479.94
2029	12,789.37	787.30	13,272.50	824.94	247.96	13,576.67	14,345.40	27,922.07
2030	12,170.52	813.06	14,135.02	931.01	312.65	12,983.59	15,378.68	28,362.27
2031	11,536.88	851.60	14,991.02	1,044.82	389.63	12,388.48	16,425.47	28,813.95
2032	10,891.63	890.41	15,834.92	1,166.84	479.95	11,782.04	17,481.71	29,263.75
2033	10,238.22	928.65	16,660.45	1,297.52	584.77	11,166.87	18,542.74	29,709.61
2034	9,580.34	966.41	17,460.92	1,437.35	705.30	10,546.75	19,603.56	30,150.31
2035	8,921.91	1,003.40	18,229.26	1,586.78	842.86	9,925.32	20,658.90	30,584.22
2036	8,267.01	1,039.37	18,958.51	1,746.29	998.82	9,306.38	21,703.62	31,010.01
2037	7,619.84	1,073.91	19,641.54	1,916.22	1,174.63	8,693.74	22,732.39	31,426.13
2038	6,984.62	1,106.92	20,270.89	2,096.92	1,371.90	8,091.54	23,739.71	31,831.24
2039	6,365.53	1,137.55	20,839.29	2,288.54	1,592.28	7,503.08	24,720.11	32,223.20
2040	5,766.61	1,165.16	21,339.49	2,491.26	1,837.53	6,931.78	25,668.28	32,600.06
2041	5,191.65	1,189.42	21,764.03	2,705.20	2,109.42	6,381.06	26,578.66	32,959.73
2042	4,644.05	1,210.04	22,106.02	2,930.03	2,409.88	5,854.09	27,445.93	33,300.02
2043	4,126.80	1,226.72	22,361.43	3,165.29	2,740.82	5,353.52	28,267.54	33,621.06
2044	3,642.33	1,238.88	22,528.13	3,410.05	3,104.21	4,881.21	29,042.39	33,923.60
2045	3,192.51	1,246.40	22,605.67	3,663.23	3,502.04	4,438.90	29,770.94	34,209.84
2046	2,778.57	1,249.54	22,596.28	3,922.85	3,936.26	4,028.12	30,455.39	34,483.51
2047	2,401.14	1,247.43	22,503.52	4,186.21	4,408.95	3,648.57	31,098.68	34,747.24
2048	2,060.20	1,240.85	22,332.26	4,450.86	4,922.08	3,301.05	31,705.20	35,006.25
2049	1,755.13	1,230.11	22,088.06	4,713.30	5,477.64	2,985.24	32,279.00	35,264.24
2050	1,484.78	1,215.66	21,776.89	4,967.81	6,077.59	2,700.44	32,822.29	35,522.73
2051	1,247.49	1,198.28	21,404.91	5,208.01	6,723.90	2,445.76	33,336.81	35,782.57
2052	1,041.21	1,178.46	20,977.14	5,428.00	7,418.26	2,219.67	33,823.40	36,043.08
2053	863.61	1,156.18	20,497.50	5,623.96	8,162.18	2,019.79	34,283.64	36,303.42
2054	712.13	1,131.46	19,969.36	5,792.48	8,956.79	1,843.60	34,718.64	36,562.23
2055	584.14	1,104.35	19,396.01	5,928.72	9,802.80	1,688.50	35,127.53	36,816.02
2056	476.98	1,074.89	18,780.73	6,028.73	10,700.40	1,551.87	35,509.86	37,061.74
2057	388.03	1,043.15	18,126.88	6,091.06	11,649.24	1,431.19	35,867.18	37,298.37
2058	314.83	1,009.22	17,437.89	6,116.70	12,648.33	1,324.05	36,202.91	37,526.96
2059	255.05	973.21	16,717.35	6,108.40	13,695.95	1,228.26	36,521.70	37,749.96

Table 22

Additional Projection Details — Present Value of Future Benefits (\$ in Millions)

Fiscal Year Ending	Current Inactives		Actives			Grand Totals		
	Retirees & Beneficiaries	Deferreds	Tier 1	Current Tier 2	Future Tier 2	Current Retirees, Beneficiaries & Deferreds	Actives	Total
2020	\$17,300.91	\$565.29	\$9,621.91	\$1,088.09	\$0.00	\$17,866.19	\$10,710.00	\$28,576.20
2021	16,914.95	587.55	10,236.70	1,133.46	100.57	17,502.50	11,470.73	28,973.22
2022	16,497.02	610.66	10,870.53	1,183.74	215.05	17,107.68	12,269.32	29,377.00
2023	16,048.21	634.57	11,522.35	1,239.28	339.11	16,682.78	13,100.74	29,783.52
2024	15,569.71	659.10	12,190.05	1,300.93	471.42	16,228.81	13,962.40	30,191.21
2025	15,062.93	684.12	12,870.47	1,368.81	613.75	15,747.06	14,853.03	30,600.09
2026	14,529.54	709.54	13,560.83	1,443.47	765.38	15,239.07	15,769.67	31,008.75
2027	13,971.37	735.32	14,258.32	1,525.56	927.13	14,706.69	16,711.00	31,417.69
2028	13,390.52	761.31	14,959.45	1,615.27	1,100.89	14,151.84	17,675.61	31,827.45
2029	12,789.37	787.30	15,660.44	1,712.70	1,287.69	13,576.67	18,660.84	32,237.51
2030	12,170.52	813.06	16,357.00	1,816.21	1,488.44	12,983.59	19,661.65	32,645.24
2031	11,536.88	851.60	17,044.74	1,925.40	1,705.60	12,388.48	20,675.75	33,064.23
2032	10,891.63	890.41	17,719.05	2,040.48	1,941.14	11,782.04	21,700.67	33,482.71
2033	10,238.22	928.65	18,374.79	2,161.73	2,196.15	11,166.87	22,732.67	33,899.54
2034	9,580.34	966.41	19,006.60	2,289.42	2,471.70	10,546.75	23,767.72	34,314.47
2035	8,921.91	1,003.40	19,608.85	2,423.80	2,769.01	9,925.32	24,801.66	34,726.98
2036	8,267.01	1,039.37	20,176.67	2,565.52	3,089.24	9,306.38	25,831.43	35,137.81
2037	7,619.84	1,073.91	20,704.46	2,714.78	3,434.08	8,693.74	26,853.32	35,547.07
2038	6,984.62	1,106.92	21,186.29	2,871.85	3,804.67	8,091.54	27,862.81	35,954.34
2039	6,365.53	1,137.55	21,616.29	3,036.81	4,202.41	7,503.08	28,855.51	36,358.59
2040	5,766.61	1,165.16	21,988.49	3,209.82	4,628.35	6,931.78	29,826.66	36,758.44
2041	5,191.65	1,189.42	22,296.97	3,390.91	5,084.04	6,381.06	30,771.92	37,152.99
2042	4,644.05	1,210.04	22,536.16	3,579.83	5,570.39	5,854.09	31,686.39	37,540.48
2043	4,126.80	1,226.72	22,702.53	3,776.21	6,088.26	5,353.52	32,567.00	37,920.51
2044	3,642.33	1,238.88	22,794.00	3,979.27	6,638.84	4,881.21	33,412.11	38,293.32
2045	3,192.51	1,246.40	22,809.63	4,188.21	7,222.66	4,438.90	34,220.50	38,659.40
2046	2,778.57	1,249.54	22,750.33	4,401.48	7,841.93	4,028.12	34,993.73	39,021.85
2047	2,401.14	1,247.43	22,618.17	4,616.97	8,497.20	3,648.57	35,732.35	39,380.91
2048	2,060.20	1,240.85	22,416.33	4,832.78	9,190.08	3,301.05	36,439.19	39,740.23
2049	1,755.13	1,230.11	22,148.80	5,046.07	9,922.58	2,985.24	37,117.44	40,102.68
2050	1,484.78	1,215.66	21,820.06	5,252.35	10,697.76	2,700.44	37,770.16	40,470.60
2051	1,247.49	1,198.28	21,435.03	5,446.71	11,515.89	2,445.76	38,397.63	40,843.39
2052	1,041.21	1,178.46	20,997.76	5,624.39	12,378.47	2,219.67	39,000.61	41,220.28
2053	863.61	1,156.18	20,511.31	5,782.39	13,285.57	2,019.79	39,579.27	41,599.05
2054	712.13	1,131.46	19,978.40	5,917.70	14,237.71	1,843.60	40,133.80	41,977.40
2055	584.14	1,104.35	19,401.76	6,025.64	15,235.30	1,688.50	40,662.71	42,351.20
2056	476.98	1,074.89	18,784.28	6,102.25	16,278.31	1,551.87	41,164.83	42,716.70
2057	388.03	1,043.15	18,128.97	6,145.80	17,366.33	1,431.19	41,641.11	43,072.30
2058	314.83	1,009.22	17,439.05	6,156.89	18,498.53	1,324.05	42,094.47	43,418.52
2059	255.05	973.21	16,717.95	6,137.70	19,673.54	1,228.26	42,529.19	43,757.45

Table 23

Additional Projection Details — Benefit Payments Including Administrative Expenses and Health Insurance Subsidy (\$ in Millions)

Fiscal Year Ending	Current Inactives		Actives			Grand Totals		
	Retirees & Beneficiaries	Deferreds	Tier 1	Current Tier 2	Future Tier 2	Current Retirees, Beneficiaries & Deferreds	Actives	Total
2021	\$1,503.84	\$15.38	\$79.15	\$46.60	\$0.00	\$1,519.23	\$125.75	\$1,644.98
2022	1,509.57	16.01	98.78	43.25	6.96	1,525.58	148.99	1,674.57
2023	1,512.16	16.76	120.91	40.36	13.51	1,528.92	174.78	1,703.70
2024	1,511.57	17.71	146.34	37.26	20.23	1,529.29	203.83	1,733.11
2025	1,507.68	18.84	175.88	34.66	27.24	1,526.52	237.79	1,764.31
2026	1,500.33	20.10	208.98	32.03	34.01	1,520.43	275.02	1,795.45
2027	1,489.46	21.40	245.45	29.33	41.17	1,510.86	315.95	1,826.82
2028	1,474.95	22.88	285.76	27.02	48.45	1,497.83	361.23	1,859.06
2029	1,456.65	24.59	329.97	25.14	55.48	1,481.24	410.59	1,891.83
2030	1,434.51	26.50	378.31	25.39	62.55	1,461.01	466.25	1,927.25
2031	1,408.39	15.82	430.57	26.44	67.86	1,424.22	524.87	1,949.09
2032	1,378.23	18.07	486.76	27.67	73.36	1,396.30	587.79	1,984.09
2033	1,343.98	21.16	547.04	29.01	79.25	1,365.14	655.30	2,020.44
2034	1,305.61	24.12	611.30	30.48	85.62	1,329.74	727.40	2,057.14
2035	1,263.17	27.33	679.44	32.10	92.46	1,290.50	803.99	2,094.49
2036	1,216.73	30.74	750.38	33.52	98.80	1,247.48	882.69	2,130.17
2037	1,166.47	34.48	824.51	35.20	105.47	1,200.95	965.18	2,166.12
2038	1,112.62	38.21	901.78	37.09	112.51	1,150.83	1,051.38	2,202.21
2039	1,055.51	42.67	981.77	39.37	119.92	1,098.17	1,141.06	2,239.24
2040	995.54	47.59	1,064.19	42.00	127.74	1,043.13	1,233.93	2,277.07
2041	933.23	52.65	1,148.62	45.10	135.96	985.88	1,329.69	2,315.56
2042	869.17	57.74	1,234.32	48.97	144.62	926.92	1,427.91	2,354.83
2043	804.03	62.92	1,319.00	53.67	153.72	866.95	1,526.39	2,393.34
2044	738.51	68.37	1,401.03	59.61	163.23	806.88	1,623.87	2,430.75
2045	673.33	73.66	1,479.22	66.73	173.23	746.99	1,719.18	2,466.17
2046	609.20	78.38	1,551.70	75.72	183.71	687.58	1,811.13	2,498.72
2047	546.83	83.69	1,617.47	86.97	194.79	630.52	1,899.23	2,529.74
2048	486.86	87.86	1,675.53	100.20	206.54	574.72	1,982.28	2,556.99
2049	429.86	91.46	1,725.33	116.18	219.08	521.32	2,060.59	2,581.91
2050	376.33	94.35	1,766.62	136.30	232.63	470.68	2,135.55	2,606.23
2051	326.67	96.25	1,799.24	160.64	247.52	422.92	2,207.40	2,630.32
2052	281.15	97.46	1,824.38	188.78	263.85	378.61	2,277.01	2,655.62
2053	239.92	98.56	1,843.18	218.72	281.93	338.48	2,343.84	2,682.32
2054	203.03	99.46	1,856.21	250.33	301.97	302.48	2,408.51	2,711.00
2055	170.40	100.16	1,863.59	285.03	324.35	270.56	2,472.97	2,743.53
2056	141.89	100.66	1,865.37	321.85	349.47	242.55	2,536.68	2,779.22
2057	117.25	100.94	1,861.56	358.33	377.76	218.19	2,597.65	2,815.85
2058	96.20	100.99	1,852.22	392.17	409.70	197.20	2,654.08	2,851.28
2059	78.43	100.79	1,837.29	421.84	445.82	179.22	2,704.95	2,884.16

Table 24

Additional Projection Details — Active Population, Covered Payroll, Employee Contributions and Normal Costs (\$ in Millions)

Fiscal Year Ending	Tier 1 Active Members				Current Tier 2 Active Members				Future Tier 2 Active Members			
	Population	Covered Payroll	Employee Contributions	Normal Cost	Population	Covered Payroll	Employee Contributions	Normal Cost	Population	Covered Payroll	Employee Contributions	Normal Cost
2020	18,118	\$1,589.74	\$143.08	\$388.03	12,079	\$677.55	\$60.98	\$76.70	0	\$0.00	\$0.00	\$0.00
2021	16,986	1,554.88	139.94	386.63	10,625	639.01	57.51	72.77	2,586	131.24	11.81	14.71
2022	16,004	1,523.27	137.09	385.14	9,616	614.56	55.31	70.62	4,577	241.93	21.77	27.42
2023	15,087	1,490.89	134.18	382.79	8,836	598.13	53.83	69.54	6,274	345.20	31.07	39.32
2024	14,224	1,456.67	131.10	379.42	8,227	587.96	52.92	69.27	7,746	443.41	39.91	50.66
2025	13,390	1,419.72	127.77	375.07	7,723	581.38	52.32	69.55	9,084	540.38	48.63	61.98
2026	12,596	1,380.57	124.25	369.75	7,309	578.24	52.04	70.31	10,293	636.00	57.24	73.24
2027	11,829	1,338.59	120.47	363.18	6,963	577.54	51.98	71.39	11,405	731.57	65.84	84.66
2028	11,075	1,292.69	116.34	355.23	6,658	578.06	52.03	72.70	12,464	829.32	74.64	96.54
2029	10,340	1,243.93	111.95	346.01	6,379	579.33	52.14	74.16	13,478	929.85	83.69	108.97
2030	9,620	1,192.00	107.28	335.61	6,121	581.53	52.34	75.78	14,457	1,033.65	93.03	121.93
2031	8,920	1,137.91	102.41	324.17	5,900	584.01	52.56	77.50	15,377	1,139.38	102.54	135.32
2032	8,236	1,081.17	97.31	311.53	5,694	585.79	52.72	79.27	16,267	1,248.88	112.40	149.35
2033	7,566	1,021.88	91.97	297.60	5,498	586.72	52.81	81.05	17,133	1,362.59	122.63	164.07
2034	6,913	959.95	86.40	282.35	5,312	586.55	52.79	82.83	17,972	1,480.37	133.23	179.52
2035	6,276	895.76	80.62	265.26	5,136	585.13	52.66	84.23	18,785	1,602.12	144.19	194.68
2036	5,657	829.44	74.65	247.01	4,970	582.30	52.41	85.58	19,570	1,727.59	155.48	210.38
2037	5,058	761.51	68.54	227.71	4,805	577.61	51.98	86.79	20,335	1,857.10	167.14	226.69
2038	4,480	692.34	62.31	207.59	4,642	571.35	51.42	87.84	21,075	1,990.34	179.13	243.60
2039	3,928	622.84	56.06	186.87	4,477	563.26	50.69	88.70	21,791	2,127.12	191.44	261.08
2040	3,404	553.46	49.81	165.39	4,316	554.22	49.88	89.41	22,477	2,266.69	204.00	279.07
2041	2,904	483.26	43.49	143.44	4,151	543.74	48.94	89.84	23,142	2,409.15	216.82	297.61
2042	2,439	415.10	37.36	122.00	3,983	531.60	47.84	89.96	23,775	2,553.48	229.81	316.58
2043	2,016	350.65	31.56	101.48	3,813	518.26	46.64	89.71	24,367	2,698.82	242.89	335.88
2044	1,634	290.12	26.11	82.49	3,636	502.87	45.26	88.99	24,926	2,844.83	256.03	355.47
2045	1,304	236.29	21.27	65.76	3,456	486.04	43.74	87.68	25,437	2,990.09	269.11	375.18
2046	1,022	189.21	17.03	51.45	3,261	465.96	41.94	85.58	25,913	3,135.75	282.22	395.13
2047	791	149.46	13.45	39.59	3,056	443.48	39.91	82.86	26,350	3,280.22	295.22	415.15
2048	602	116.28	10.47	29.99	2,846	419.24	37.73	79.41	26,749	3,423.68	308.13	435.23
2049	452	89.35	8.04	22.40	2,619	391.45	35.23	74.70	27,127	3,566.73	321.01	455.45
2050	334	67.70	6.09	16.50	2,357	357.26	32.15	68.57	27,506	3,711.10	334.00	476.00
2051	243	50.55	4.55	11.95	2,081	319.46	28.75	61.47	27,873	3,854.39	346.90	496.62
2052	174	37.06	3.34	8.49	1,794	278.84	25.10	53.77	28,229	3,997.07	359.74	517.31
2053	122	26.62	2.40	5.92	1,518	238.80	21.49	46.02	28,558	4,137.35	372.36	537.81
2054	83	18.80	1.69	4.04	1,257	200.18	18.02	38.46	28,856	4,274.80	384.73	558.02
2055	56	12.90	1.16	2.70	1,018	164.03	14.76	31.32	29,123	4,408.96	396.81	577.80
2056	36	8.68	0.78	1.76	803	130.95	11.79	24.79	29,357	4,539.35	408.54	597.04
2057	23	5.65	0.51	1.11	617	101.74	9.16	19.04	29,557	4,665.53	419.90	615.63
2058	14	3.50	0.31	0.67	461	76.90	6.92	14.20	29,722	4,787.06	430.84	633.45
2059	8	2.07	0.19	0.38	336	56.73	5.11	10.42	29,853	4,903.54	441.32	650.40

Employee contributions and normal cost are for the following year.

Normal cost includes administrative expenses and the health insurance subsidy of \$65 million annually.

Covered payroll is capped for members hired after December 31, 2010, as defined by Public Act 96-0889.

Fiscal year ending June 30, 2020 includes 106 members expected to be hired to replace retirements and terminations that occurred in June 2020.



APPENDIX 1: STRESS TESTING SCENARIOS



May 20, 2020

Board of Trustees
Public School Teachers' Pension and Retirement Fund of Chicago
203 North LaSalle Street, Suite 2600
Chicago, Illinois 60601

Re: Stress Testing Scenarios Based on Actuarial Valuation Results as of June 30, 2019

Dear Members of the Board:

At your request, we have performed stress testing of the required employer contributions and funded ratio for the Public School Teachers' Pension and Retirement Fund of Chicago (commonly known as the Chicago Teachers' Pension Fund, "CTPF" or "Fund") based on the results of the June 30, 2019, actuarial valuation. This stress testing was performed to illustrate the potential impact on total required employer contributions and the CTPF funded ratio in the event there is a significant and/or sustained market downturn or significant volatility in investment returns or volatility in future active population.

GRS has prepared this analysis exclusively for the Trustees of the Chicago Teachers' Pension Fund; GRS is not responsible for reliance upon this report by any other party. This report may be provided to parties other than the CTPF only in its entirety and only with the permission of the CTPF and the Board.

Description of Investment Return Stress Testing Scenarios

GRS performed six alternative investment return stress testing scenarios. A summary of the results of these six scenarios can be found in Exhibit I(A) and the details of the results of these scenarios can be found in Exhibits II through VII. Due to the market volatility and uncertainty related to COVID-19, we asked CTPF staff for estimates of potential returns for fiscal years 2020 and 2021, in order to reflect the potential adverse impact on near-term contribution requirements and funded ratios. Based on input from CTPF staff, we used returns of 4.50 percent in 2020 and 7.00 percent in 2021 for all alternative investment return scenarios. The alternative investment return stress testing scenarios are based on the Fund earning (on average) three different rates of investment return through fiscal year 2059. The three different rates of average investment returns are: 1) earning the actuarial valuation assumption of 7.00 percent; 2) earning the 25th percentile return (4.12 percent) from the GRS 2019 Investment Return Assumption Review; and 3) earning the 40th percentile return (5.81 percent) from the GRS 2019 Investment Return Assumption Review. The percentile returns are the averages from the 14 investment

consultants with shorter term horizons from the GRS 2019 Investment Return Assumption Review dated September 10, 2019.

In order to demonstrate the risk and volatility of the returns, a static return scenario and a volatile return scenario were completed for each of the three rates of return. We provided results assuming volatile returns with a geometric average equal to the actuarial valuation assumption, the 25th percentile return (75 percent probability that the average return will be higher than the 25th percentile return) and the 40th percentile return (60 percent probability that the average return will be higher than the 40th percentile return). Please note that each volatile return scenario represents one possible trial that generates the targeted average geometric return, and that another equally likely trial that produces the same targeted average geometric return could produce significantly different contribution and funded ratio patterns. Following is a summary table illustrating the alternate investment return stress testing scenarios:

Scenario	Returns in FY 2020 and 2021	Returns in FY 2022 through 2059	Basis for Returns
Baseline	7.00%, 7.00%	Static return of 7.00%	June 30, 2019, actuarial valuation results
1	4.50%, 7.00%	Static return of 7.00%	Valuation assumption
2	4.50%, 7.00%	Volatile returns, 40-year average of 7.00% from 2020 through 2059	Valuation assumption
3	4.50%, 7.00%	Static return of 4.12%	25 th percentile return
4	4.50%, 7.00%	Volatile returns, 40-year average of 4.12% from 2020 through 2059	25 th percentile return
5	4.50%, 7.00%	Static return of 5.81%	40 th percentile return
6	4.50%, 7.00%	Volatile returns, 40-year average of 5.81% from 2020 through 2059	40 th percentile return

Description of Future Active Population Stress Testing Scenarios

In addition to the six investment return stress testing scenarios, GRS performed stress testing scenarios based on variations in the future active population. A summary of these scenarios can be found in Exhibit I(B) and the details of the results of these scenarios can be found in Exhibits VIII and IX. Based on information reviewed during the most recent experience study, there was an average decrease in the CTPF active member population between 2007 and 2016 of approximately 1.2 percent per year. There was an additional decrease between 2016 and 2017 and increases between 2017 to 2018 and 2018 to 2019. To demonstrate the risk and volatility associated with changes in the number of future active CTPF members, the table on the following page illustrates the future active population stress testing scenarios that we examined.

Scenario	Population Growth Assumption	Returns in FY 2020 and 2021	Returns in FY 2022 through 2059
7	1.0 percent reduction in the number of active members per year for the next ten years, then remains constant	4.50%, 7.00%	Static return of 7.00%
8	1.0 percent increase in the number of active members per year for the next ten years, then remains constant	4.50%, 7.00%	Static return of 7.00%

The future active population used in Scenarios 7 and 8 can be found in Exhibit I(B).

In each scenario, the discount rate used to determine liabilities remains at 7.00 percent, and average future uncapped salary growth or wage inflation remains at 3.00 percent per year. The future active population remains constant at 29,437 (29,295 active members as of June 30, 2019 plus 142 members assumed to be hired after June 30 to replace recent retirees) in the investment return stress testing scenarios.

The following graphs are included in the analysis:

- Exhibit X: Projected Annual Rates of Return used in the Stress Testing Scenarios
- Exhibit XI: Projected Total Required Employer Contributions under the Stress Testing Scenarios
- Exhibit XII: Projected Funded Ratio under the Stress Testing Scenarios

GRS believes that these scenarios provide a reasonable illustration of the potential future volatility of investment returns, population and the resulting employer contribution requirements based on the current actuarial assumptions and methods. These scenarios are not intended to represent the full range of all possible outcomes. Annual returns will likely be significantly different from the returns shown in Exhibit I(A) and the 40-year geometric average of actual returns may be either higher or lower than the assumption of 7.00 percent.

Analysis of Stress Testing Scenario Results

Under the projected results from the actuarial valuation as of June 30, 2019, in which all future actuarial assumptions are assumed to be realized, the statutory contribution dollar amount increases at the rate of projected payroll growth. There is an increase in the funded ratio from 47.4 percent as of June 30, 2019, to 90 percent as of June 30, 2059, with a significant portion of the increase occurring after 2049. The funded ratio increases from about 47.4 percent to 62.9 percent between 2019 and 2049, and then to 90.0 percent by 2059.



The following table presents a summary of the combined State and Board of Education contributions ("total required employer contributions"), and the present value of the total required employer contributions during the fiscal years 2020 through 2059 under the alternative stress testing scenarios. In addition, we have summarized the maximum and minimum percentage change in the annual total required employer contribution and the funded ratio for each scenario. The second table compares the results of the static and volatile return and the increasing and decreasing population scenarios.

	Contributions* FY 2020-2059 (\$ in Millions)		Annual Total Required Contribution \$ Change		Minimum Funded Ratio		Annual Funded Ratio (AVA) Change	
	Total	PV	Maximum	Maximum	(AVA)	(MVA)	Maximum	Maximum
	Contributions	Contributions ¹	Increase	Decrease			Increase	Decrease
Baseline (7.00% Static)	\$54,694.662	\$15,789.915	5.68%	1.97%	47.25%	47.24%	3.76%	-0.26%
Scenario 1 (7.00% Static, 4.50% FY 2020)	55,646.009	16,044.368	5.68%	1.97%	46.15%	46.08%	3.85%	-0.51%
Scenario 2 (7.00% Volatile)	60,837.876	17,850.729	249.55%	-67.94%	28.32%	25.62%	11.26%	-3.70%
Scenario 3 (4.12% Static)	80,394.454	19,691.504	15.13%	2.57%	35.50%	34.00%	8.89%	-1.13%
Scenario 4 (4.12% Volatile)	78,310.347	19,586.218	41.87%	-50.60%	26.16%	22.36%	9.25%	-3.75%
Scenario 5 (5.81% Static)	67,740.109	17,784.763	10.69%	2.57%	42.93%	42.15%	6.40%	-0.64%
Scenario 6 (5.81% Volatile)	66,932.786	16,240.381	71.06%	-17.81%	45.11%	42.13%	12.52%	-3.49%
Scenario 7 (1% Population Decrease)	54,821.695	16,065.743	11.59%	1.36%	46.55%	46.08%	3.92%	-0.46%
Scenario 8 (1% Population Increase)	56,427.558	16,020.181	5.68%	-2.30%	45.42%	45.42%	3.76%	-0.72%

	Total Contributions* FY 2020-2059 (\$ in Millions)			PV Contributions* FY 2020-2059 (\$ in Millions)		
	Static/ Decrease	Volatile/ Increase	Difference	Static/ Decrease	Volatile/ Increase	Difference
Baseline/Scenario 1** (7.00%)	\$54,694.662	\$55,646.009	\$951.347	\$15,789.915	\$16,044.368	\$254.453
Scenario 1/Scenario 2 (7.00%)	55,646.009	60,837.876	5,191.867	16,044.368	17,850.729	1,806.361
Scenario 3/Scenario 4 (4.12%)	80,394.454	78,310.347	-2,084.107	19,691.504	19,586.218	-105.286
Scenario 5/Scenario 6 (5.81%)	67,740.109	66,932.786	-807.323	17,784.763	16,240.381	-1,544.382
Scenario 7/Scenario 8 (Decrease/Increase)	54,821.695	56,427.558	1,605.863	16,065.743	16,020.181	-45.562

* Combined State and Board of Education contributions.

** Baseline assumes a return of 7.00% in FY 2020 and Scenario 1 assumes a return of 4.50% in FY 2020.

¹ In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.

Based on the scenarios that are summarized above, the total required employer contributions and the present value of the total required employer contributions are dependent on both the annualized investment returns and the pattern of the investment returns. In the scenarios in which the average return is 7.00 percent (Baseline, Scenario 1, and Scenario 2), the total required employer contributions are approximately \$6.1 billion higher under Scenario 2 compared to the Baseline, with the present value of contributions about \$2.1 billion higher under Scenario 2 based on lower average returns in the first 10 years of the projection period. Due to the difference in the fiscal year 2020 return (7.00 percent in the Baseline and 4.50 percent in Scenario 1), the total required employer contributions are approximately \$950 million higher under Scenario 1 compared to the Baseline, with the present value of contributions about \$250 million higher under Scenario 1. Under Scenarios 3 and 4, in which the average annualized return is 4.12 percent, the volatile returns result in lower total nominal contributions and lower contributions on a present value basis, when compared to the static return results. Similarly, under Scenarios 5 and 6, in which the average annualized return is 5.81 percent, the volatile returns result in lower total nominal contributions and lower contributions on a present value basis, when compared to the static return results.



For the period 2020 through 2059, an average annualized return approximately equal to 5.81 percent would result in higher required employer contributions of about \$500 million to \$2 billion (on a present value basis), when compared to an average annualized return of 7.00 percent. For the same period, an average annualized return approximately equal to 4.12 percent would result in higher required employer contributions of about \$3 billion to \$4 billion (on a present value basis), when compared to an average annualized return of 7.00 percent.

Because of the volatility and deviation of the returns from the actuarial assumption of 7.00 percent and the fact that the required employer contribution for fiscal year 2059 is determined based on the actuarial valuation results as of June 30, 2057, the funded ratios on both an actuarial value of assets basis and a market value of assets basis do not equal 90 percent in 2059. In Scenarios 3 through 5 the funded ratio is less than 90 percent in 2059 and in Scenarios 2 and 6 the funded ratios are greater than 90 percent.

Under Scenarios 7 and 8, in which the future active population decreases/increases 1.00 percent per year for 10 years then remains constant through 2059, the decreasing population scenario results in slightly higher total nominal contributions and higher contributions on a present value basis, when compared to the Scenario 1 results (with the same returns). Incorporating an assumption of a decreasing active population is projected to result in an increase in the required employer contribution rate of over three percent of pay. In addition, the required employer contribution (as a dollar amount) is projected to be higher than the baseline through the fiscal year 2038 contribution. A decreasing active population (that is not reflected in the actuarial assumptions) is expected to result in losses due to the Fund receiving lower contributions than expected (based on lower payroll) and a continuously increasing employer rate as a percentage of payroll if the actual payroll is lower than had been projected. The increasing population scenario results in higher total nominal contributions and higher contributions on a present value basis, when compared to the baseline results.

In each scenario, we have assumed that the State and the Board of Education will make the required contribution when due. However, some scenarios result in very high contributions for extended periods of time which may be difficult to afford. We are not able to opine on the sponsor's ability to pay such high contribution requirements.

Due to the volatility of the investment returns, the minimum market value of assets funded ratio during the projection period is less than 30 percent in Scenarios 2 and 4, less than 40 percent in Scenario 3, and less than 50 percent in Scenarios 1, 5, and 6. The stress testing does not take into account the implication of potential required changes in asset allocation in order to maintain sufficient liquidity to make all required benefit payments.

To the best of our knowledge, this actuarial statement is complete and accurate, fairly presents the actuarial position of CTPF as of June 30, 2019, under the alternative stress testing scenarios and has been prepared in accordance with generally accepted actuarial principles and practices, with the Actuarial Standards of Practice issued by the Actuarial Standards Board and with applicable statutes.



Future actuarial measurements may differ significantly from the current measurements presented in this report due to such factors as the following: plan experience differing from that anticipated by the economic or demographic assumptions; changes in economic or demographic assumptions; increases or decreases expected as part of the natural operation of the methodology used for these measurements (such as the end of an amortization period or additional cost or contribution requirements based on the plan's funded status); and changes in plan provisions, contribution amounts or applicable law. Due to the limited scope of the actuary's assignment, the actuary did not perform an analysis of the potential range of such future measurements in this report.

This letter is part of the CTPF actuarial valuation as of June 30, 2019, and is subject to the same actuarial assumptions and disclosures as used in the presentation and the actuarial valuation report. The stress testing scenarios used future investment returns and active populations as shown in Exhibits I(A) and I(B). All other assumptions and methods were the same as those used in the CTPF actuarial valuation as of June 30, 2019, including the 7.00 percent discount rate used to determine liabilities in all stress testing scenarios.

In each projection scenario, the total required employer contribution in each year has been projected as though an actuarial valuation in each of those years was performed. The market value of assets at each projected valuation is assumed to be based on the rates of investment return and total required employer contributions in the preceding years, according to the Scenario being modeled, and the valuation interest rate (7.00 percent) going forward. At each projected actuarial valuation, an additional 20 percent of the investment gains and losses are recognized in the actuarial value of assets. This iterative process is followed for each projection year through 2059.

Numerous additional assumptions could be varied in stress testing scenarios, such as payroll growth, inflation and other member behavior. The stress testing analysis performed focused on variations in investment return, which is likely one of the most significant stressors on the Fund and population growth. GRS can perform additional stress testing scenarios at the Board's request.

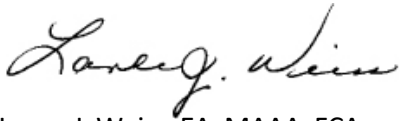
The statutory funding method generates a contribution requirement that is less than a reasonable actuarially determined contribution. Meeting the statutory requirement does not mean that the undersigned agree that adequate actuarial funding has been achieved; we recommend the adherence to a funding policy, such as the Board policy used to calculate the actuarially determined contribution under GASB Statement Nos. 67 and 68, which funds the normal cost of the plan as well as an amortization payment that seeks to pay off any unfunded accrued liability over a closed period of 30 years, beginning July 1, 2013.



Board of Trustees
Public School Teachers' Pension and Retirement Fund of Chicago
May 20, 2020
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The signing actuaries are independent of the plan sponsor.

Lance Weiss and Amy Williams are Members of the American Academy of Actuaries ("MAAA") and meet the Qualification Standards of the American Academy of Actuaries to render the actuarial opinion herein. Sincerely,



Lance J. Weiss, EA, MAAA, FCA
Senior Consultant and Team Leader



Amy Williams, ASA, MAAA, FCA
Senior Consultant

LJW/AW:ah
Enclosure

cc: Kristen Brundirks, GRS



³In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.

Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Assumed Investment Returns and Active Population Used in Stress testing Scenarios
and Total Required Employer Contributions
Based on Actuarial Valuation as of June 30, 2019

Year	Baseline - Population remains constant at 2019 level	Scenario 7 - Population decreases 1% per year for 10 years, then constant	Scenario 8 - Population increases 1% per year for 10 years, then constant	Total Required Employer Contributions (\$ in millions)			Total Required Employer Contributions (% of Payroll)			Funded Ratio (AVA)		
	Active Population	Active Population	Active Population	Baseline	Scenario 7	Scenario 8	Baseline	Scenario 7	Scenario 8	Baseline	Scenario 7	Scenario 8
2020	29,437	29,143	29,731	\$854.500	\$854.500	\$854.500	38.787%	38.787%	38.787%	47.65%	47.37%	47.37%
2021	29,437	28,851	30,029	885.893	880.097	891.690	39.076%	39.076%	39.076%	47.52%	46.91%	46.96%
2022	29,437	28,563	30,329	909.101	982.098	871.187	39.076%	42.781%	36.952%	47.26%	46.62%	46.24%
2023	29,437	28,277	30,632	932.439	1,000.335	899.817	39.076%	42.781%	36.952%	47.25%	46.55%	45.76%
2024	29,437	27,994	30,939	955.936	1,018.224	929.169	39.076%	42.781%	36.952%	47.27%	46.74%	45.58%
2025	29,437	27,714	31,248	979.383	1,035.554	959.089	39.076%	42.781%	36.952%	47.32%	46.95%	45.47%
2026	29,437	27,437	31,560	1,002.665	1,052.177	989.456	39.076%	42.781%	36.952%	47.41%	47.18%	45.42%
2027	29,437	27,163	31,876	1,025.946	1,068.263	1,020.474	39.076%	42.781%	36.952%	47.53%	47.43%	45.44%
2028	29,437	26,891	32,195	1,049.294	1,083.872	1,052.259	39.076%	42.781%	36.952%	47.68%	47.68%	45.52%
2029	29,437	26,622	32,517	1,072.631	1,098.866	1,084.756	39.076%	42.781%	36.952%	47.86%	47.92%	45.65%
2030	29,437	26,622	32,517	1,096.466	1,113.807	1,118.484	39.076%	42.781%	36.952%	48.09%	48.19%	45.88%
2031	29,437	26,622	32,517	1,120.278	1,135.812	1,144.851	39.076%	42.781%	36.952%	48.35%	48.48%	46.15%
2032	29,437	26,622	32,517	1,144.501	1,158.186	1,171.679	39.076%	42.781%	36.952%	48.65%	48.81%	46.46%
2033	29,437	26,622	32,517	1,168.991	1,180.758	1,198.847	39.076%	42.781%	36.952%	48.98%	49.16%	46.81%
2034	29,437	26,622	32,517	1,193.698	1,203.462	1,226.318	39.076%	42.781%	36.952%	49.35%	49.54%	47.20%
2035	29,437	26,622	32,517	1,218.420	1,226.065	1,253.913	39.076%	42.781%	36.952%	49.76%	49.96%	47.65%
2036	29,437	26,622	32,517	1,243.297	1,248.729	1,281.757	39.076%	42.781%	36.952%	50.23%	50.42%	48.16%
2037	29,437	26,622	32,517	1,268.158	1,271.300	1,309.662	39.076%	42.781%	36.952%	50.74%	50.92%	48.72%
2038	29,437	26,622	32,517	1,293.261	1,294.056	1,337.874	39.076%	42.781%	36.952%	51.30%	51.47%	49.35%
2039	29,437	26,622	32,517	1,318.690	1,317.107	1,366.458	39.076%	42.781%	36.952%	51.93%	52.07%	50.04%
2040	29,437	26,622	32,517	1,344.763	1,340.813	1,395.701	39.076%	42.781%	36.952%	52.61%	52.72%	50.80%
2041	29,437	26,622	32,517	1,371.630	1,365.357	1,425.730	39.076%	42.781%	36.952%	53.36%	53.44%	51.64%
2042	29,437	26,622	32,517	1,398.692	1,390.122	1,455.939	39.076%	42.781%	36.952%	54.19%	54.23%	52.56%
2043	29,437	26,622	32,517	1,426.402	1,415.642	1,486.718	39.076%	42.781%	36.952%	55.10%	55.09%	53.57%
2044	29,437	26,622	32,517	1,455.366	1,442.578	1,518.642	39.076%	42.781%	36.952%	56.10%	56.03%	54.68%
2045	29,437	26,622	32,517	1,485.462	1,470.801	1,551.592	39.076%	42.781%	36.952%	57.20%	57.08%	55.89%
2046	29,437	26,622	32,517	1,517.174	1,500.860	1,586.003	39.076%	42.781%	36.952%	58.42%	58.24%	57.23%
2047	29,437	26,622	32,517	1,550.012	1,532.227	1,621.406	39.076%	42.781%	36.952%	59.76%	59.53%	58.70%
2048	29,437	26,622	32,517	1,583.970	1,564.915	1,657.779	39.076%	42.781%	36.952%	61.25%	60.96%	60.31%
2049	29,437	26,622	32,517	1,618.789	1,598.649	1,694.868	39.076%	42.781%	36.952%	62.89%	62.56%	62.08%
2050	29,437	26,622	32,517	1,654.313	1,633.245	1,732.536	39.076%	42.781%	36.952%	64.70%	64.32%	64.01%
2051	29,437	26,622	32,517	1,690.032	1,668.140	1,770.307	39.076%	42.781%	36.952%	66.68%	66.26%	66.11%
2052	29,437	26,622	32,517	1,726.186	1,703.593	1,808.413	39.076%	42.781%	36.952%	68.84%	68.40%	68.39%
2053	29,437	26,622	32,517	1,762.598	1,739.405	1,846.688	39.076%	42.781%	36.952%	71.19%	70.74%	70.86%
2054	29,437	26,622	32,517	1,799.801	1,776.164	1,885.638	39.076%	42.781%	36.952%	73.75%	73.30%	73.51%
2055	29,437	26,622	32,517	1,837.638	1,813.698	1,925.107	39.076%	42.781%	36.952%	76.52%	76.10%	76.37%
2056	29,437	26,622	32,517	1,876.006	1,851.901	1,964.997	39.076%	42.781%	36.952%	79.51%	79.15%	79.43%
2057	29,437	26,622	32,517	1,914.827	1,890.692	2,005.227	39.076%	42.781%	36.952%	82.75%	82.47%	82.72%
2058	29,437	26,622	32,517	1,954.024	1,929.994	2,045.719	39.076%	42.781%	36.952%	86.24%	86.08%	86.23%
2059	29,437	26,622	32,517	1,993.429	1,969.631	2,086.308	39.076%	42.781%	36.952%	90.00%	90.00%	90.00%
Total Contributions FY 2020-2030				\$10,764.254	\$11,187.793	\$10,670.881						
Total Contributions FY 2031-2059				\$43,930.408	\$43,633.902	\$45,756.677						
Total Contributions FY 2020-2059				\$54,694.662	\$54,821.695	\$56,427.558						
Total Present Value FY 2020-2059				\$15,789.915	\$16,065.743	\$16,020.181						

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.

There were 29,295 active members as of June 30, 2019, plus 142 members assumed to be hired after June 30 to replace recent June retirees.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 1 (4.50% Return in FY 2020, 7.00% Thereafter)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions														
June 30, 2019, Valuation Baseline							Stress Test Scenario 1							
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Increase in AVA Funded Ratio
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio	
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%			
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%	
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%	
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	7.00%	6.16%	925.434	46.43%	46.08%	4.46%	-0.51%	
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	7.00%	6.37%	949.191	46.15%	46.08%	2.57%	-0.28%	
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	7.00%	6.50%	973.110	46.15%	46.12%	2.52%	0.00%	
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	7.00%	6.58%	996.978	46.20%	46.19%	2.45%	0.05%	
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	7.00%	6.64%	1,020.678	46.29%	46.29%	2.38%	0.09%	
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	7.00%	6.68%	1,044.377	46.42%	46.41%	2.32%	0.13%	
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	7.00%	6.72%	1,068.145	46.58%	46.58%	2.28%	0.16%	
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	7.00%	6.75%	1,091.901	46.77%	46.77%	2.22%	0.19%	
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	7.00%	6.77%	1,116.165	47.01%	47.01%	2.22%	0.24%	
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	7.00%	6.79%	1,140.405	47.29%	47.29%	2.17%	0.28%	
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	7.00%	6.81%	1,165.062	47.60%	47.60%	2.16%	0.31%	
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	7.00%	6.82%	1,189.992	47.95%	47.95%	2.14%	0.35%	
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	7.00%	6.83%	1,215.144	48.33%	48.33%	2.11%	0.39%	
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	7.00%	6.84%	1,240.310	48.76%	48.76%	2.07%	0.43%	
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	7.00%	6.85%	1,265.633	49.24%	49.24%	2.04%	0.48%	
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	7.00%	6.86%	1,290.941	49.77%	49.77%	2.00%	0.53%	
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	7.00%	6.87%	1,316.495	50.36%	50.36%	1.98%	0.59%	
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	7.00%	6.87%	1,342.381	51.00%	51.00%	1.97%	0.64%	
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	7.00%	6.88%	1,368.922	51.71%	51.71%	1.98%	0.71%	
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	7.00%	6.89%	1,396.272	52.49%	52.49%	2.00%	0.78%	
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	7.00%	6.89%	1,423.820	53.34%	53.34%	1.97%	0.85%	
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	7.00%	6.89%	1,452.028	54.27%	54.27%	1.98%	0.93%	
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	7.00%	6.90%	1,481.512	55.30%	55.30%	2.03%	1.03%	
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	7.00%	6.90%	1,512.149	56.43%	56.43%	2.07%	1.13%	
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	7.00%	6.91%	1,544.431	57.68%	57.68%	2.13%	1.25%	
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	7.00%	6.91%	1,577.858	59.05%	59.05%	2.16%	1.38%	
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	7.00%	6.91%	1,612.427	60.58%	60.58%	2.19%	1.52%	
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	7.00%	6.92%	1,647.872	62.26%	62.26%	2.20%	1.68%	
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	7.00%	6.92%	1,684.034	64.11%	64.11%	2.19%	1.85%	
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	7.00%	6.92%	1,720.394	66.14%	66.14%	2.16%	2.03%	
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	7.00%	6.92%	1,757.197	68.35%	68.35%	2.14%	2.21%	
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	7.00%	6.93%	1,794.263	70.76%	70.76%	2.11%	2.41%	
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	7.00%	6.93%	1,832.135	73.37%	73.37%	2.11%	2.62%	
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	7.00%	6.93%	1,870.652	76.21%	76.21%	2.10%	2.83%	
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	7.00%	6.93%	1,909.709	79.27%	79.27%	2.09%	3.06%	
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	7.00%	6.93%	1,949.228	82.58%	82.58%	2.07%	3.31%	
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	7.00%	6.94%	1,989.129	86.15%	86.15%	2.05%	3.57%	
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	7.00%	6.94%	2,029.242	90.00%	90.00%	2.02%	3.85%	
Total Contributions FY 2020-2059		\$54,694.662							\$55,646.009					
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$16,044.368					
Difference from June 30, 2019, Valuation														
Total Contributions FY 2020-2059		\$0.000							\$951.347					
Total Present Value FY 2020-2059 ¹		\$0.000							\$254.453					
Maximum Annual Increase					5.68%	3.76%								
Maximum Annual Decrease					1.97%	-0.26%								

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 2 (Volatile Returns Averaging 7.00%)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions														
June 30, 2019, Valuation Baseline							Stress Test Scenario 2							
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio	
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%			
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%	
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%	
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	12.63%	7.99%	925.441	47.03%	48.51%	4.46%	0.10%	
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	-4.05%	4.84%	949.198	46.16%	43.61%	2.57%	-0.88%	
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	10.66%	5.98%	935.164	45.72%	44.86%	-1.48%	-0.44%	
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	1.21%	5.17%	1,038.959	45.01%	42.60%	11.10%	-0.71%	
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	3.25%	4.89%	1,039.021	43.24%	41.09%	0.01%	-1.77%	
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	-12.95%	2.48%	1,104.896	40.80%	33.57%	6.34%	-2.44%	
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	-1.92%	1.98%	1,156.455	37.29%	30.64%	4.67%	-3.51%	
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	-8.19%	0.91%	1,321.858	33.59%	26.60%	14.30%	-3.70%	
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	2.01%	1.01%	1,403.327	29.98%	25.62%	6.16%	-3.62%	
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	7.57%	1.54%	1,517.159	28.32%	26.24%	8.11%	-1.65%	
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	32.40%	3.64%	1,574.274	28.96%	32.90%	3.76%	0.64%	
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	14.44%	4.37%	1,605.173	31.37%	36.13%	1.96%	2.41%	
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	4.45%	4.38%	1,507.318	33.73%	36.00%	-6.10%	2.35%	
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	19.46%	5.26%	1,487.319	37.15%	40.73%	-1.33%	3.42%	
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	15.39%	5.83%	1,537.773	40.27%	44.84%	3.39%	3.12%	
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	10.49%	6.09%	1,466.752	43.08%	47.15%	-4.62%	2.81%	
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	1.24%	5.83%	1,414.960	45.34%	45.36%	-3.53%	2.26%	
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	10.62%	6.06%	1,404.062	46.98%	47.45%	-0.77%	1.64%	
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	-7.06%	5.40%	1,502.200	46.57%	42.11%	6.99%	-0.41%	
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	4.53%	5.36%	1,487.939	45.30%	41.64%	-0.95%	-1.27%	
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	-3.69%	4.95%	1,706.297	44.01%	38.56%	14.68%	-1.29%	
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	0.11%	4.74%	1,770.963	41.72%	37.10%	3.79%	-2.29%	
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	6.26%	4.80%	1,944.687	41.11%	38.20%	9.81%	-0.60%	
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	3.49%	4.75%	2,072.320	40.74%	38.68%	6.56%	-0.37%	
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	4.11%	4.72%	2,126.069	41.22%	39.48%	2.59%	0.48%	
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	7.29%	4.82%	2,221.561	42.63%	41.68%	4.49%	1.41%	
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	4.72%	4.81%	2,314.266	44.24%	43.23%	4.17%	1.61%	
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	32.14%	5.62%	2,360.409	48.90%	56.02%	1.99%	4.67%	
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	14.07%	5.89%	2,456.089	55.40%	63.38%	4.05%	6.50%	
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	27.74%	6.51%	1,951.854	64.12%	78.55%	-20.53%	8.72%	
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	8.91%	6.58%	1,765.460	73.61%	82.99%	-9.55%	9.49%	
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	15.00%	6.82%	943.483	80.68%	90.40%	-46.56%	7.07%	
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	5.01%	6.77%	849.655	86.56%	89.86%	-9.94%	5.88%	
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	-2.16%	6.51%	272.374	86.10%	81.53%	-67.94%	-0.45%	
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	2.58%	6.40%	475.485	84.60%	77.67%	74.57%	-1.50%	
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	3.25%	6.32%	1,662.057	83.57%	77.34%	249.55%	-1.03%	
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	16.54%	6.57%	2,438.741	86.41%	88.85%	46.73%	2.83%	
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	25.30%	7.00%	3,386.465	97.66%	112.08%	38.86%	11.26%	
Total Contributions FY 2020-2059		\$54,694.662							\$60,837.876					
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$17,850.729					
Difference from June 30, 2019, Valuation														
Total Contributions FY 2020-2059		\$0.000							\$6,143.214					
Total Present Value FY 2020-2059 ¹		\$0.000							\$2,060.814					
Maximum Annual Increase					5.68%	3.76%								
Maximum Annual Decrease					1.97%	-0.26%								

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 3 (Static Returns of 4.12%)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions													
June 30, 2019, Valuation Baseline							Stress Test Scenario 3						
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in	
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%		
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	4.12%	5.20%	925.441	46.12%	44.83%	4.46%	-0.82%
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	4.12%	4.93%	949.198	45.20%	43.57%	2.57%	-0.92%
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	4.12%	4.77%	992.519	44.29%	42.38%	4.56%	-0.90%
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	4.12%	4.66%	1,036.345	43.17%	41.27%	4.42%	-1.13%
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	4.12%	4.58%	1,080.483	42.10%	40.24%	4.26%	-1.07%
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	4.12%	4.52%	1,125.099	41.10%	39.28%	4.13%	-1.00%
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	4.12%	4.48%	1,170.398	40.17%	38.40%	4.03%	-0.93%
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	4.12%	4.44%	1,216.176	39.32%	37.59%	3.91%	-0.85%
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	4.12%	4.41%	1,263.145	38.57%	36.88%	3.86%	-0.75%
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	4.12%	4.39%	1,310.657	37.90%	36.25%	3.76%	-0.67%
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	4.12%	4.37%	1,359.375	37.31%	35.69%	3.72%	-0.59%
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	4.12%	4.35%	1,409.161	36.79%	35.20%	3.66%	-0.52%
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	4.12%	4.33%	1,460.015	36.35%	34.79%	3.61%	-0.44%
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	4.12%	4.32%	1,511.739	36.00%	34.46%	3.54%	-0.35%
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	4.12%	4.31%	1,564.602	35.74%	34.22%	3.50%	-0.26%
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	4.12%	4.30%	1,618.511	35.57%	34.06%	3.45%	-0.17%
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	4.12%	4.29%	1,673.737	35.50%	34.00%	3.41%	-0.07%
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	4.12%	4.28%	1,730.702	35.52%	34.03%	3.40%	0.02%
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	4.12%	4.27%	1,789.823	35.65%	34.16%	3.42%	0.13%
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	4.12%	4.27%	1,851.713	35.89%	34.40%	3.46%	0.24%
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	4.12%	4.26%	1,915.391	36.26%	34.75%	3.44%	0.36%
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	4.12%	4.25%	1,982.037	36.75%	35.23%	3.48%	0.49%
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	4.12%	4.25%	2,052.740	37.39%	35.86%	3.57%	0.64%
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	4.12%	4.24%	2,127.706	38.20%	36.64%	3.65%	0.81%
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	4.12%	4.24%	2,208.021	39.21%	37.61%	3.77%	1.00%
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	4.12%	4.24%	2,293.696	40.43%	38.80%	3.88%	1.22%
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	4.12%	4.23%	2,385.704	41.90%	40.22%	4.01%	1.47%
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	4.12%	4.23%	2,484.003	43.64%	41.90%	4.12%	1.74%
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	4.12%	4.22%	2,590.320	45.69%	43.88%	4.28%	2.05%
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	4.12%	4.22%	2,705.187	48.07%	46.18%	4.43%	2.38%
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	4.12%	4.22%	2,831.426	50.83%	48.84%	4.67%	2.76%
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	4.12%	4.21%	2,971.181	54.00%	51.90%	4.94%	3.17%
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	4.12%	4.21%	3,132.625	57.65%	55.42%	5.43%	3.65%
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	4.12%	4.21%	3,320.663	61.86%	59.48%	6.00%	4.21%
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	4.12%	4.21%	3,552.715	66.76%	64.20%	6.99%	4.90%
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	4.12%	4.20%	3,848.486	72.53%	69.77%	8.33%	5.77%
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	4.12%	4.20%	4,282.590	79.57%	76.57%	11.28%	7.03%
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	4.12%	4.20%	4,930.731	88.45%	85.18%	15.13%	8.89%
Total Contributions FY 2020-2059		\$54,694.662							\$80,394.454				
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$19,691.504				
Difference from June 30, 2019, Valuation													
Total Contributions FY 2020-2059		\$0.000							\$25,699.792				
Total Present Value FY 2020-2059 ¹		\$0.000							\$3,901.589				
Maximum Annual Increase					5.68%	3.76%						15.13%	8.89%
Maximum Annual Decrease					1.97%	-0.26%						2.57%	-1.13%

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 4 (Volatile Returns Averaging 4.12%)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions													
June 30, 2019, Valuation Baseline							Stress Test Scenario 4						
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in	
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%		
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	26.54%	12.26%	925.441	48.54%	54.53%	4.46%	1.60%
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	0.84%	9.29%	949.198	49.76%	51.80%	2.57%	1.22%
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	-9.68%	5.20%	841.002	48.80%	43.49%	-11.40%	-0.96%
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	21.50%	7.76%	912.757	49.43%	48.99%	8.53%	0.63%
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	6.91%	7.64%	1,069.664	48.64%	49.38%	17.19%	-0.79%
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	-8.95%	5.41%	993.124	46.29%	42.09%	-7.16%	-2.35%
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	4.53%	5.31%	1,016.428	45.44%	40.87%	2.35%	-0.86%
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	-13.53%	3.26%	1,174.536	41.74%	33.27%	15.56%	-3.69%
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	8.05%	3.68%	1,219.011	37.99%	33.51%	3.79%	-3.75%
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	6.35%	3.90%	1,398.299	36.25%	33.79%	14.71%	-1.74%
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	-19.58%	1.87%	1,421.979	32.66%	26.05%	1.69%	-3.59%
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	-0.69%	1.69%	1,456.605	30.31%	24.36%	2.44%	-2.35%
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	-5.14%	1.22%	1,667.085	27.79%	22.36%	14.45%	-2.52%
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	24.12%	2.52%	1,742.419	26.23%	26.46%	4.52%	-1.56%
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	-4.00%	2.12%	1,840.336	26.16%	24.96%	5.62%	-0.07%
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	17.23%	2.91%	1,793.822	26.95%	28.06%	-2.53%	0.79%
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	16.58%	3.59%	1,895.803	29.36%	31.71%	5.69%	2.41%
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	32.69%	4.88%	1,872.221	32.92%	40.43%	-1.24%	3.56%
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	9.64%	5.10%	1,841.767	37.45%	42.98%	-1.63%	4.54%
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	-19.98%	3.80%	1,662.373	38.46%	33.27%	-9.74%	1.01%
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	-11.06%	3.11%	1,664.936	37.39%	28.23%	0.15%	-1.07%
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	2.22%	3.07%	2,040.252	35.03%	28.20%	22.54%	-2.36%
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	17.49%	3.61%	2,265.015	33.33%	32.64%	11.02%	-1.70%
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	7.57%	3.76%	2,355.219	34.28%	35.01%	3.98%	0.94%
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	4.72%	3.80%	2,305.472	36.31%	36.40%	-2.11%	2.03%
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	16.37%	4.22%	2,348.271	39.76%	41.82%	1.86%	3.45%
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	-0.26%	4.06%	2,430.934	42.38%	41.76%	3.52%	2.62%
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	-6.26%	3.70%	2,339.561	43.51%	39.06%	-3.76%	1.14%
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	6.34%	3.79%	2,532.798	45.24%	41.59%	8.26%	1.73%
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	-5.39%	3.49%	2,868.797	45.92%	40.58%	13.27%	0.68%
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	13.87%	3.79%	2,944.461	48.02%	47.06%	2.64%	2.10%
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	1.02%	3.70%	3,334.753	51.84%	49.67%	13.26%	3.81%
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	23.86%	4.23%	3,198.213	57.53%	62.55%	-4.09%	5.69%
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	1.56%	4.16%	3,515.581	64.83%	65.84%	9.92%	7.30%
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	-2.54%	3.97%	2,683.913	68.27%	64.30%	-23.66%	3.45%
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	19.20%	4.34%	3,221.050	75.74%	77.50%	20.01%	7.47%
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	5.70%	4.38%	4,569.654	84.99%	86.48%	41.87%	9.25%
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	-5.50%	4.12%	2,257.204	86.90%	80.52%	-50.60%	1.91%
Total Contributions FY 2020-2059		\$54,694.662								\$78,310.347			
Total Present Value FY 2020-2059 ¹		\$15,789.915								\$19,586.218			
Difference from June 30, 2019, Valuation													
Total Contributions FY 2020-2059		\$0.000								\$23,615.685			
Total Present Value FY 2020-2059 ¹		\$0.000								\$3,796.303			
Maximum Annual Increase					5.68%	3.76%						41.87%	9.25%
Maximum Annual Decrease					1.97%	-0.26%						-50.60%	-3.75%

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 5 (Static Returns of 5.81%)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions														
June 30, 2019, Valuation Baseline							Stress Test Scenario 5							
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio	
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%			
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%	
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%	
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	5.81%	5.77%	925.441	46.30%	45.56%	4.46%	-0.64%	
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	5.81%	5.78%	949.198	45.75%	45.03%	2.57%	-0.55%	
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	5.81%	5.78%	981.129	45.37%	44.55%	3.36%	-0.38%	
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	5.81%	5.79%	1,013.381	44.92%	44.10%	3.29%	-0.45%	
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	5.81%	5.79%	1,045.818	44.51%	43.70%	3.20%	-0.41%	
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	5.81%	5.79%	1,078.625	44.15%	43.34%	3.14%	-0.37%	
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	5.81%	5.80%	1,111.874	43.82%	43.02%	3.08%	-0.33%	
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	5.81%	5.80%	1,145.501	43.54%	42.74%	3.02%	-0.29%	
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	5.81%	5.80%	1,180.111	43.32%	42.53%	3.02%	-0.22%	
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	5.81%	5.80%	1,215.089	43.15%	42.37%	2.96%	-0.17%	
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	5.81%	5.80%	1,251.026	43.03%	42.25%	2.96%	-0.12%	
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	5.81%	5.80%	1,287.743	42.95%	42.18%	2.93%	-0.08%	
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	5.81%	5.80%	1,325.219	42.93%	42.15%	2.91%	-0.02%	
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	5.81%	5.80%	1,363.282	42.96%	42.19%	2.87%	0.03%	
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	5.81%	5.80%	1,402.134	43.06%	42.28%	2.85%	0.10%	
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	5.81%	5.80%	1,441.612	43.22%	42.44%	2.82%	0.16%	
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	5.81%	5.80%	1,482.098	43.45%	42.67%	2.81%	0.23%	
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	5.81%	5.80%	1,523.638	43.75%	42.97%	2.80%	0.30%	
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	5.81%	5.80%	1,566.792	44.14%	43.35%	2.83%	0.38%	
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	5.81%	5.80%	1,611.844	44.61%	43.82%	2.88%	0.47%	
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	5.81%	5.80%	1,658.015	45.18%	44.38%	2.86%	0.57%	
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	5.81%	5.80%	1,706.234	45.85%	45.04%	2.91%	0.67%	
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	5.81%	5.80%	1,757.071	46.64%	45.81%	2.98%	0.79%	
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	5.81%	5.80%	1,811.341	47.56%	46.73%	3.09%	0.93%	
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	5.81%	5.81%	1,868.510	48.65%	47.79%	3.16%	1.08%	
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	5.81%	5.81%	1,928.042	49.90%	49.03%	3.19%	1.25%	
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	5.81%	5.81%	1,993.496	51.35%	50.46%	3.39%	1.45%	
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	5.81%	5.81%	2,061.945	53.03%	52.11%	3.43%	1.67%	
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	5.81%	5.81%	2,134.433	54.94%	53.99%	3.52%	1.91%	
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	5.81%	5.81%	2,211.711	57.12%	56.13%	3.62%	2.18%	
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	5.81%	5.81%	2,294.322	59.58%	58.56%	3.74%	2.46%	
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	5.81%	5.81%	2,384.134	62.36%	61.30%	3.91%	2.78%	
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	5.81%	5.81%	2,484.700	65.49%	64.38%	4.22%	3.13%	
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	5.81%	5.81%	2,598.732	69.03%	67.86%	4.59%	3.54%	
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	5.81%	5.81%	2,732.581	73.03%	71.80%	5.15%	4.00%	
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	5.81%	5.81%	2,894.882	77.60%	76.30%	5.94%	4.57%	
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	5.81%	5.81%	3,122.157	82.93%	81.55%	7.85%	5.33%	
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	5.81%	5.81%	3,455.855	89.32%	87.85%	10.69%	6.40%	
Total Contributions FY 2020-2059		\$54,694.662							\$67,740.109					
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$17,784.763					
Difference from June 30, 2019, Valuation														
Total Contributions FY 2020-2059		\$0.000							\$13,045.447					
Total Present Value FY 2020-2059 ¹		\$0.000							\$1,994.848					
Maximum Annual Increase					5.68%	3.76%						10.69%	6.40%	
Maximum Annual Decrease					1.97%	-0.26%						2.57%	-0.64%	

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 6 (Volatile Returns Averaging 5.81%)
Based on Actuarial Valuation as of June 30, 2019

								\$ in Millions						
June 30, 2019, Valuation Baseline								Stress Test Scenario 6						
				Increase in								Increase in		
Year Ending	Projected	Total	Actuarial	Market	Total	Increase in	Annual Rate	Cumulative	Total	Actuarial	Market	Total	Increase in	
June 30	Investment	Required	Value of	Value of	Required	AVA Funded	of	Geometric	Required	Value of	Value of	Required	AVA Funded	
June 30	Return	Employer	Assets (AVA)	Assets (MVA)	Employer	Ratio	Investment	Return	Employer	Assets (AVA)	Assets (MVA)	Employer	AVA Funded	
		Contribution	Funded Ratio	Funded Ratio	Contribution		Return	(from 2020)	Contribution	Funded Ratio	Funded Ratio	Contribution	Ratio	
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%			
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%	
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	885.893	46.94%	46.11%	3.67%	-0.44%	
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	28.97%	12.98%	925.441	48.80%	55.58%	4.46%	1.87%	
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	13.54%	13.12%	949.198	51.96%	59.51%	2.57%	3.16%	
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	-6.68%	8.85%	824.557	53.14%	51.95%	-13.13%	1.18%	
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	12.60%	9.47%	789.322	54.85%	54.26%	-4.27%	1.71%	
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	13.38%	10.02%	936.317	55.72%	57.72%	18.62%	0.88%	
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	-8.11%	7.57%	910.981	53.55%	49.70%	-2.71%	-2.18%	
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	18.05%	8.69%	873.474	54.01%	54.48%	-4.12%	0.46%	
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	12.06%	9.02%	1,044.126	55.08%	57.51%	19.54%	1.08%	
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	-13.10%	6.79%	969.379	52.46%	46.88%	-7.16%	-2.62%	
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	4.34%	6.59%	939.175	51.09%	45.27%	-3.12%	-1.37%	
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	1.34%	6.17%	1,181.334	48.60%	43.14%	25.78%	-2.49%	
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	3.87%	6.01%	1,231.258	45.11%	42.13%	4.23%	-3.49%	
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	31.29%	7.53%	1,309.629	46.55%	52.10%	6.37%	1.44%	
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	4.00%	7.31%	1,365.323	48.27%	51.66%	4.25%	1.71%	
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	4.55%	7.14%	1,167.943	49.78%	50.79%	-14.46%	1.51%	
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	-5.05%	6.43%	1,227.342	50.45%	45.45%	5.09%	0.67%	
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	17.57%	6.99%	1,281.957	50.11%	50.20%	4.45%	-0.34%	
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	22.52%	7.71%	1,459.706	52.35%	58.54%	13.87%	2.24%	
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	5.14%	7.59%	1,363.980	54.43%	58.62%	-6.56%	2.08%	
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	-5.11%	6.98%	1,179.030	55.75%	52.42%	-13.56%	1.31%	
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	22.99%	7.63%	1,233.434	58.18%	60.62%	4.61%	2.43%	
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	9.52%	7.71%	1,471.226	60.04%	63.48%	19.28%	1.86%	
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	10.57%	7.82%	1,236.556	62.02%	66.50%	-15.95%	1.98%	
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	-13.69%	6.90%	1,211.235	62.35%	54.34%	-2.05%	0.33%	
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	3.31%	6.76%	1,156.453	60.04%	52.46%	-4.52%	-2.30%	
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	23.00%	7.31%	1,702.569	60.61%	61.81%	47.22%	0.56%	
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	5.89%	7.26%	1,820.412	60.79%	63.44%	6.92%	0.18%	
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	5.62%	7.20%	1,496.157	62.98%	64.04%	-17.81%	2.19%	
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	2.71%	7.05%	1,561.383	65.41%	63.03%	4.36%	2.42%	
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	4.50%	6.97%	1,640.841	65.95%	63.27%	5.09%	0.54%	
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	9.96%	7.06%	1,832.714	67.56%	67.30%	11.69%	1.61%	
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	-14.67%	6.35%	1,977.275	66.29%	56.23%	7.89%	-1.26%	
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	-8.96%	5.88%	1,877.679	62.99%	49.57%	-5.04%	-3.30%	
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	-2.21%	5.65%	3,212.043	62.11%	50.20%	71.06%	-0.87%	
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	7.90%	5.71%	4,248.059	63.40%	58.24%	32.25%	1.28%	
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	3.98%	5.66%	4,974.040	69.48%	66.47%	17.09%	6.08%	
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	10.82%	5.79%	4,985.220	78.68%	79.24%	0.22%	9.20%	
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	6.55%	5.81%	5,625.625	91.21%	91.58%	12.85%	12.52%	
Total Contributions FY 2020-2059		\$54,694.662								\$66,932.786				
Total Present Value FY 2020-2059 ¹		\$15,789.915								\$16,240.381				
Difference from June 30, 2019, Valuation														
Total Contributions FY 2020-2059		\$0.000								\$12,238.124				
Total Present Value FY 2020-2059 ¹		\$0.000								\$450.466				
Maximum Annual Increase					5.68%	3.76%								
Maximum Annual Decrease					1.97%	-0.26%								

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 7 (Population Decreases 1% Per Year for 10 years)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions													
June 30, 2019, Valuation Baseline							Stress Test Scenario 7						
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in	
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%		
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	880.097	46.91%	46.08%	3.00%	-0.46%
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	7.00%	6.16%	982.098	46.62%	46.27%	11.59%	-0.29%
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	7.00%	6.37%	1,000.335	46.55%	46.48%	1.86%	-0.07%
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	7.00%	6.50%	1,018.224	46.74%	46.71%	1.79%	0.19%
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	7.00%	6.58%	1,035.554	46.95%	46.94%	1.70%	0.21%
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	7.00%	6.64%	1,052.177	47.18%	47.18%	1.61%	0.23%
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	7.00%	6.68%	1,068.263	47.43%	47.43%	1.53%	0.24%
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	7.00%	6.72%	1,083.872	47.68%	47.68%	1.46%	0.25%
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	7.00%	6.75%	1,098.866	47.92%	47.92%	1.38%	0.25%
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	7.00%	6.77%	1,113.807	48.19%	48.19%	1.36%	0.27%
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	7.00%	6.79%	1,135.812	48.48%	48.48%	1.98%	0.29%
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	7.00%	6.81%	1,158.186	48.81%	48.81%	1.97%	0.32%
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	7.00%	6.82%	1,180.758	49.16%	49.16%	1.95%	0.35%
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	7.00%	6.83%	1,203.462	49.54%	49.54%	1.92%	0.38%
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	7.00%	6.84%	1,226.065	49.96%	49.96%	1.88%	0.42%
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	7.00%	6.85%	1,248.729	50.42%	50.42%	1.85%	0.46%
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	7.00%	6.86%	1,271.300	50.92%	50.92%	1.81%	0.50%
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	7.00%	6.87%	1,294.056	51.47%	51.47%	1.79%	0.55%
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	7.00%	6.87%	1,317.107	52.07%	52.07%	1.78%	0.60%
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	7.00%	6.88%	1,340.813	52.72%	52.72%	1.80%	0.66%
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	7.00%	6.89%	1,365.357	53.44%	53.44%	1.83%	0.72%
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	7.00%	6.89%	1,390.122	54.23%	54.23%	1.81%	0.78%
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	7.00%	6.89%	1,415.642	55.09%	55.09%	1.84%	0.86%
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	7.00%	6.90%	1,442.578	56.03%	56.03%	1.90%	0.95%
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	7.00%	6.90%	1,470.801	57.08%	57.08%	1.96%	1.05%
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	7.00%	6.91%	1,500.860	58.24%	58.24%	2.04%	1.16%
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	7.00%	6.91%	1,532.227	59.53%	59.53%	2.09%	1.29%
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	7.00%	6.91%	1,564.915	60.96%	60.96%	2.13%	1.43%
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	7.00%	6.92%	1,598.649	62.56%	62.56%	2.16%	1.59%
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	7.00%	6.92%	1,633.245	64.32%	64.32%	2.16%	1.76%
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	7.00%	6.92%	1,668.140	66.26%	66.26%	2.14%	1.94%
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	7.00%	6.92%	1,703.593	68.40%	68.40%	2.13%	2.14%
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	7.00%	6.93%	1,739.405	70.74%	70.74%	2.10%	2.34%
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	7.00%	6.93%	1,776.164	73.30%	73.30%	2.11%	2.56%
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	7.00%	6.93%	1,813.698	76.10%	76.10%	2.11%	2.80%
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	7.00%	6.93%	1,851.901	79.15%	79.15%	2.11%	3.05%
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	7.00%	6.93%	1,890.692	82.47%	82.47%	2.09%	3.32%
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	7.00%	6.94%	1,929.994	86.08%	86.08%	2.08%	3.61%
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	7.00%	6.94%	1,969.631	90.00%	90.00%	2.05%	3.92%
Total Contributions FY 2020-2059		\$54,694.662							\$54,821.695				
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$16,065.743				
Difference from June 30, 2019, Valuation													
Total Contributions FY 2020-2059		\$0.000							\$127.033				
Total Present Value FY 2020-2059 ¹		\$0.000							\$275.828				
Maximum Annual Increase					5.68%	3.76%						11.59%	3.92%
Maximum Annual Decrease					1.97%	-0.26%						1.36%	-0.46%

¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



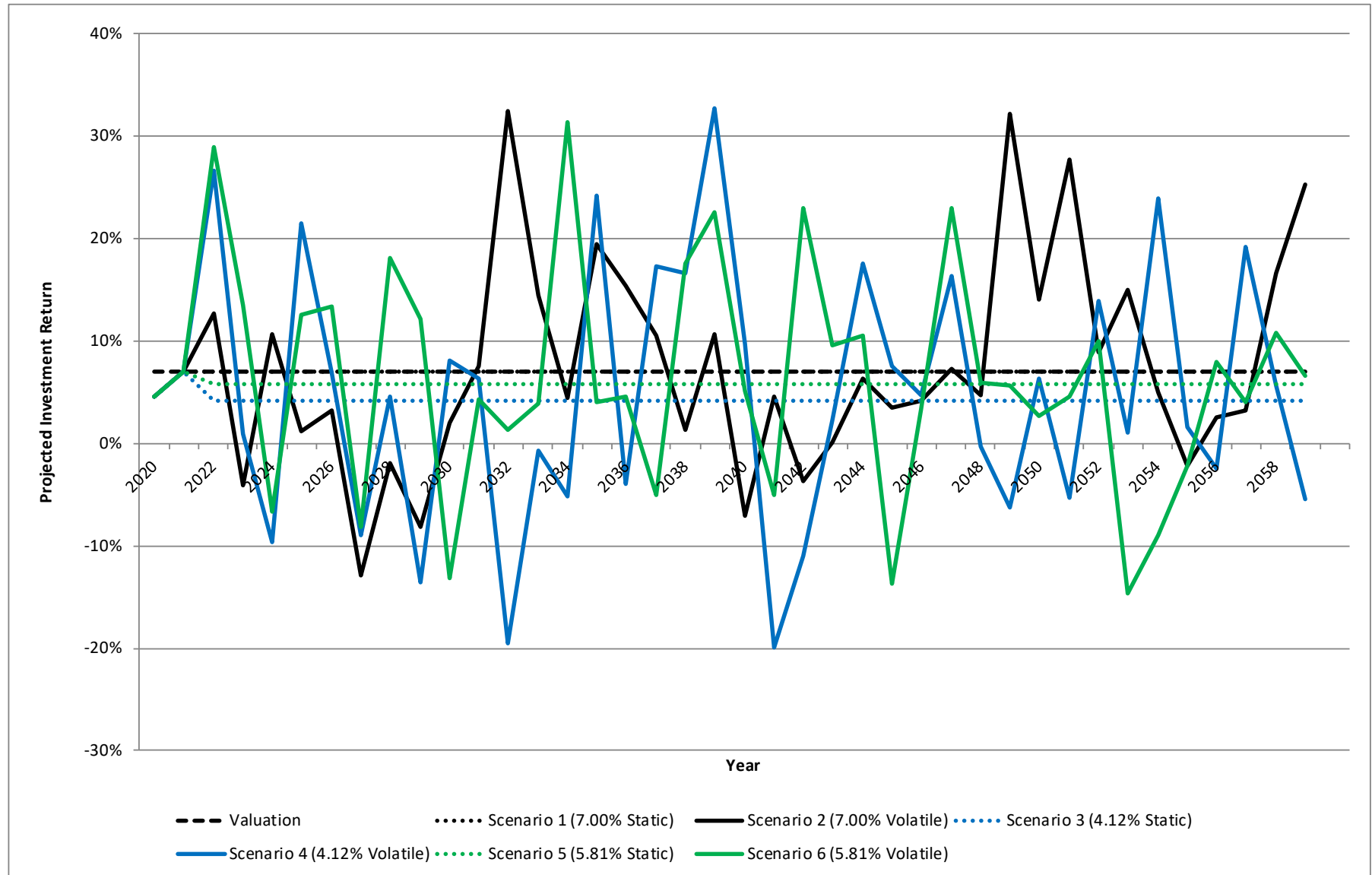
Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Actuarial Valuation Results and Stress Testing Scenario 8 (Population Increases 1% Per Year for 10 years)
Based on Actuarial Valuation as of June 30, 2019

\$ in Millions													
June 30, 2019, Valuation Baseline							Stress Test Scenario 8						
Year Ending June 30	Projected Investment Return	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in		Annual Rate of Investment Return	Cumulative Geometric Return (from 2020)	Total Required Employer Contribution	Actuarial Value of Assets (AVA) Funded Ratio	Market Value of Assets (MVA) Funded Ratio	Increase in	
					Total Required Employer Contribution	Increase in AVA Funded Ratio						Total Required Employer Contribution	Increase in AVA Funded Ratio
2019		\$808.570	47.40%	47.47%					\$808.570	47.40%	47.47%		
2020	7.00%	854.500	47.65%	47.34%	5.68%	0.25%	4.50%	4.50%	854.500	47.37%	46.23%	5.68%	-0.03%
2021	7.00%	885.893	47.52%	47.28%	3.67%	-0.14%	7.00%	5.74%	891.690	46.96%	46.13%	4.35%	-0.41%
2022	7.00%	909.101	47.26%	47.24%	2.62%	-0.26%	7.00%	6.16%	871.187	46.24%	45.89%	-2.30%	-0.72%
2023	7.00%	932.439	47.25%	47.24%	2.57%	-0.01%	7.00%	6.37%	899.817	45.76%	45.70%	3.29%	-0.48%
2024	7.00%	955.936	47.27%	47.26%	2.52%	0.02%	7.00%	6.50%	929.169	45.58%	45.55%	3.26%	-0.18%
2025	7.00%	979.383	47.32%	47.32%	2.45%	0.05%	7.00%	6.58%	959.089	45.47%	45.46%	3.22%	-0.11%
2026	7.00%	1,002.665	47.41%	47.41%	2.38%	0.09%	7.00%	6.64%	989.456	45.42%	45.42%	3.17%	-0.05%
2027	7.00%	1,025.946	47.53%	47.53%	2.32%	0.12%	7.00%	6.68%	1,020.474	45.44%	45.44%	3.13%	0.02%
2028	7.00%	1,049.294	47.68%	47.68%	2.28%	0.15%	7.00%	6.72%	1,052.259	45.52%	45.52%	3.11%	0.08%
2029	7.00%	1,072.631	47.86%	47.86%	2.22%	0.18%	7.00%	6.75%	1,084.756	45.65%	45.65%	3.09%	0.14%
2030	7.00%	1,096.466	48.09%	48.09%	2.22%	0.23%	7.00%	6.77%	1,118.484	45.88%	45.88%	3.11%	0.23%
2031	7.00%	1,120.278	48.35%	48.35%	2.17%	0.26%	7.00%	6.79%	1,144.851	46.15%	46.15%	2.36%	0.27%
2032	7.00%	1,144.501	48.65%	48.65%	2.16%	0.30%	7.00%	6.81%	1,171.679	46.46%	46.46%	2.34%	0.31%
2033	7.00%	1,168.991	48.98%	48.98%	2.14%	0.33%	7.00%	6.82%	1,198.847	46.81%	46.81%	2.32%	0.35%
2034	7.00%	1,193.698	49.35%	49.35%	2.11%	0.37%	7.00%	6.83%	1,226.318	47.20%	47.20%	2.29%	0.40%
2035	7.00%	1,218.420	49.76%	49.76%	2.07%	0.41%	7.00%	6.84%	1,253.913	47.65%	47.65%	2.25%	0.45%
2036	7.00%	1,243.297	50.23%	50.23%	2.04%	0.46%	7.00%	6.85%	1,281.757	48.16%	48.16%	2.22%	0.50%
2037	7.00%	1,268.158	50.74%	50.74%	2.00%	0.51%	7.00%	6.86%	1,309.662	48.72%	48.72%	2.18%	0.56%
2038	7.00%	1,293.261	51.30%	51.30%	1.98%	0.57%	7.00%	6.87%	1,337.874	49.35%	49.35%	2.15%	0.63%
2039	7.00%	1,318.690	51.93%	51.93%	1.97%	0.62%	7.00%	6.87%	1,366.458	50.04%	50.04%	2.14%	0.69%
2040	7.00%	1,344.763	52.61%	52.61%	1.98%	0.69%	7.00%	6.88%	1,395.701	50.80%	50.80%	2.14%	0.76%
2041	7.00%	1,371.630	53.36%	53.36%	2.00%	0.75%	7.00%	6.89%	1,425.730	51.64%	51.64%	2.15%	0.84%
2042	7.00%	1,398.692	54.19%	54.19%	1.97%	0.83%	7.00%	6.89%	1,455.939	52.56%	52.56%	2.12%	0.92%
2043	7.00%	1,426.402	55.10%	55.10%	1.98%	0.91%	7.00%	6.89%	1,486.718	53.57%	53.57%	2.11%	1.01%
2044	7.00%	1,455.366	56.10%	56.10%	2.03%	1.00%	7.00%	6.90%	1,518.642	54.68%	54.68%	2.15%	1.11%
2045	7.00%	1,485.462	57.20%	57.20%	2.07%	1.10%	7.00%	6.90%	1,551.592	55.89%	55.89%	2.17%	1.22%
2046	7.00%	1,517.174	58.42%	58.42%	2.13%	1.22%	7.00%	6.91%	1,586.003	57.23%	57.23%	2.22%	1.34%
2047	7.00%	1,550.012	59.76%	59.76%	2.16%	1.35%	7.00%	6.91%	1,621.406	58.70%	58.70%	2.23%	1.47%
2048	7.00%	1,583.970	61.25%	61.25%	2.19%	1.49%	7.00%	6.91%	1,657.779	60.31%	60.31%	2.24%	1.61%
2049	7.00%	1,618.789	62.89%	62.89%	2.20%	1.64%	7.00%	6.92%	1,694.868	62.08%	62.08%	2.24%	1.77%
2050	7.00%	1,654.313	64.70%	64.70%	2.19%	1.81%	7.00%	6.92%	1,732.536	64.01%	64.01%	2.22%	1.93%
2051	7.00%	1,690.032	66.68%	66.68%	2.16%	1.98%	7.00%	6.92%	1,770.307	66.11%	66.11%	2.18%	2.10%
2052	7.00%	1,726.186	68.84%	68.84%	2.14%	2.16%	7.00%	6.92%	1,808.413	68.39%	68.39%	2.15%	2.28%
2053	7.00%	1,762.598	71.19%	71.19%	2.11%	2.35%	7.00%	6.93%	1,846.688	70.86%	70.86%	2.12%	2.46%
2054	7.00%	1,799.801	73.75%	73.75%	2.11%	2.56%	7.00%	6.93%	1,885.638	73.51%	73.51%	2.11%	2.66%
2055	7.00%	1,837.638	76.52%	76.52%	2.10%	2.77%	7.00%	6.93%	1,925.107	76.37%	76.37%	2.09%	2.86%
2056	7.00%	1,876.006	79.51%	79.51%	2.09%	2.99%	7.00%	6.93%	1,964.997	79.43%	79.43%	2.07%	3.06%
2057	7.00%	1,914.827	82.75%	82.75%	2.07%	3.23%	7.00%	6.93%	2,005.227	82.72%	82.72%	2.05%	3.28%
2058	7.00%	1,954.024	86.24%	86.24%	2.05%	3.49%	7.00%	6.94%	2,045.719	86.23%	86.23%	2.02%	3.52%
2059	7.00%	1,993.429	90.00%	90.00%	2.02%	3.76%	7.00%	6.94%	2,086.308	90.00%	90.00%	1.98%	3.76%
Total Contributions FY 2020-2059		\$54,694.662							\$56,427.558				
Total Present Value FY 2020-2059 ¹		\$15,789.915							\$16,020.181				
Difference from June 30, 2019, Valuation													
Total Contributions FY 2020-2059		\$0.000							\$1,732.896				
Total Present Value FY 2020-2059 ¹		\$0.000							\$230.266				
Maximum Annual Increase					5.68%	3.76%						5.68%	3.76%
Maximum Annual Decrease					1.97%	-0.26%						-2.30%	-0.72%

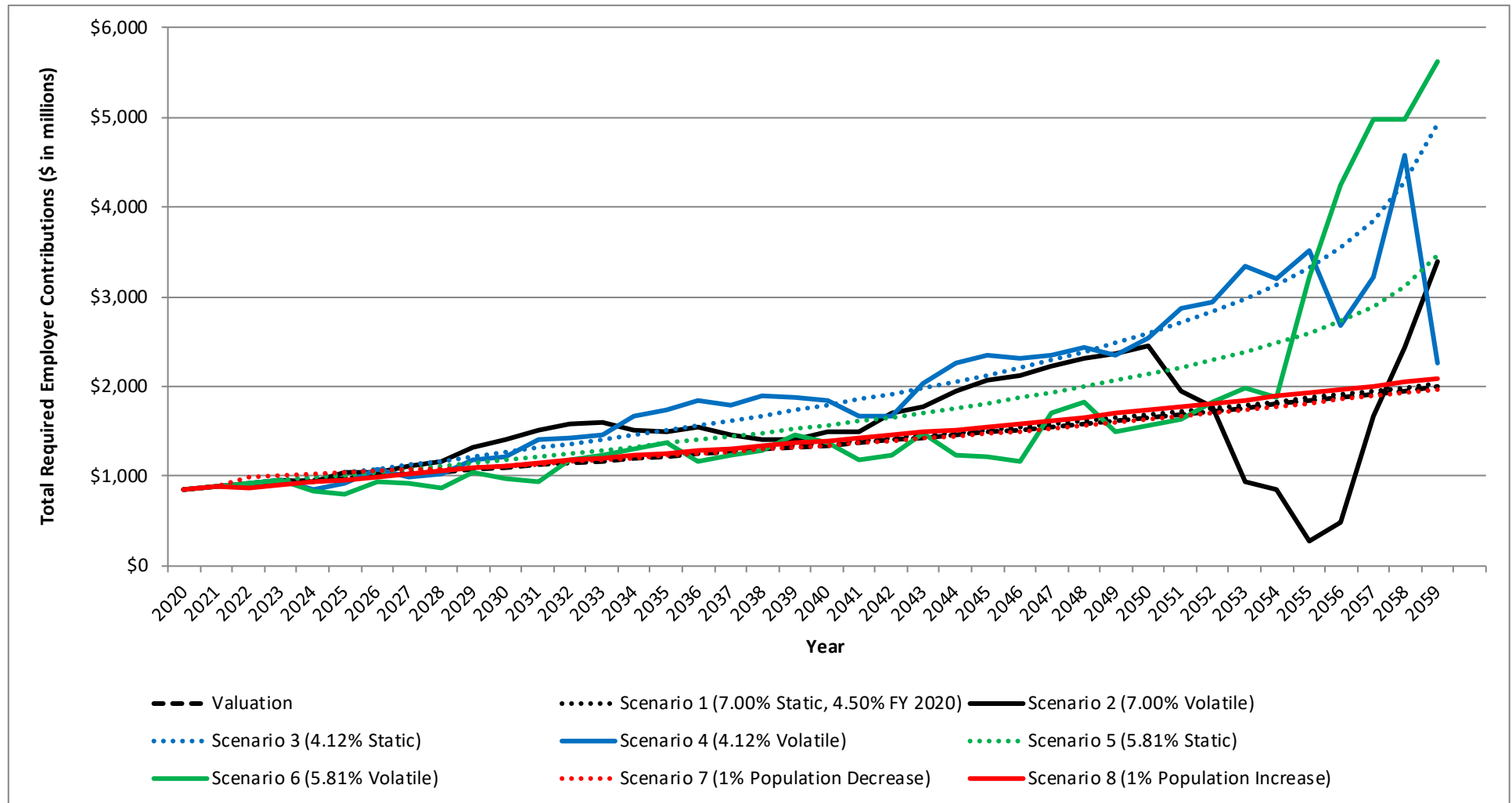
¹In calculating the present value of contributions, total required employer contributions were assumed to occur mid-year.



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Projected Rates of Investment Return
Assuming Valuation Assumptions (Annual Return of 7.00% and Constant Active Population)
and Alternate Rates of Return and Active Population Changes Under the Stress Testing Scenarios



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Total Required Employer Contributions
Assuming Valuation Assumptions (Annual Return of 7.00% and Constant Active Population)
and Alternate Rates of Return and Active Population Changes Under the Stress Testing Scenarios



Public School Teachers' Pension and Retirement Fund of Chicago
Comparison of Funded Ratio
Assuming Valuation Assumptions (Annual Return of 7.00% and Constant Active Population)
and Alternate Rates of Return and Active Population Changes Under the Stress Testing Scenarios

